Diary Data Analysis

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Primary Analyses

1. **Prediction 1:** *Training will reduce antisocial behaviours reported byadolescents with CP at two-weeks post-training.*
2. **Prediction 2:** *Training will reduce antisocial behaviour in adolescents with CP as reported by teachers at two-weeks post-training.*

Analysis procedure preregistered for our models models: (1) We will run a separate linear mixed effects model for teacher and child data. (2) Our primary analysis will be an intention to treat analysis including all participants. (3) We will report the unstandardised coefficients, 95% confidence intervals (CIs), t-statistics, and p-values for each comparison.

I have cleaned child data for CP children who took part in the Prosocial Project as follows:

1. Removed all participants who are on our project exclusion list (e.g. for having a clinical diagnosis)
2. Removed anyone who intentionally did a one off session (e.g. to boost baseline task data or because the school could not do the full project)).
3. Removed all participants who did not have primary endpoint data - i.e. data for D1 or D2. This could be for the following reasons: (1) they dropped out during the project (2) they did not have enough data for imputation at at least one timepoint.

Otherwise, all participants have been kept in the dataset for the intention to treat analysis.

I have cleaned teacher data for CP children who took part in the Prosocial Project as follows:

1. Removed all participants who are on our project exclusion list (e.g. for having a clinical diagnosis)
2. Removed anyone for whom the school could not take part in the full project
3. Removed anyone who dropped out of the project before training but have an accidental teacher diary
4. Removed all participants who did not have primary endpoint data - i.e. data for D1 or D2. This could be for the following reasons: (1) they dropped out during the project (2) they did not have enough data for imputation at at least one timepoint.

Otherwise, all participants have been kept in the dataset for the intention to treat analysis.

Note some teachers swapped pupils between screening and screening and diaries. I will run a sensitivity analysis with and without these pupils.

Sensitivity Analyses

0) Tables of Group Numbers

Printing tables of Ns for each analyses so can confirm analysis is being run on correct data. Note the Ns should be as follows:

- Main analyses
  - Child asb = 82
  - Teach asb = 102
- Sensitivity Analyses 1 & 2 (controlling for training group size and number of sessions completed)
  - These analyses are conducted on the original sample; therefore, the sample sizes (Ns) are the same as in the main analysis.
  - They are included at the end of the main analysis PDF alongside the main analyses.

Analysis:

```
## [1] "Main analysis (+ covariates and sensitivit 1 & 2 at end of document."
```

Table 1 : child ASB

Training_Condition	n
CONTROL	44
TRAIN	38

```
## [1] "n participants total = "
```

```
## [1] 82
```

Table 2: teach ASB

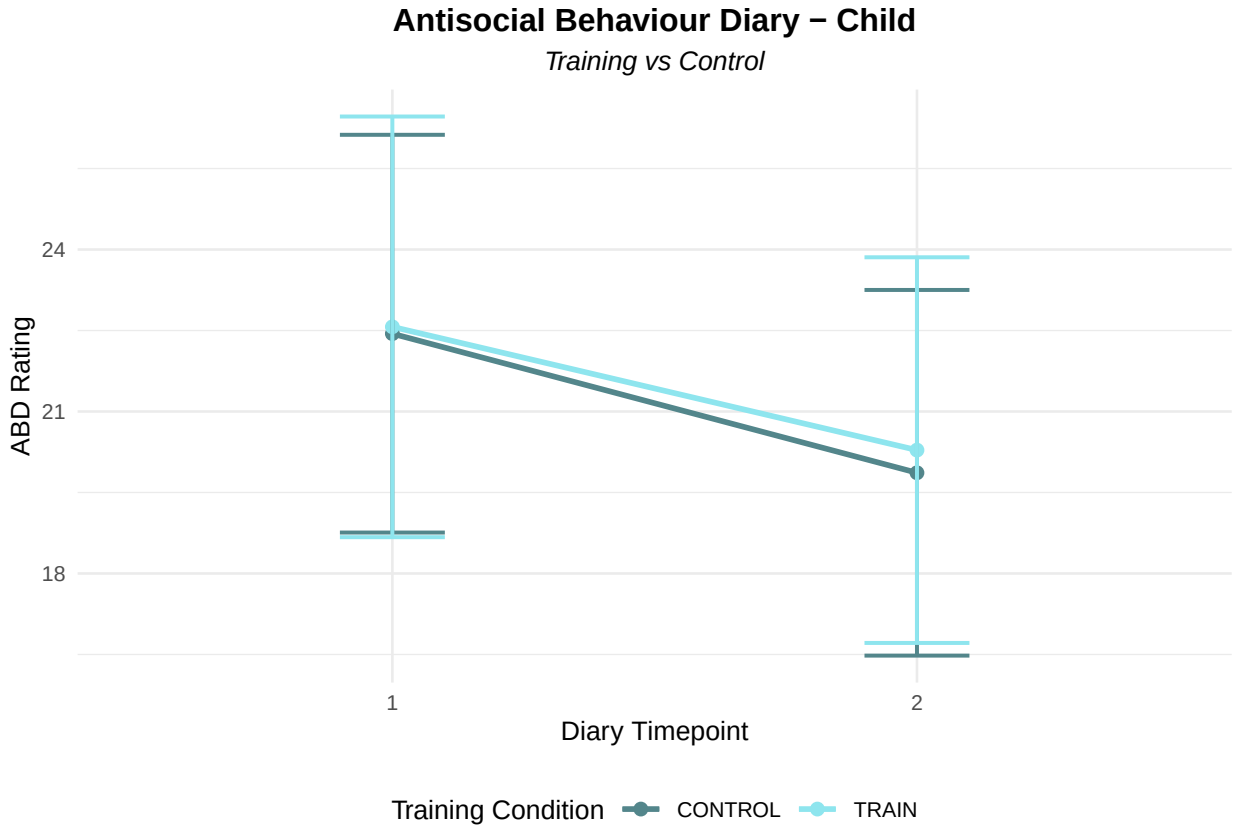
Training_Condition	n
CONTROL	53
TRAIN	49

```
## [1] "n participants total = "
```

```
## [1] 102
```

1) Impact of Training on Child Rated Antisocial Behaviour at Two Weeks

1.1) Plotting Child Antisocial Behaviour Diary Ratings at D1 & D2



Difference score plot:



1.2) Model specification in preregistration

Model specification in SAP:

$$\text{Child_ASB}_j = (\beta_0 + u_k) + \beta_1 \text{ASB.0}_j + \beta_2 \text{Train}_j + \varepsilon_j$$

Where:

- Child_ASB_j: Child-rated antisocial behaviour at D2 for participant j.
- $j = 1, \ldots, N$ identifies the participants.
- β_0 is a fixed intercept representing the average baseline level of the dependent variable (Child_ASB_j) across all participants when all predictors are zero.
- $u_k \sim \mathcal{N}(0, \sigma_u^2)$ is the random effect for schools, accounting for variability between schools.
- $\beta_1 \text{ASB.0}_j$: effect of child-rated antisocial behaviour score at baseline (D1) on ASB at D2.
- β_2 is the fixed effect of training condition (Train; an indicator variable coded as 0 for controls or 1 for training group) on the outcome, indicating how being in the training group influences the outcome.
- $\varepsilon_j \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is random error.

1.3) Preregistered LMM - model returns singular fit

```
model_c_asb_1 <- lmer(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition +
                      (1|School.ID),
                      data =  child_asb_d1d2_wide)
```

```
## boundary (singular) fit: see help('isSingular')
```

1.2) Linear Regression

Removing the random effect and running a simple regression model. This shows no impact of training on diary ratings at D2. However, diagnostics plot showed **simple linear model is not a good fit to data**.

```
model_c_asb_2 <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition,
                    data =  child_asb_d1d2_wide)

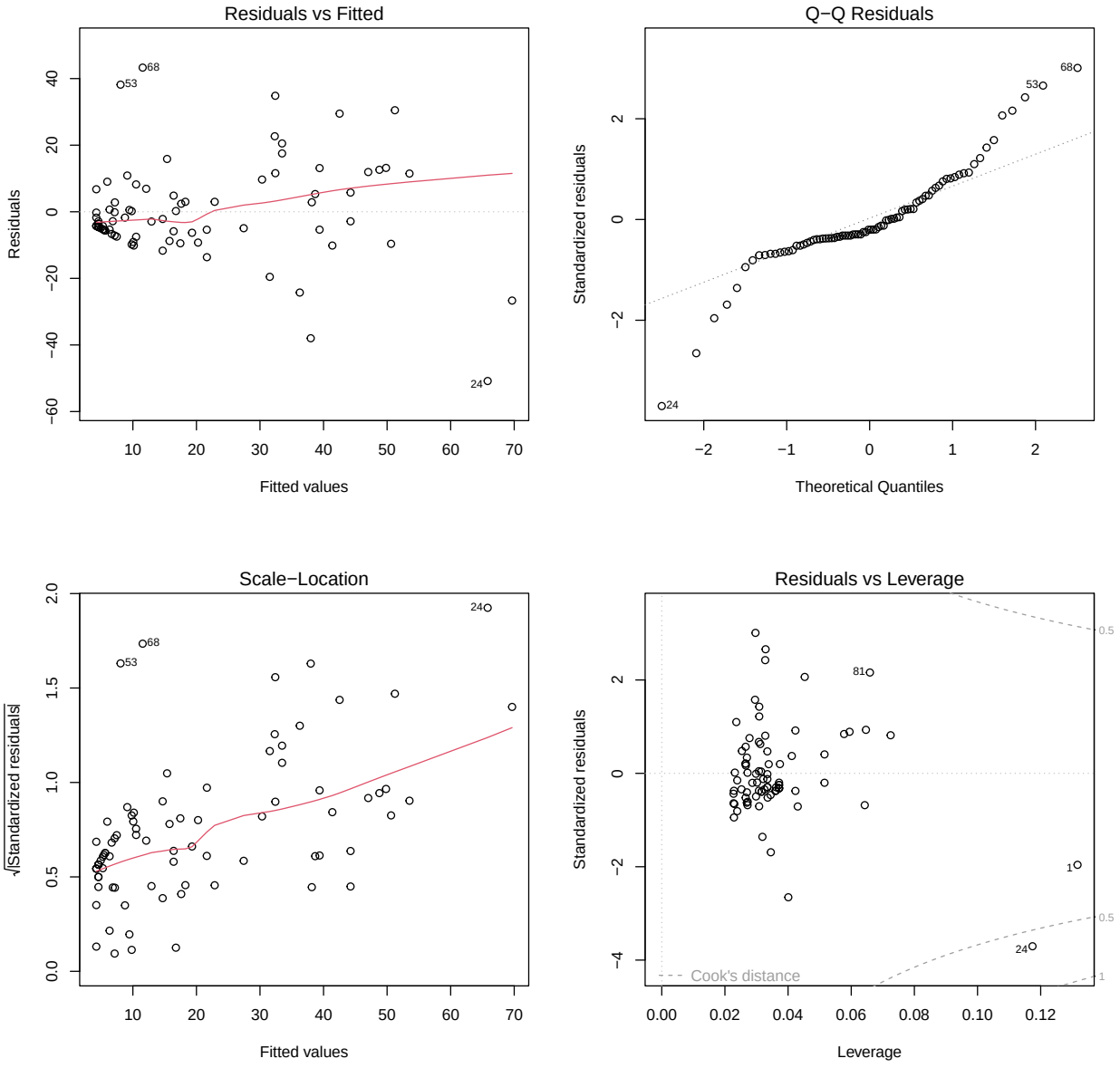
summary(model_c_asb_2)
```

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition, data = child_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.826  -5.804  -2.900   6.508  43.292
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               4.24588    2.67110   1.590    0.116
## ASB_sum_D1                0.69600    0.06737  10.330 2.54e-16 ***
## Training_ConditionTRAIN   0.33193    3.23469   0.103    0.919
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.61 on 79 degrees of freedom
## Multiple R-squared:  0.5747, Adjusted R-squared:  0.5639
## F-statistic: 53.37 on 2 and 79 DF,  p-value: 2.163e-15
```

1.2.1) Assumption check LM model

Here I am following this tutorial on diagnostics for lm models https://www.sthda.com/english/articles/39-regression-model-diagnostics/161-linear-regression-assumptions-and-diagnostics-in-r-essentials/. Diagnostic plots reveal the following:

1. *Fitted residual plot:* Relationship between fitted values and residuals **does not appear to be linear** (non-horizontal line, patterns in data - tend to be clustered to the left)
2. *QQ plot:* Residuals **do not appear to be normally distributed**. They are not following the dashed line.
3. *Scale location plot:* variance **does not appear to be homogenous**. The line is not horizontal, and the spread of the points is not equal.
4. *Residuals vs leverage plot:* There appear to be several influential values. These are being checked. For now leaving in as outliers could represent real clinical phenomena.



1.3) Final model: Boostrapped LM model

This indicates that the initial behaviour score contributes significantly, but the training condition does not. The confidence intervals are wide for the training, and the bias is moderate.

```
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide,
                      statistic = boot_coef_c_asb,
                      R = 5000)
```

```
child_asb_d1d2_wide$Training_Condition_standardised <- standardise(child_asb_d1d2_wide$Training_Conditi
child_asb_d1d2_wide$ASB_sum_D1_standardised  <- standardise(child_asb_d1d2_wide$ASB_sum_D1)
child_asb_d1d2_wide$ASB_sum_D2_standardised  <- standardise(child_asb_d1d2_wide$ASB_sum_D2)

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb_standardised <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2_standardised ~ ASB_sum_D1_standardised + Training_Condition_standardised,
                    data =  d)
  return(coef(fit))
}

# get standardised parameters
set.seed(12345)  # For reproducibility
boot_results_c_asb_standardised <- boot(data = child_asb_d1d2_wide,
                      statistic = boot_coef_c_asb_standardised,
                      R = 5000)
```

1.3.1) Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##      original       bias    std. error
## t1* 4.2458848 -0.193714977   2.2206125
```

```
## t2* 0.6959967  0.009131192   0.1034412
## t3* 0.3319348  0.154816227   3.2770690
```

Standardised results:

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb_standardised,
##     R = 5000)
##
##
## Bootstrap Statistics :
##         original       bias    std. error
## t1* -0.006954687 -0.002003789  0.08487495
## t2*  0.758006346  0.009332537  0.10999801
## t3*  0.015007483  0.009583799  0.14868526
```

1.3.2) Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_standardised, type = "bca",
##     index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.1732,  0.1633 )
## Calculations and Intervals on Original Scale
```

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_standardised, type = "bca",
##     index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.5136,  0.9467 )
## Calculations and Intervals on Original Scale
```

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_standardised, type = "bca",
##     index = 3)
```

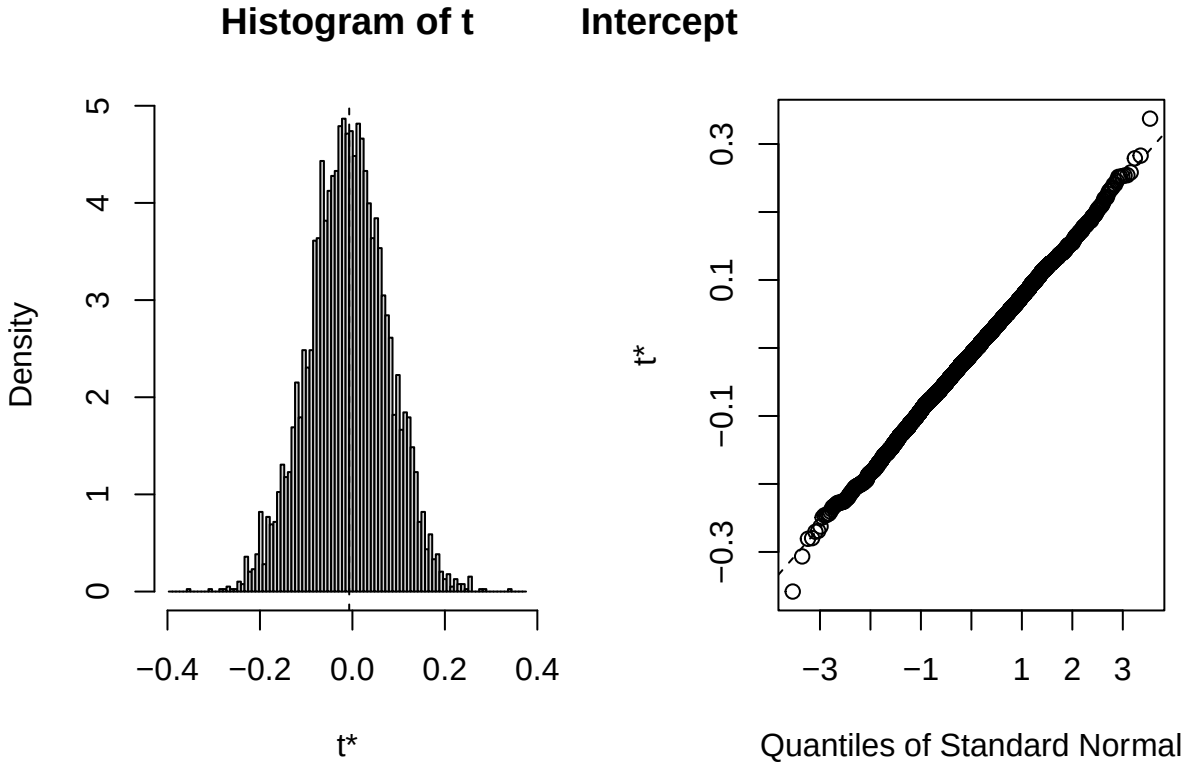
```
##
## Intervals :
## Level       BCa
## 95%   (-0.2956,  0.2932 )
## Calculations and Intervals on Original Scale

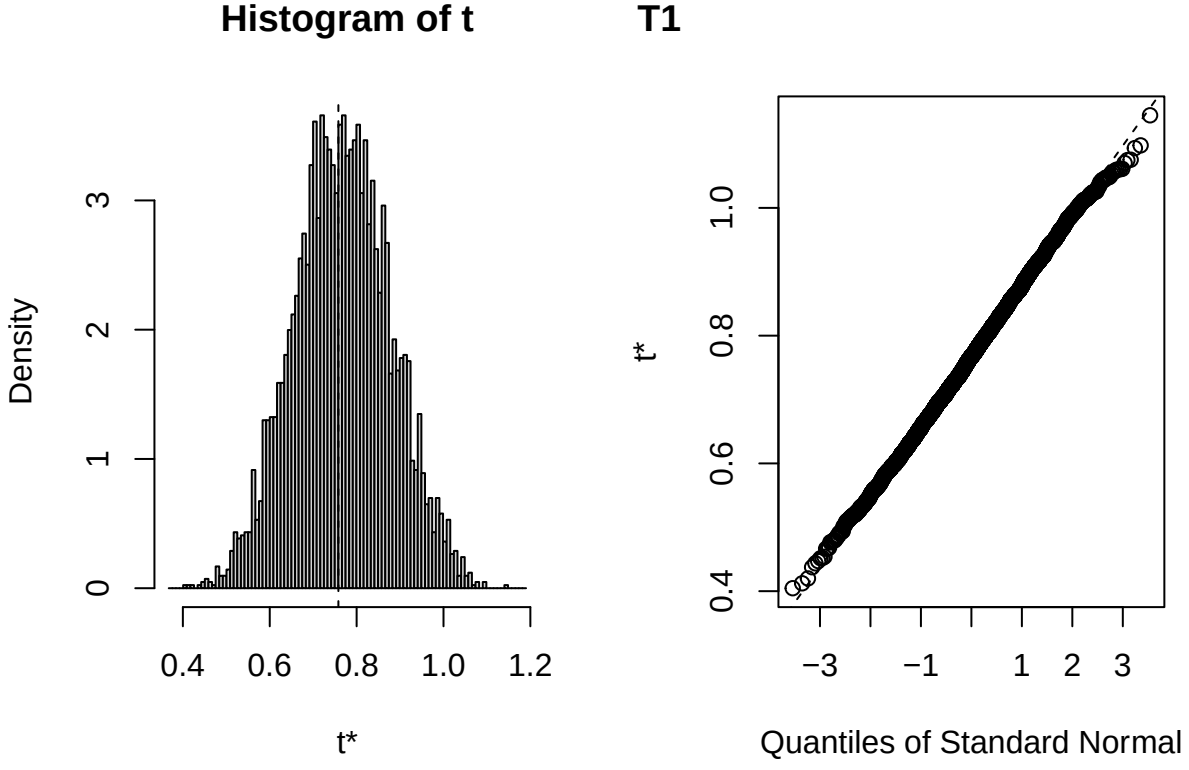
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 0.721,  9.953 )
## Calculations and Intervals on Original Scale

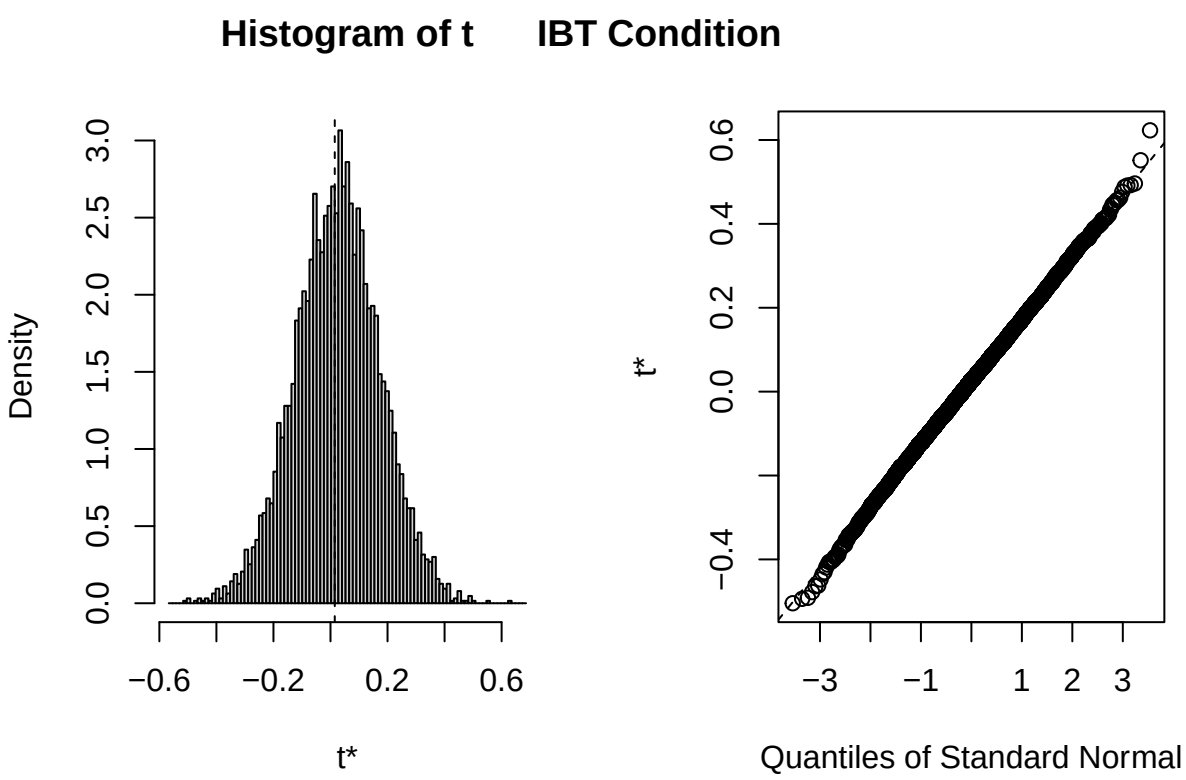
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4727,  0.8796 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.3036,  6.4673 )
## Calculations and Intervals on Original Scale
```

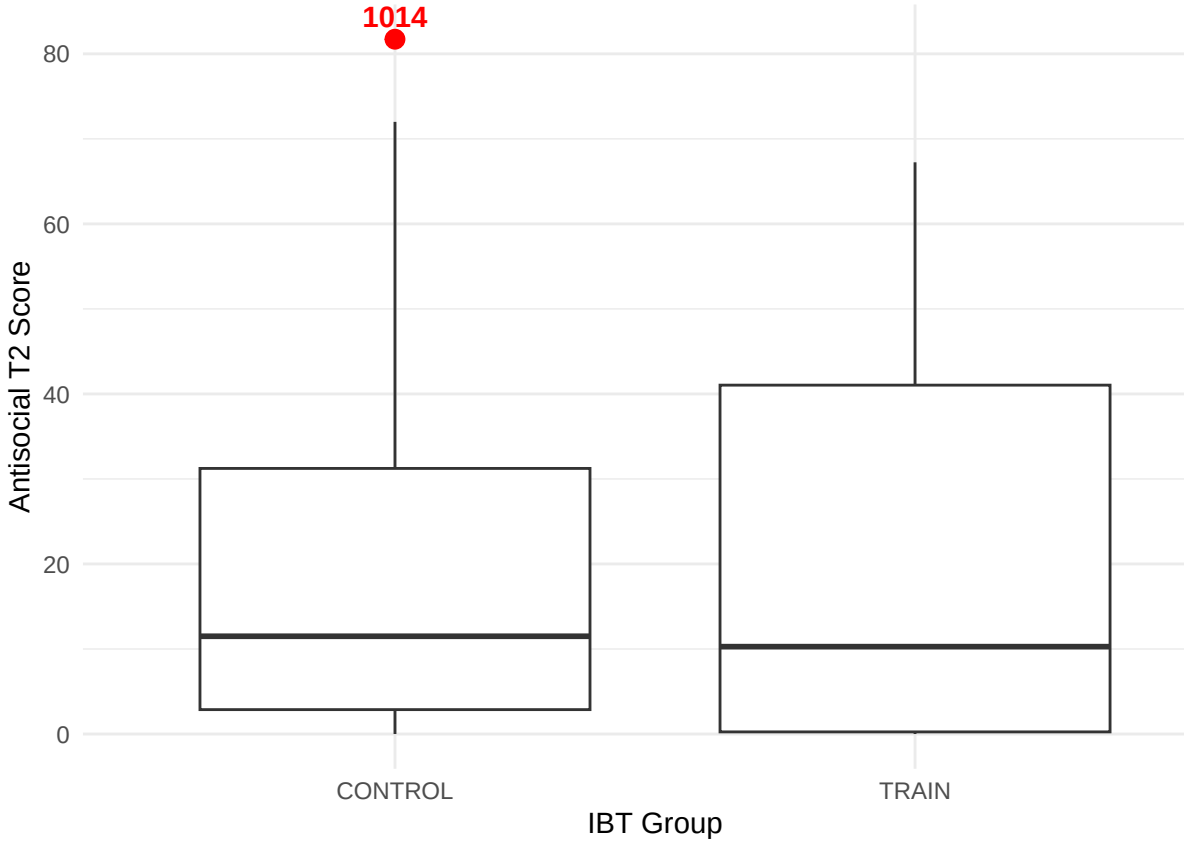
1.3.3) Bootstrap Diagnostics






Histogram of t IBT Condition
Density
0.0
0.5
1.0
1.5
2.0
2.5
3.0
−0.6
−0.2
0.2
0.6
t*
t*
0.6
0.4
0.2
0.0
−0.4
−3
−1
1
2
3
Quantiles of Standard Normal

1.4) Sensitivity check: Identifying outliers (analysis will be rerun without this participant - see sensitivity 5.pdf)



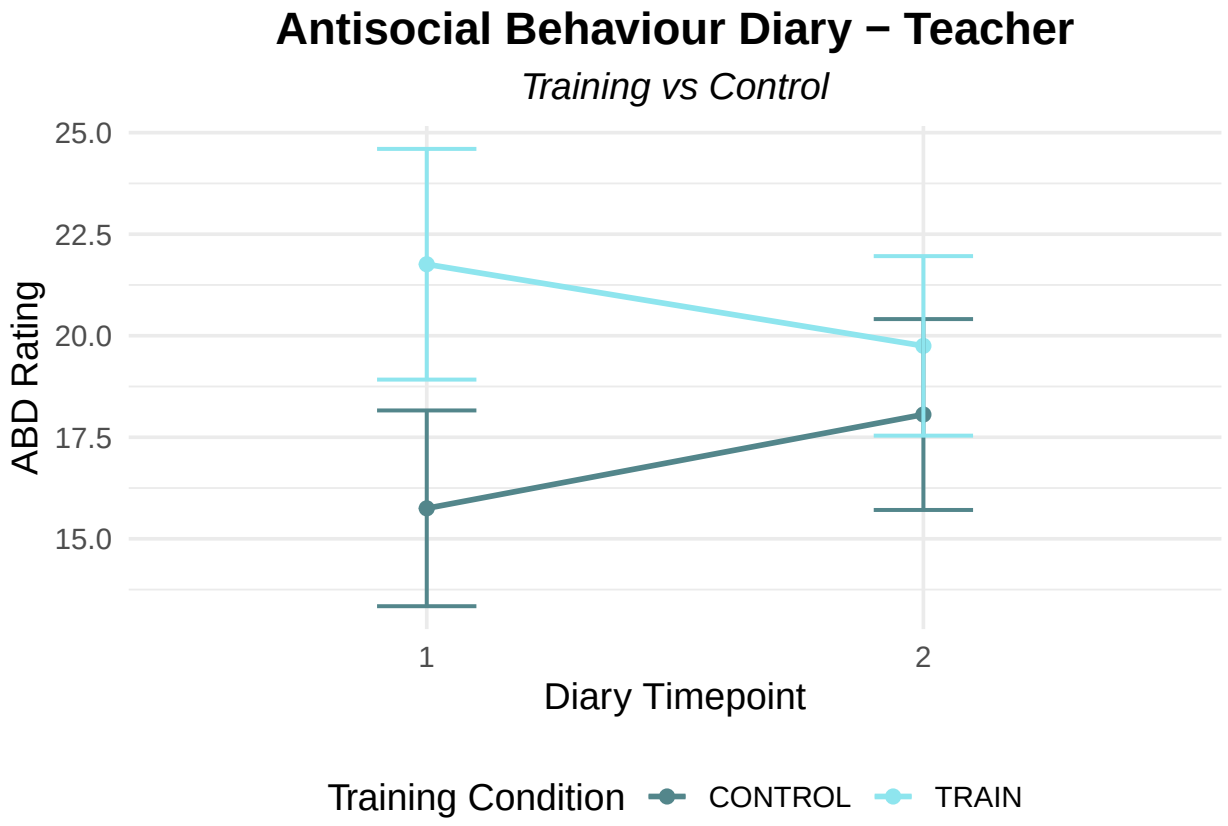
2) Impact of Training on Teacher Rated Antisocial Behaviour at Two Weeks

Important Note: We asked teachers to rate behaviour on a sliding scale from 0-100 one time per week as this was advised by teachers in focus group. They said it would be difficult to report on individual instances of behaviour as they see students infreuqently each week.

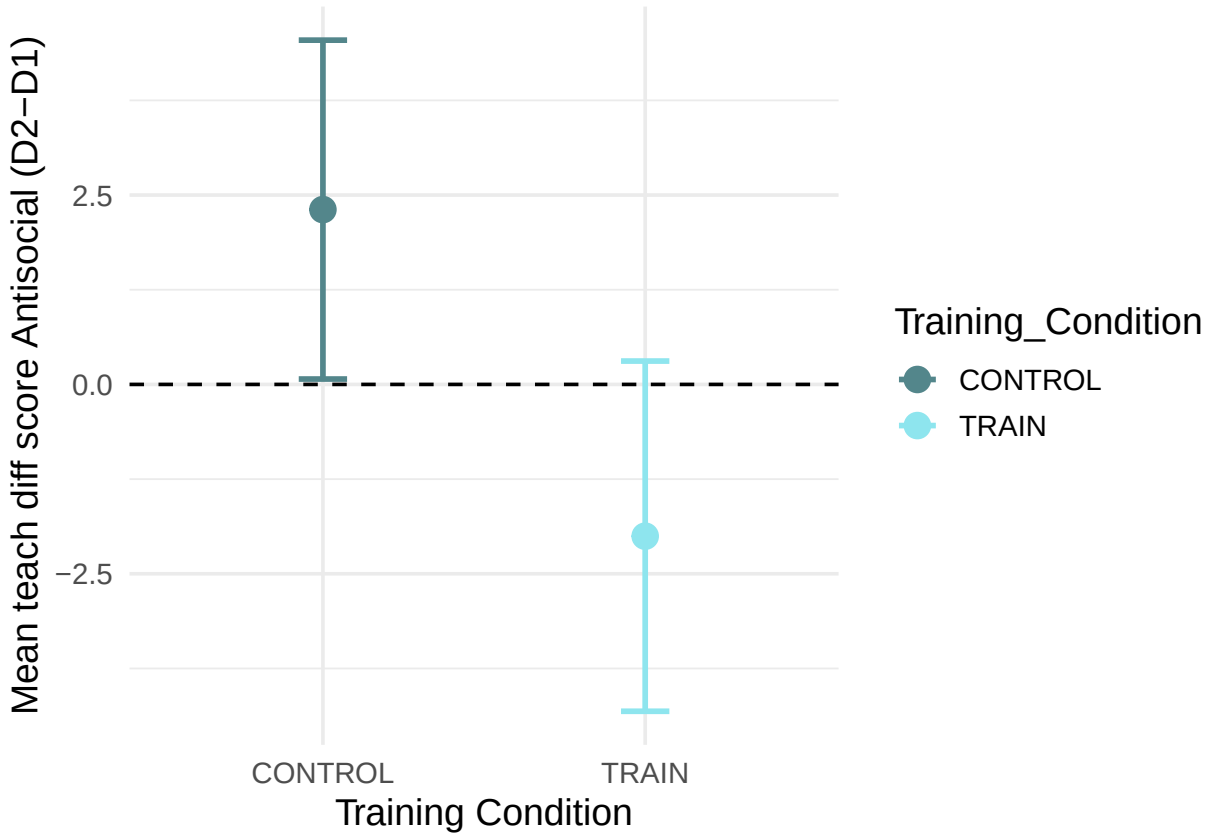
In teacher interviews/focus groups we got consistent feedback from teachers that they found the slider rating difficult to use to report on behaviour. We also got consistent feedback that it was extremely difficult to report on these indices of student behaviour as they see the students for such little time each week.

However, teacher and participant diary ratings correlate quite strongly at all but one timepoint, indicating that teachers were agreeing with children on the behaivours reported (good interrater reliability).

2.1) Plotting Teacher Antisocial Behaviour Diary Ratings at D1 & D2



Plotting Difference Scores:


Mean teach diff score Antisocial (D2−D1)
2.5
0.0
−2.5
CONTROL
TRAIN
Training Condition
Training_Condition
CONTROL
TRAIN

2.2) Model Specification in preregistration

$$\text{Teach_ASB}_j = (\beta_0 + u_k + u_{i(k)}) + \beta_1 \text{ASB.0}_j + \beta_2 \text{Train}_j + \varepsilon_j$$

where:

- Teach_ASB_j: Teacher-rated antisocial behaviour (at D2) for participant j.
- $j = 1, \ldots, N$ identifies the participants.
- $u_k \sim \mathcal{N}(0, \sigma_u^2)$ is the random effect for schools, accounting for variability between schools.
- $u_{i(k)} \sim \mathcal{N}(0, \sigma_u^2)$ is the random effect for teachers, nested within schools.
- β_0 is a fixed intercept representing the average baseline level of the dependent variable (Teach_ASB_j) across all participants when all predictors are zero.
- $\beta_1 \text{ASB.0}_j$: effect of teacher-rated antisocial behaviour score at baseline (D1) on later antisocial behaviour.
- β_2 is the fixed effect of Training condition (Train; an indicator variable coded as 0 for controls or 1 for training group) on the outcome, indicating how being in the training group influences the outcome.
- $\varepsilon_j \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is the random error term.

Note I have included the teachUUID for diaries diaries, not screening as 12 pupils had teacher changes between diaries and screening. One pupil had a teacher change between D1 and D2 which meant I could not his data here and reshape the data how I wanted. I have removed him from further diary analyses.

2.3) Preregistered LMM - this gives singular fit

Once again model has a singular fit. It looks like this is because teacherUUID is not contirbuting to variance.
I am therefore going to try running the model with only a random effect of school.

```
model_t_asb_1 <- lmer(ASB_mean_D2~ ASB_mean_D1 + Training_Condition +
                      (1|School.ID/teachUUID_diaries),  ## note 3 children changed teachers during diar
                      data =  teach_asb_d1d2_wide) # note adding CASI does not change singular fit
```

```
## boundary (singular) fit: see help('isSingular')
```

2.4) Simplified LMM (without teacher random effect)

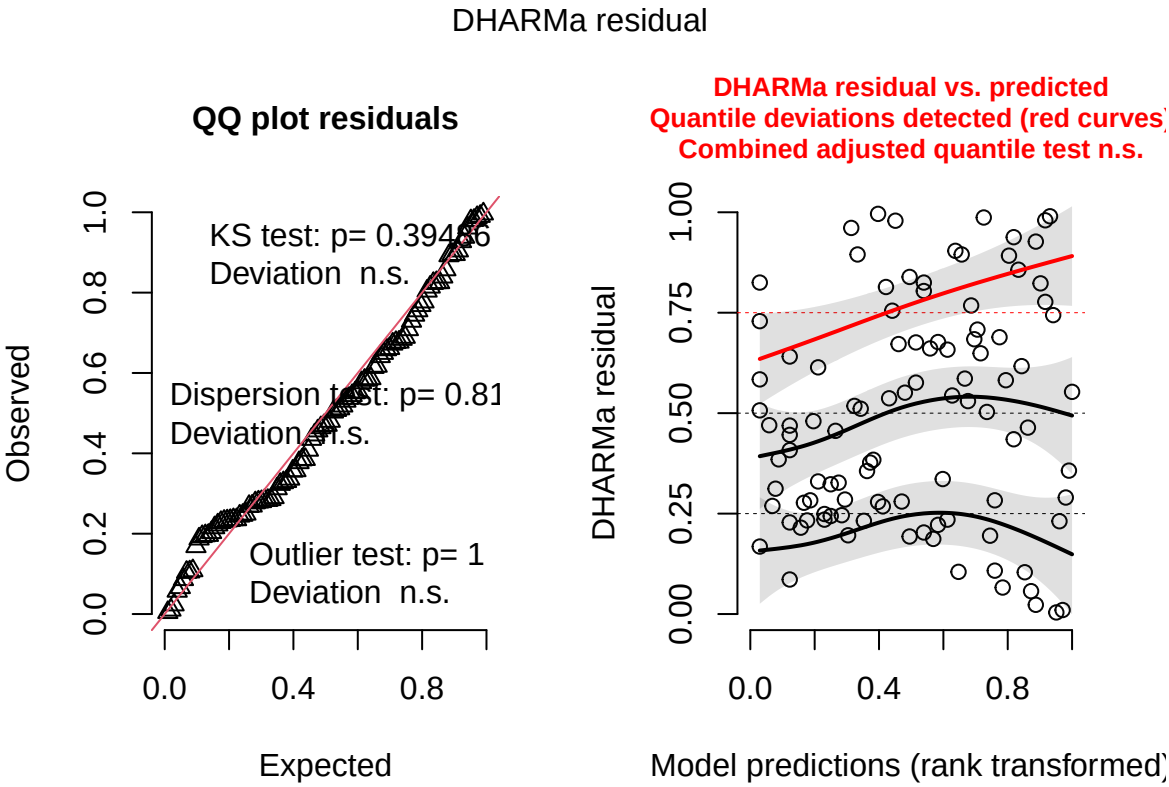
This model does fit. Here we see no impact of training on behaviour ratings from D1 to D2, when adjusted for baseline ASB ratings. **Assumption checks** indicate non-linearity and possible heteroscadicity.

```
model_t_asb_2 <- lmer(ASB_mean_D2~ ASB_mean_D1 + Training_Condition +
                      (1|School.ID),
                      data =  teach_asb_d1d2_wide)

summary(model_t_asb_2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + (1 | School.ID)
##    Data: teach_asb_d1d2_wide
##
## REML criterion at convergence: 811.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.63254 -0.65275 -0.09209  0.47413  2.99398
##
## Random effects:
##  Groups    Name        Variance Std.Dev.
##  School.ID (Intercept)  16.15    4.019
##  Residual              165.85   12.878
## Number of obs: 102, groups:  School.ID, 16
##
## Fixed effects:
##                           Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)              10.60724    2.44643 29.54430   4.336 0.000155 ***
## ASB_mean_D1               0.50271    0.07385 96.76401   6.807 8.36e-10 ***
## Training_ConditionTRAIN -1.36745    2.61535 92.54249  -0.523 0.602325
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) ASB__D
## ASB_mean_D1 -0.464
## Trnn_CTRAIN -0.441 -0.143
```

2.4.1) Assumption Checks



2.5) Bootstrapped simlified LMM - this gives singular fit error on many iterations

```
# REB bootstrap advised when assumptions are violated
boot_results <- lmeresampler::bootstrap(model_t_asb_2, .f = fixef, type = "reb", B = 5000)

boot_results
```

```
## Bootstrap type: reb0
##
## Number of resamples: 5000
##
##                       term   observed   rep.mean         se         bias
## 1              (Intercept) 10.6072441  9.8947663 2.83953818 -0.712477837
## 2              ASB_mean_D1  0.5027094  0.5007977 0.07116341 -0.001911677
## 3 Training_ConditionTRAIN -1.3674509 -1.3405481 2.41603138  0.026902836
##
## There were 89 messages, 0 warnings, and 0 errors.
##
## The most commonly occurring message was: boundary (singular) fit: see help('isSingular')
```

2.6) Simple Linear Regression

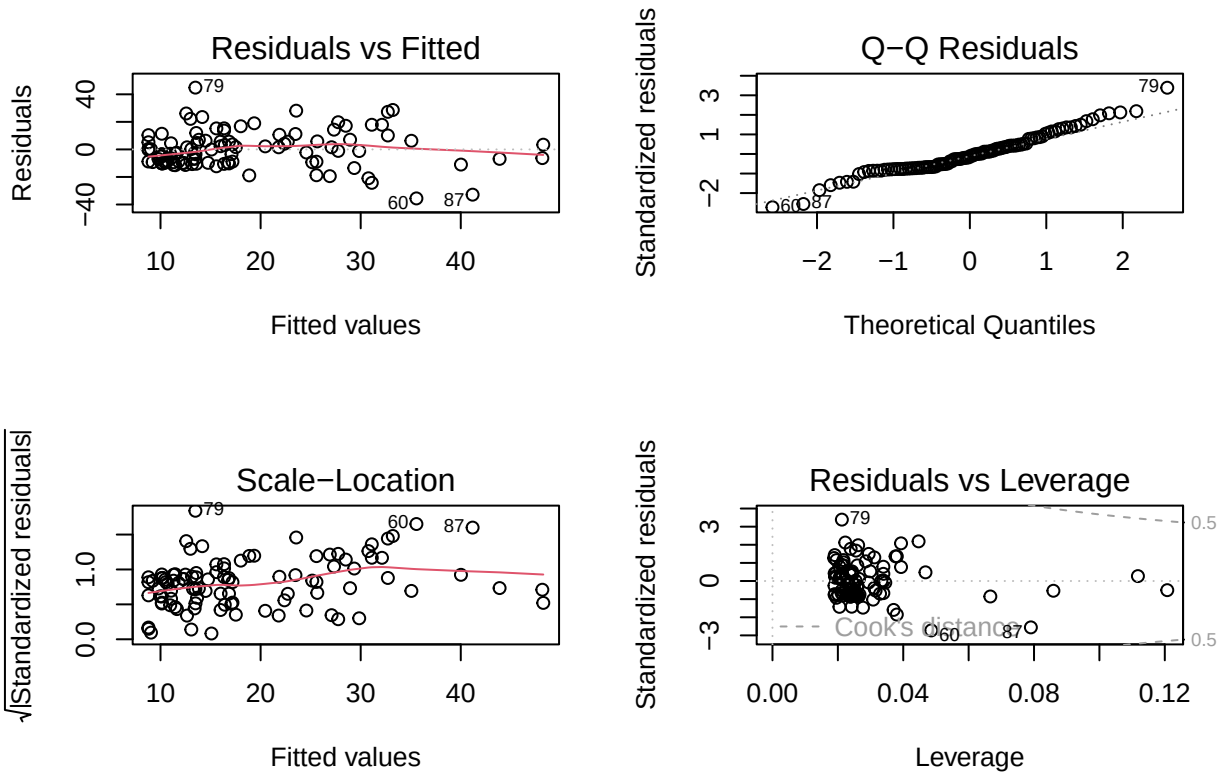
As LMM gave singular fit, I am instead running simple linear regression.

```
model_t_asb_3 <- lm(ASB_mean_D2~ ASB_mean_D1 + Training_Condition,
                      data =  teach_asb_d1d2_wide)

summary(model_t_asb_3)
```

```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition,
##     data = teach_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -35.571  -9.323  -1.714   6.314  44.847
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              10.12878    2.15577   4.698 8.47e-06 ***
## ASB_mean_D1               0.50360    0.07149   7.044 2.51e-10 ***
## Training_ConditionTRAIN -1.33300    2.68646  -0.496    0.621
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.38 on 99 degrees of freedom
## Multiple R-squared:  0.3357, Adjusted R-squared:  0.3222
## F-statistic: 25.01 on 2 and 99 DF,  p-value: 1.614e-09
```

2.6.1) Assumption check



2.7) Final Model: Bootstrapped Linear Regression

```r
# non-parametric case bootstrap: https://journal.r-project.org/articles/RN-2002-017/RN-2002-017.pdf
# argued to be better  for heteroskedastic data: https://library.virginia.edu/data/articles/bootstrappi
boot_coef_t_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
set.seed(123)  # For reproducibility
boot_results_t_asb <- boot(data = teach_asb_d1d2_wide,
                     statistic = boot_coef_t_asb,
                     R = 5000)
```

2.7.1) Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_t_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 10.1287765 -0.018738431  1.96739948
## t2*  0.5036026  0.003892104  0.08593524
## t3* -1.3330020 -0.052889559  2.73816659
```

Standardised results:

```
teach_asb_d1d2_wide$Training_Condition_standardised <- standardise(teach_asb_d1d2_wide$Training_Conditi
teach_asb_d1d2_wide$ASB_mean_D1_standardised  <- standardise(teach_asb_d1d2_wide$ASB_mean_D1)
teach_asb_d1d2_wide$ASB_mean_D2_standardised  <- standardise(teach_asb_d1d2_wide$ASB_mean_D2)

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_t_asb_standardised <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2_standardised ~ ASB_mean_D1_standardised + Training_Condition_standardised,
                    data =  d)
  return(coef(fit))
}

# get standardised parameters
set.seed(12345)  # For reproducibility
boot_results_t_asb_standardised <- boot(data = teach_asb_d1d2_wide,
                      statistic = boot_coef_t_asb_standardised,
                      R = 5000)
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_t_asb_standardised,
##     R = 5000)
##
##
## Bootstrap Statistics :
##        original       bias    std. error
## t1*  0.03939621  0.003995150  0.12634699
## t2*  0.58451808  0.006333759  0.09730645
## t3* -0.08200845 -0.005769813  0.16787578
```

2.7.2) Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_standardised, type = "bca",
##     index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.1988,  0.2958 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_standardised, type = "bca",
##     index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3844,  0.7653 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_standardised, type = "bca",
##     index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-0.4062,  0.2560 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 6.93, 14.97 )
## Calculations and Intervals on Original Scale

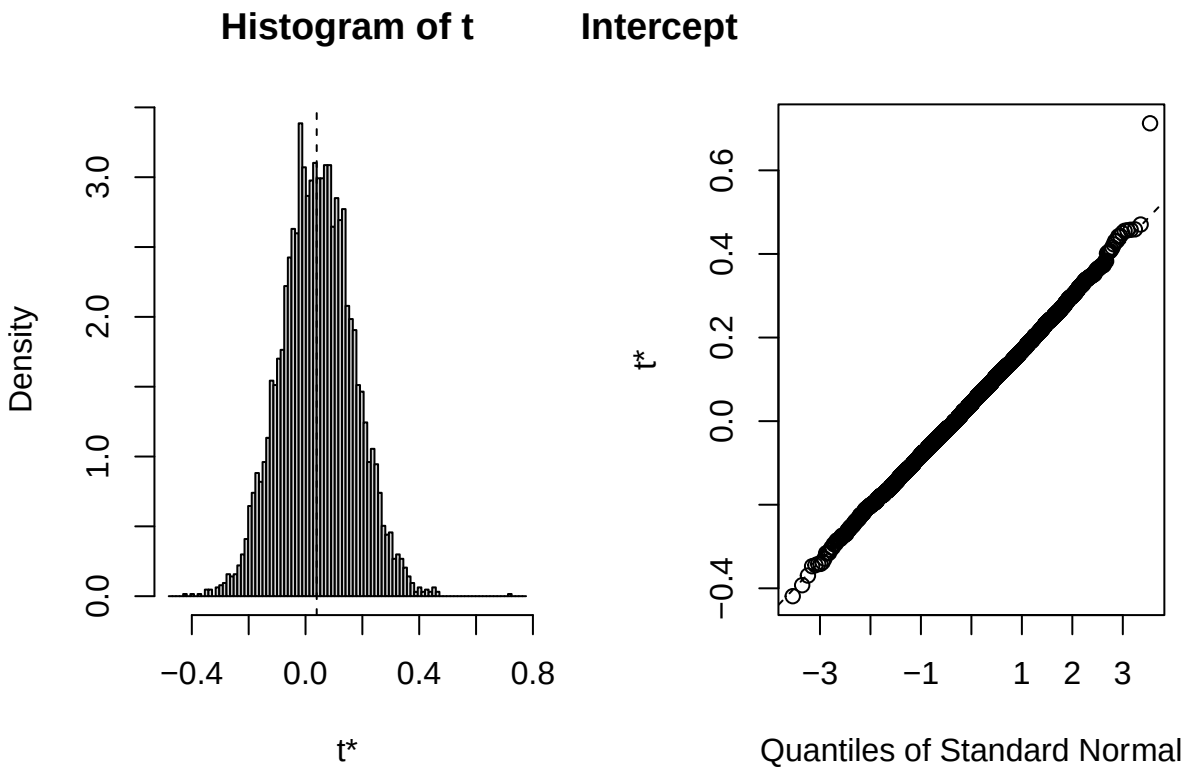
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
```

```
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3245,  0.6683 )
## Calculations and Intervals on Original Scale

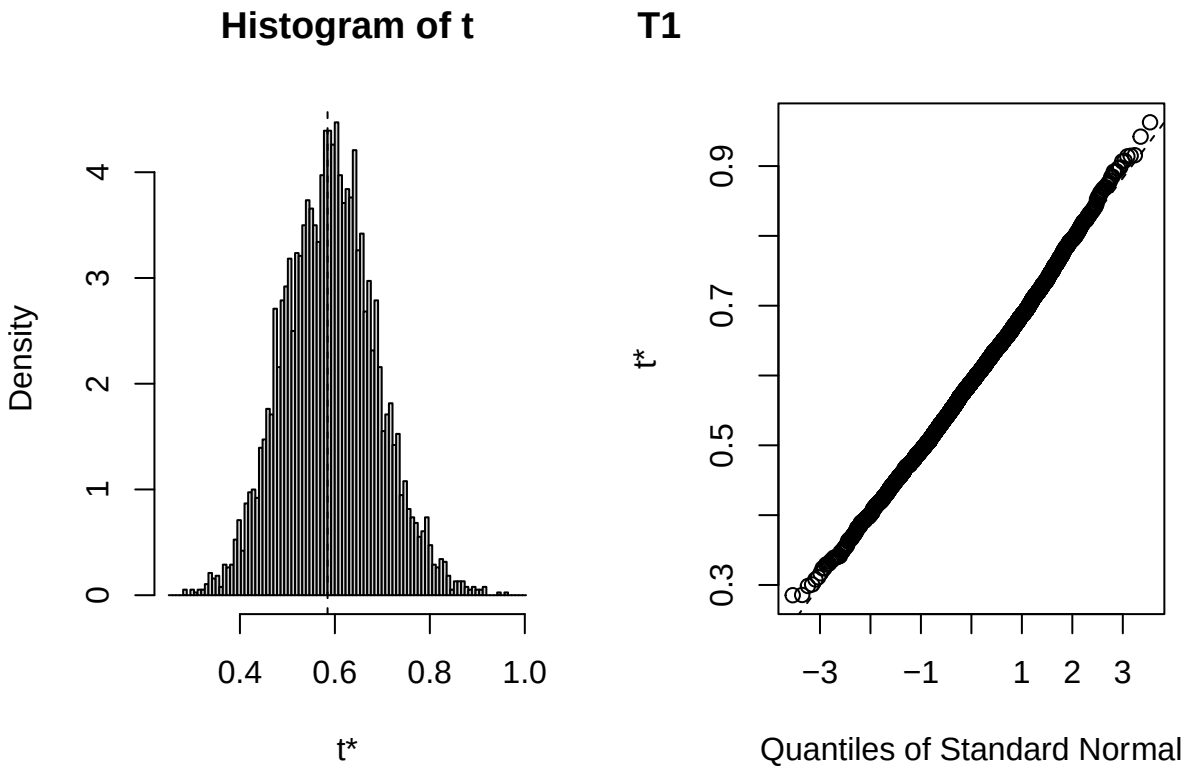
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.598,  4.157 )
## Calculations and Intervals on Original Scale
```

2.7.3) Bootstrap Distribution

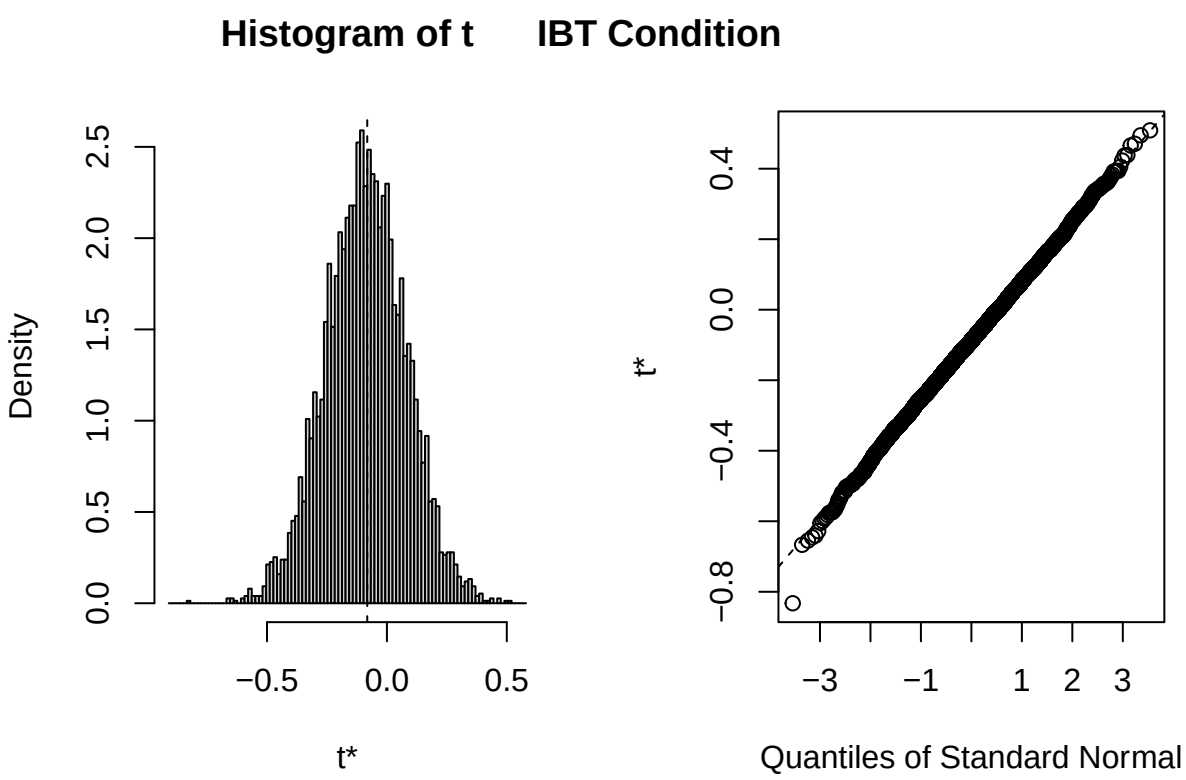
```
# Visualize bootstrap distributions
plot(boot_results_t_asb_standardised, index = 1, main = "Intercept")  # For binary predictor
title("Intercept")
```



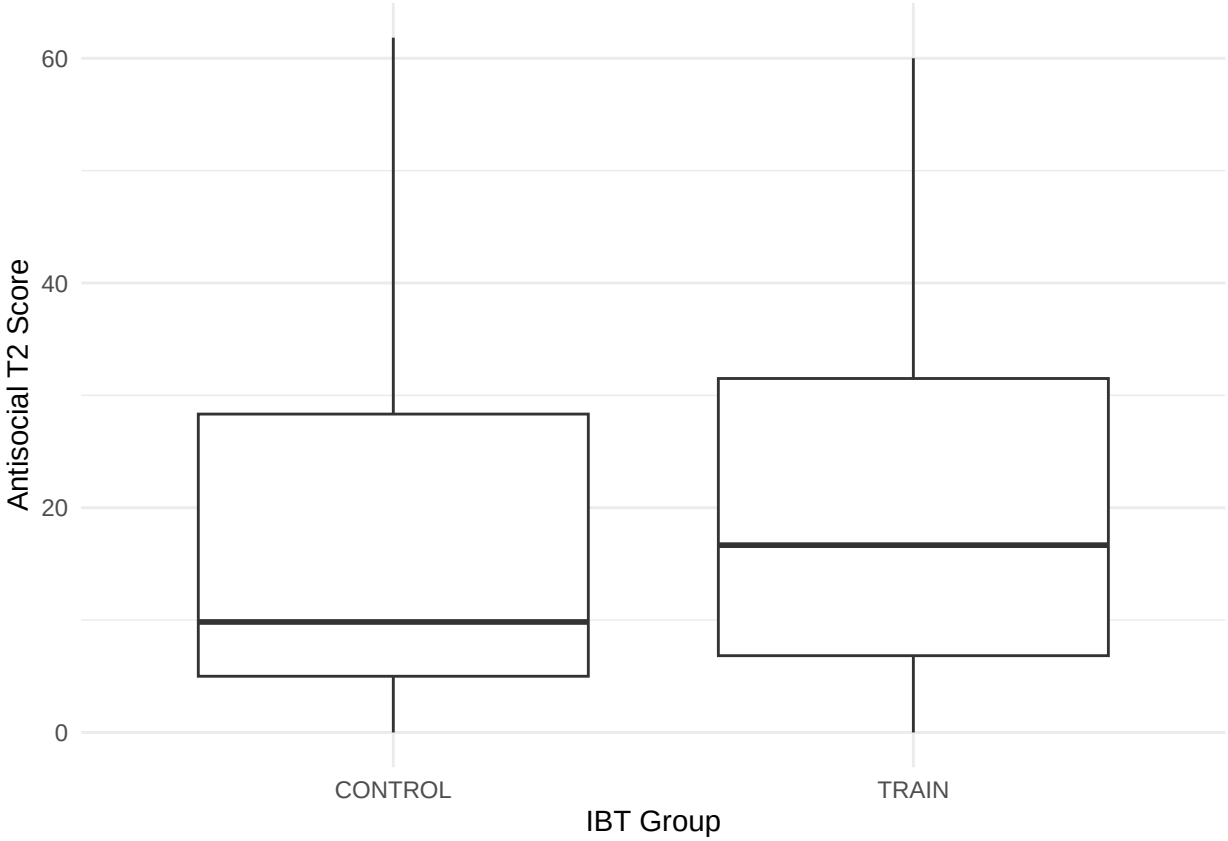
```
plot(boot_results_t_asb_standardised, index = 2, main = "T1 Behaviour Diary")  # For continuous predict
title("T1")
```




```
plot(boot_results_t_asb_standardised, index = 3, main = "IBT Condition")  # For binary predictor
title("IBT Condition")
```


Histogram of t IBT Condition
Density
0.0
0.5
1.0
1.5
2.0
2.5
−0.5
0.0
0.5
t*
t*
−0.8
−0.4
0.0
0.4
−3
−1
1
2
3
Quantiles of Standard Normal

2.7.4) Sensitivity check: Identifying outliers - none identified



3) Sensitivity Models

3.1) Models with Group Size for Training as a covariate (note that not every participant has this)

3.2.1) Child

1) Define and summarise model

```
model_c_asb_2_with_groupNtrain <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + group_n_training,
                             data =  child_asb_d1d2_wide)


summary(model_c_asb_2_with_groupNtrain)
```

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + group_n_training,
##     data = child_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.929  -5.993  -2.489   6.430  42.861
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               5.17390    3.86329   1.339    0.184
## ASB_sum_D1                0.69569    0.06776  10.267 3.89e-16 ***
## Training_ConditionTRAIN   0.29367    3.25504   0.090    0.928
## group_n_training         -0.15178    0.45412  -0.334    0.739
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.69 on 78 degrees of freedom
## Multiple R-squared:  0.5753, Adjusted R-squared:  0.5589
## F-statistic: 35.22 on 3 and 78 DF,  p-value: 1.708e-14
```

```
confint(model_c_asb_2_with_groupNtrain)
```

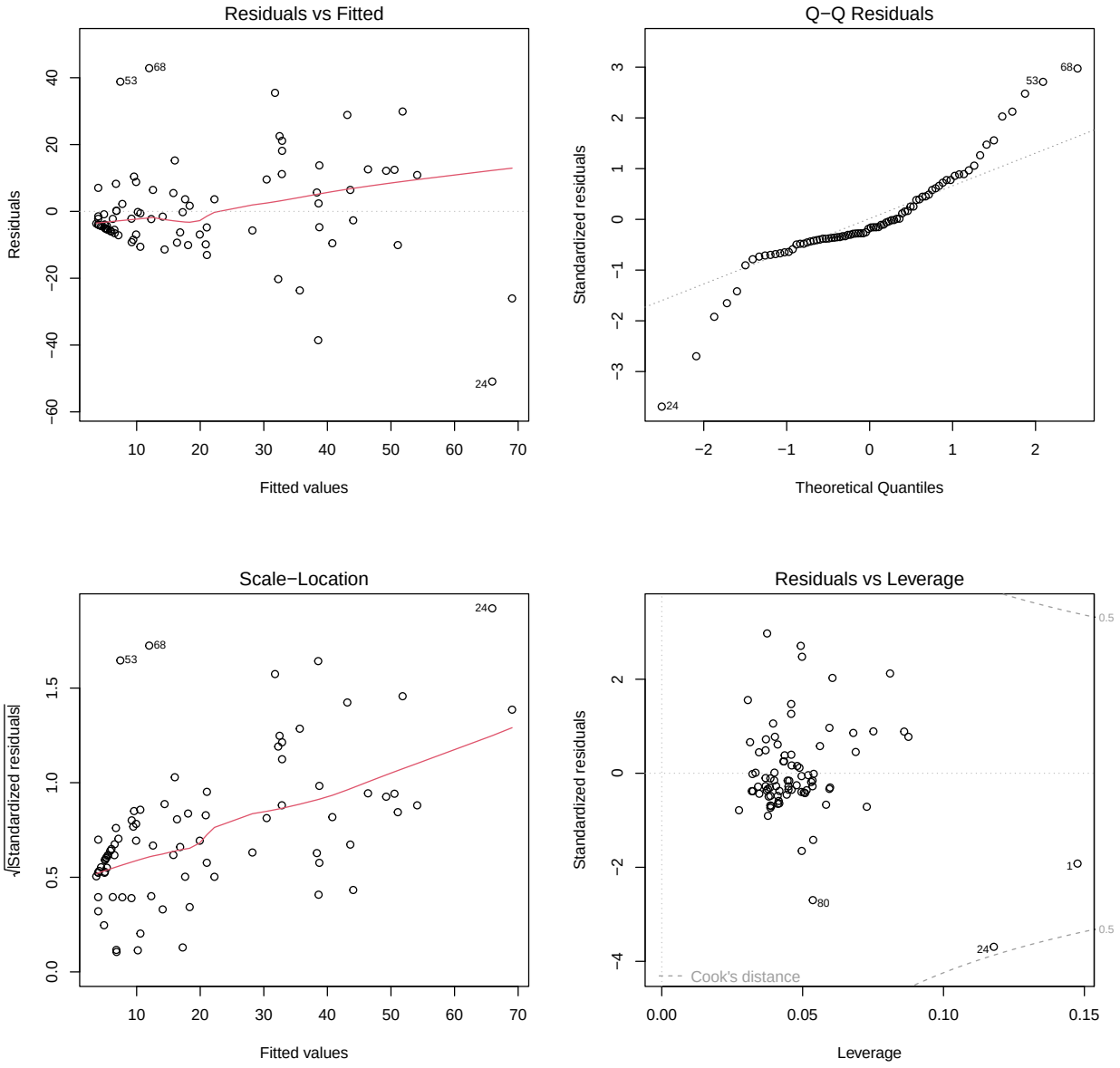
```
##                              2.5 %     97.5 %
## (Intercept)             -2.517324 12.8651288
## ASB_sum_D1               0.560782  0.8305898
## Training_ConditionTRAIN -6.186622  6.7739529
## group_n_training        -1.055859  0.7522942
```

2) compare to original model - this cannot be done as some participants are missing the group n for training col.

```
## [1] "Number of participants for whom we do not have the group n for training:"
```

```
## [1] 0
```

3) Get Diagnostics



4) Bootstrap:

```
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition+ group_n_training,
                    data =  d)
  return(coef(fit))
}
```

```r
# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide,
                     statistic = boot_coef_c_asb,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original        bias    std. error
## t1*  5.1739025 -0.1795798511   3.3692546
## t2*  0.6956859  0.0085124039   0.1036899
## t3*  0.2936654  0.1426650063   3.3213409
## t4* -0.1517823 -0.0008073691   0.4422322
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.602, 13.046 )
## Calculations and Intervals on Original Scale

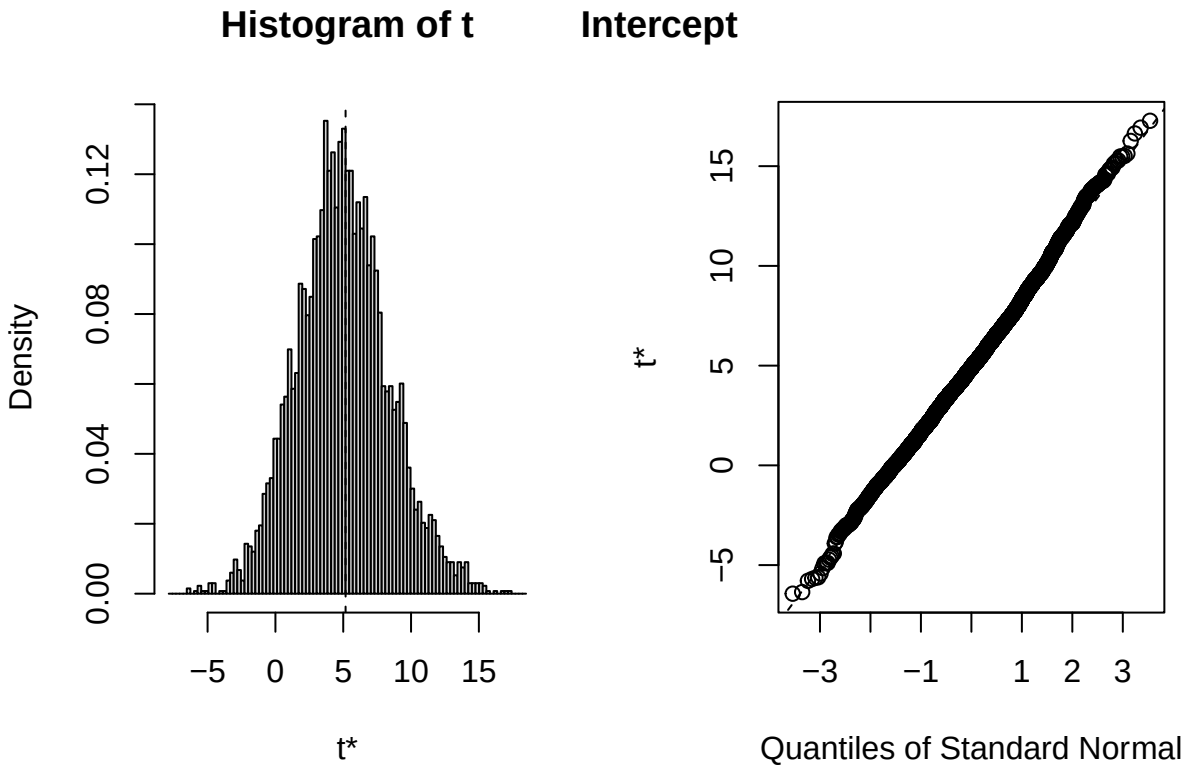
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4673,  0.8799 )
## Calculations and Intervals on Original Scale

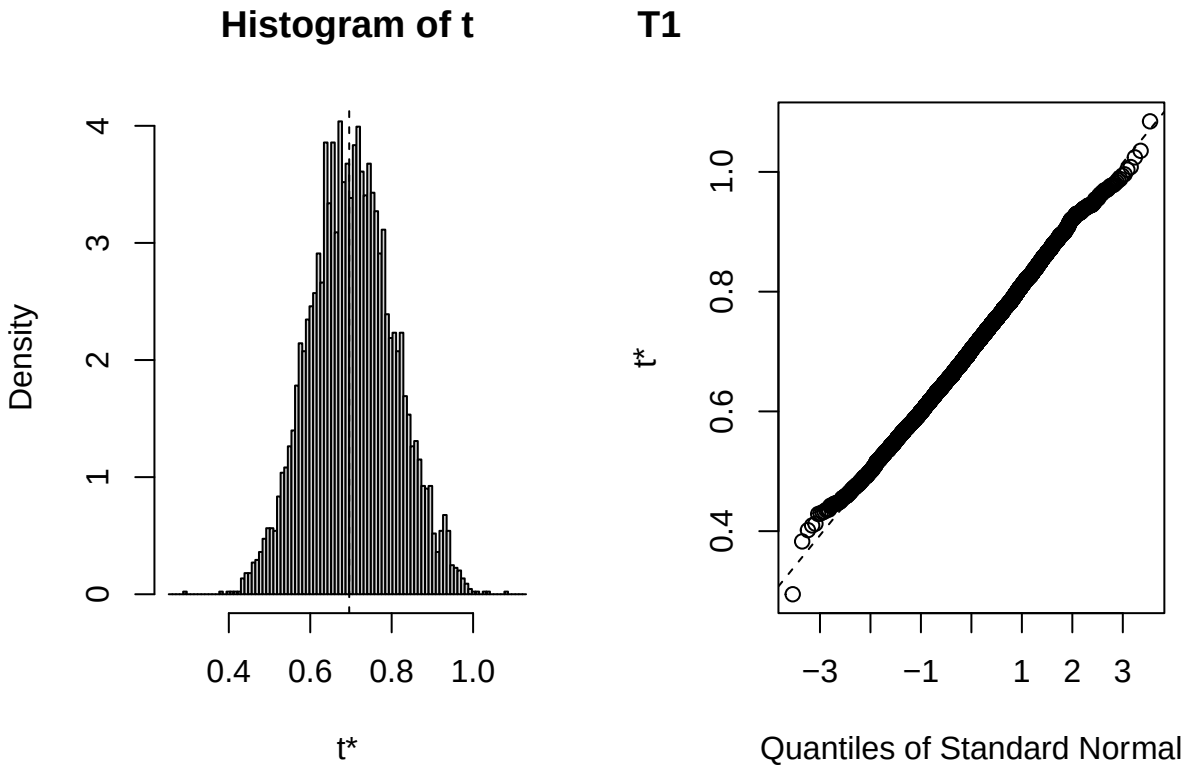
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
```

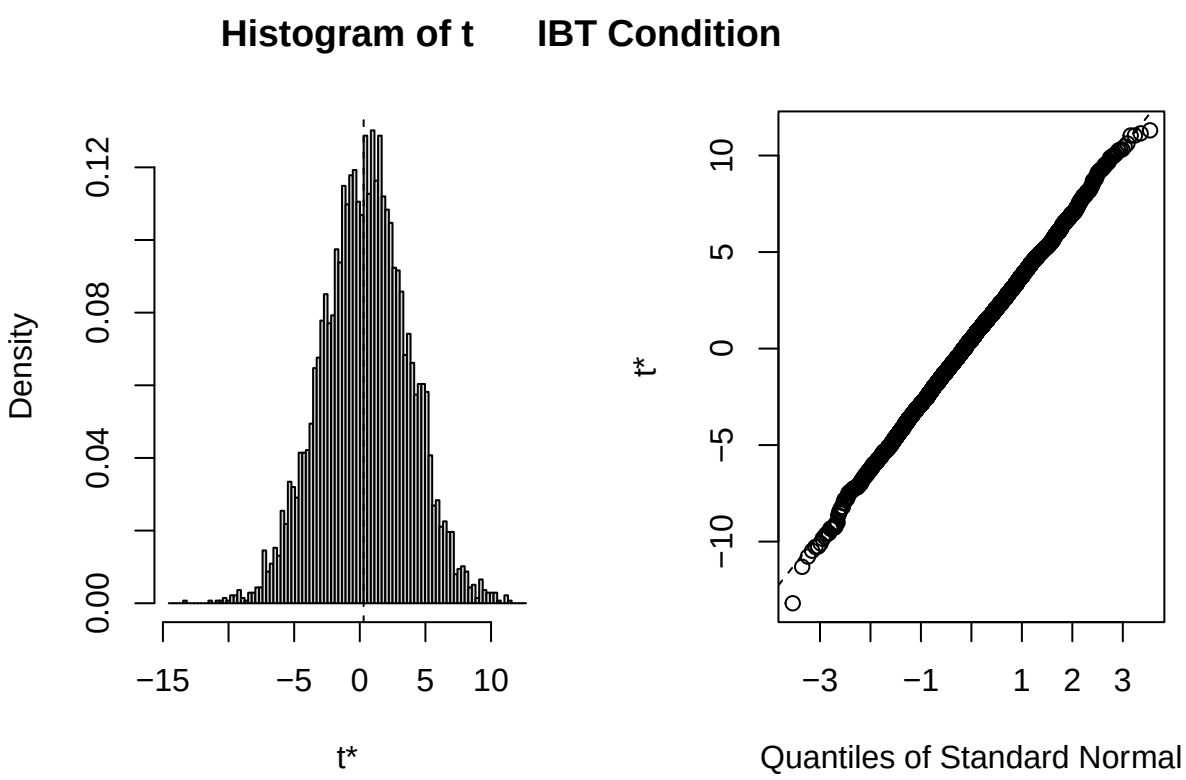
```
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.5426,  6.5334 )
## Calculations and Intervals on Original Scale

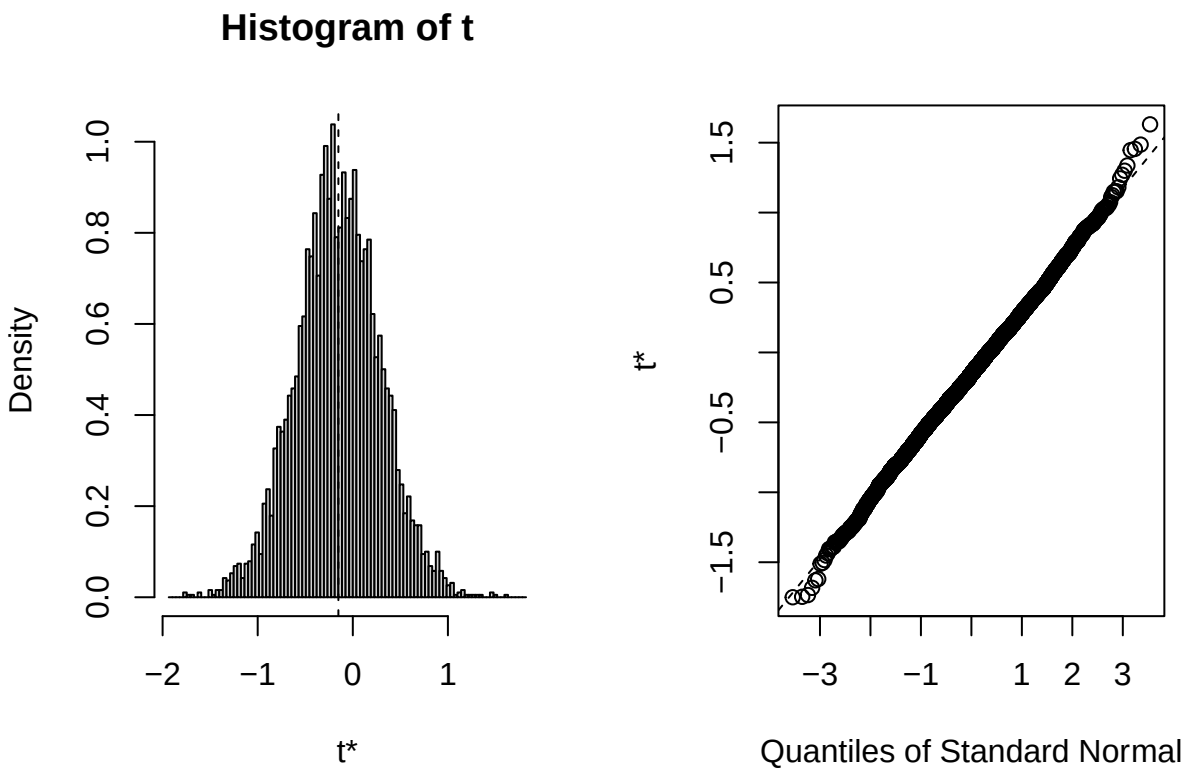
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-1.0036,  0.7559 )
## Calculations and Intervals on Original Scale
```

Bootstrap Diagnostics




Histogram of t
Density
0
1
2
3
4
0.4
0.6
0.8
1.0
t*
T1
t*
0.4
0.6
0.8
1.0
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t IBT Condition
Density
0.00
0.04
0.08
0.12
−15
−5
0
5
10
t*
t*
−10
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.2
0.4
0.6
0.8
1.0
−2
−1
0
1
t*
t*
−1.5
−0.5
0.5
1.5
−3
−1
1
2
3
Quantiles of Standard Normal

3.1.2) Teacher

1) Define and summarise model

```
model_t_asb_3_with_groupNtrain <- lm(ASB_mean_D2~ ASB_mean_D1 + Training_Condition + group_n_training,
                                data =  teach_asb_d1d2_wide)


summary(model_t_asb_3_with_groupNtrain)
```

```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition +
##     group_n_training, data = teach_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.712  -9.146  -1.786   6.864  45.104
##
## Coefficients:
##                           Estimate Std. Error t value Pr(>|t|)
## (Intercept)              12.02924    3.16194   3.804 0.000247 ***
## ASB_mean_D1               0.51338    0.07259   7.072 2.28e-10 ***
## Training_ConditionTRAIN -1.43775    2.69386  -0.534 0.594749
## group_n_training        -0.31746    0.38584  -0.823 0.412635
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.4 on 98 degrees of freedom
## Multiple R-squared:  0.3402, Adjusted R-squared:   0.32
## F-statistic: 16.85 on 3 and 98 DF,  p-value: 6.687e-09
```

```
confint(model_t_asb_3_with_groupNtrain)
```

```
##                               2.5 %     97.5 %
## (Intercept)              5.7544716 18.3040100
## ASB_mean_D1              0.3693254  0.6574333
## Training_ConditionTRAIN -6.7836365  3.9081350
## group_n_training        -1.0831448  0.4482283
```

4) Bootstrap:

```
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition+ group_n_training,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
```

```r
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = teach_asb_d1d2_wide,
                     statistic = boot_coef_c_asb,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 12.0292408 -0.074417660  2.73479206
## t2*  0.5133793  0.003340180  0.08363614
## t3* -1.4377508 -0.018265077  2.72865553
## t4* -0.3174582  0.006784738  0.32803464
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 7.34, 18.27 )
## Calculations and Intervals on Original Scale

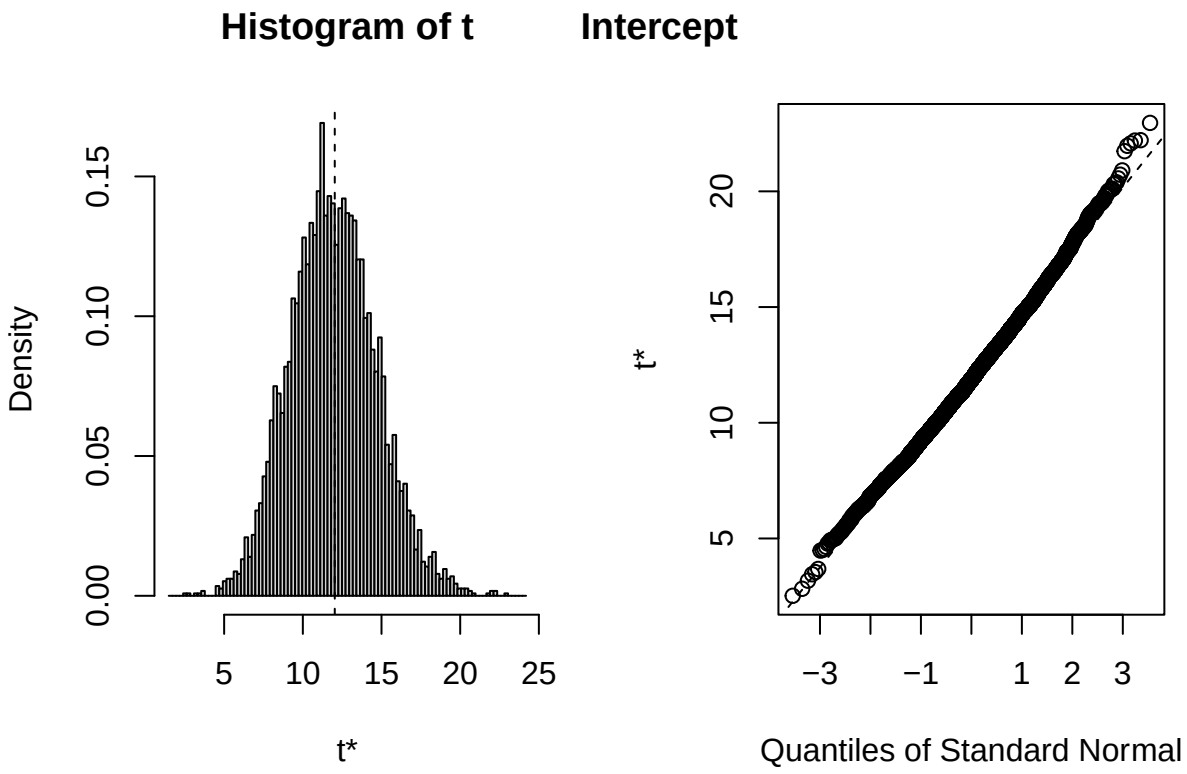
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3440,  0.6759 )
## Calculations and Intervals on Original Scale

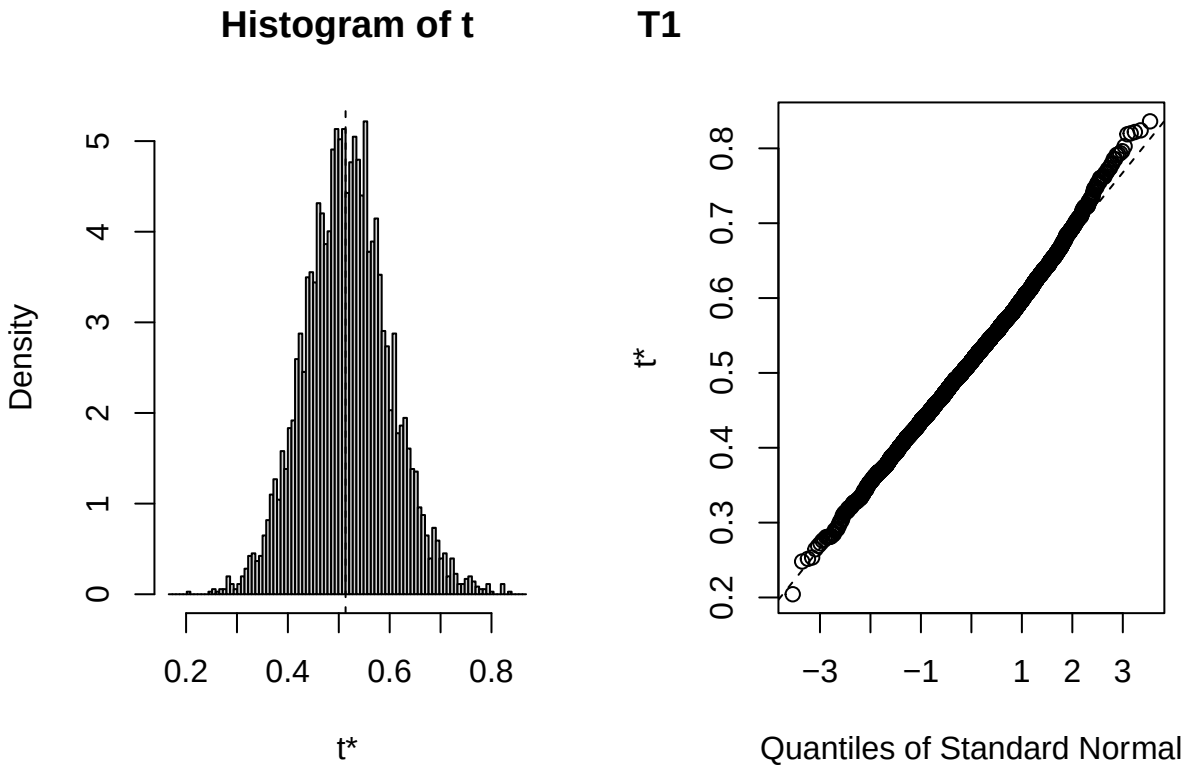
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
```

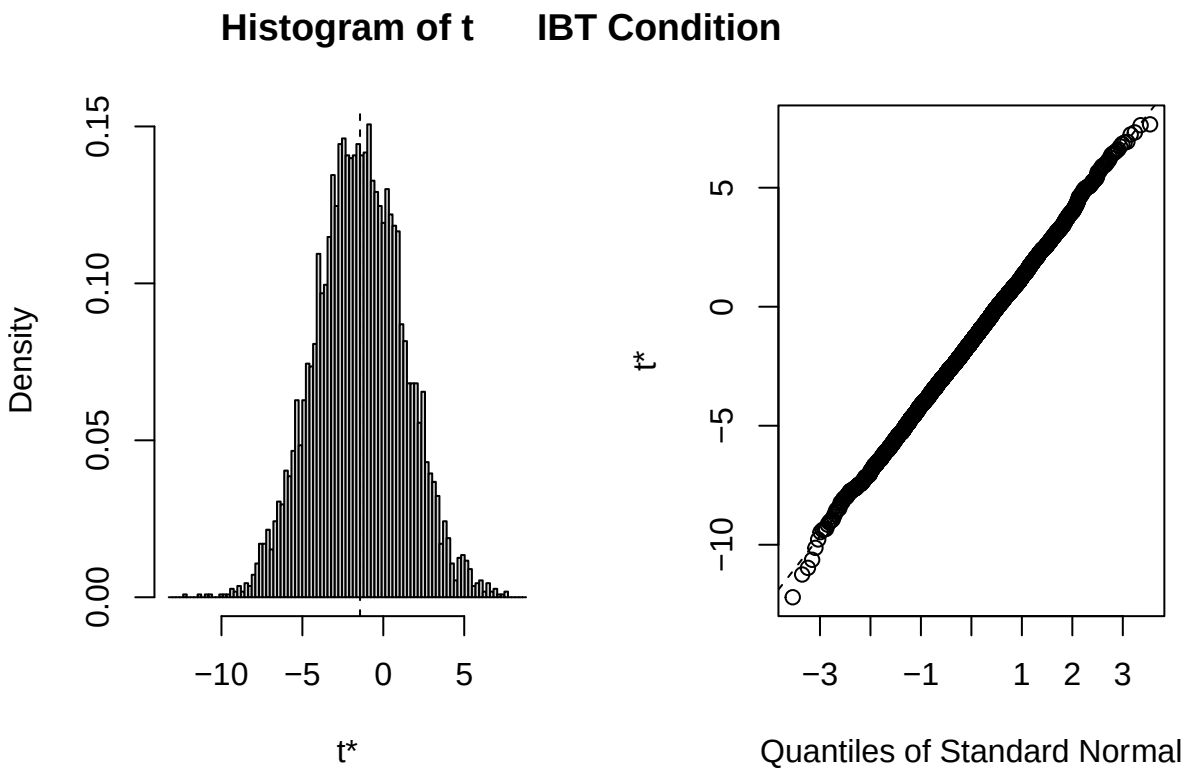
```
##
## Intervals :
## Level       BCa
## 95%   (-6.736,  3.952 )
## Calculations and Intervals on Original Scale

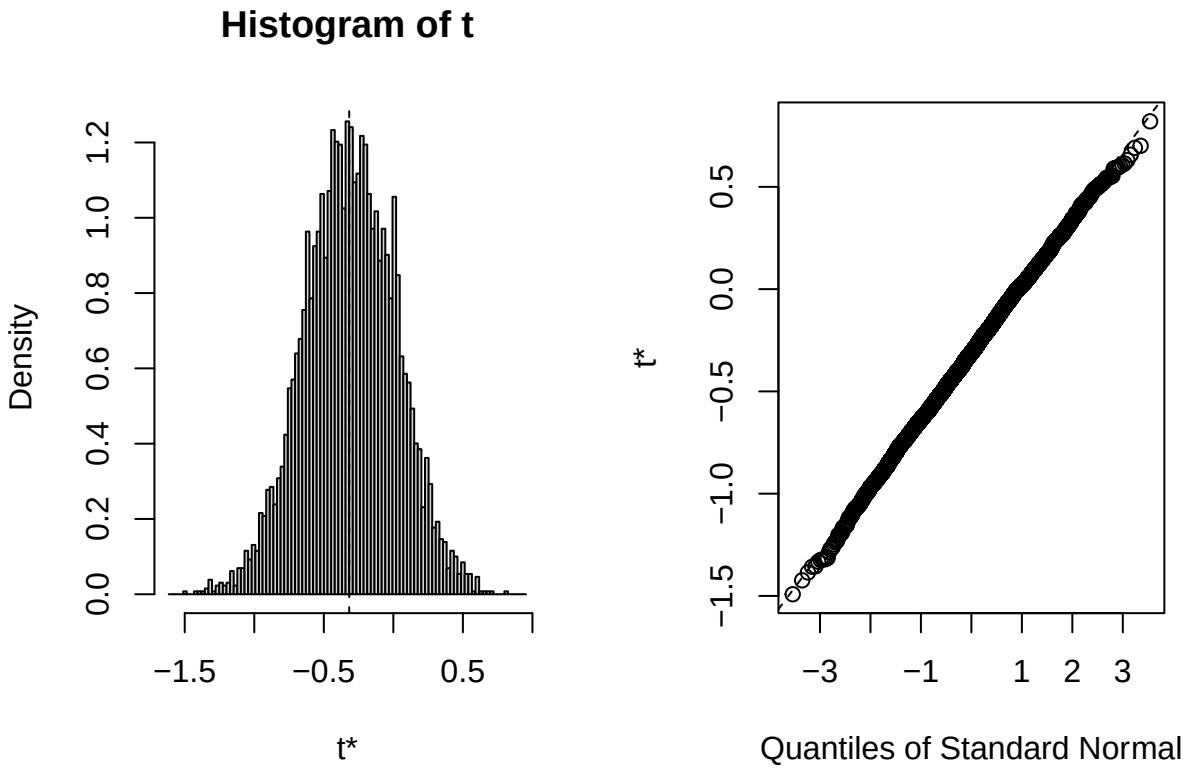
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-0.9832,  0.3012 )
## Calculations and Intervals on Original Scale
```

Bootstrap Diagnostics




Histogram of t
Density
t*
0
1
2
3
4
5
0.2
0.4
0.6
0.8
T1
t*
Quantiles of Standard Normal
0.2
0.3
0.4
0.5
0.6
0.7
0.8
−3
−1
1
2
3


Histogram of t IBT Condition
Density
0.00
0.05
0.10
0.15
−10
−5
0
5
t*
t*
5
0
−5
−10
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
−1.5
−0.5
0.5
t*
t*
−1.5
−1.0
−0.5
0.0
0.5
−3
−1
1
2
3
Quantiles of Standard Normal

4) Covariate analyses

Primary and Secondary models will first be run unadjusted. Models will then be adjusted for sociodemographic variables (age, IQ, SES). Finally, we will run models adjusted for individual level variables (substance use, SDQ variables as rated by teachers).

If sufficient parent data is returned, we will run an additional model adjusting for any differences in parent rated SDQ variables, as a sensitivity analysis.

4.1) IQ, Age, SES

4.1.1) Child

1) Define and summarise model

```
#making important variable factor:


model_c_asb_2_with_cov_1 <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + IQ_score + age_screening +
                               data =  child_asb_d1d2_wide)


summary(model_c_asb_2_with_cov_1)
```

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + IQ_score +
##     age_screening + FSM_eligibility, data = child_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -47.190  -6.457  -2.306   5.750  44.421
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              20.47502   28.60780   0.716    0.477
## ASB_sum_D1                0.65876    0.07677   8.581 2.42e-12 ***
## Training_ConditionTRAIN   0.87907    3.67760   0.239    0.812
## IQ_score                  0.05320    0.16600   0.320    0.750
## age_screening            -1.40668    1.91057  -0.736    0.464
## FSM_eligibilityYes       -1.92876    3.75989  -0.513    0.610
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.38 on 66 degrees of freedom
##   (10 observations deleted due to missingness)
## Multiple R-squared:  0.5531, Adjusted R-squared:  0.5192
## F-statistic: 16.34 on 5 and 66 DF,  p-value: 1.841e-10
```

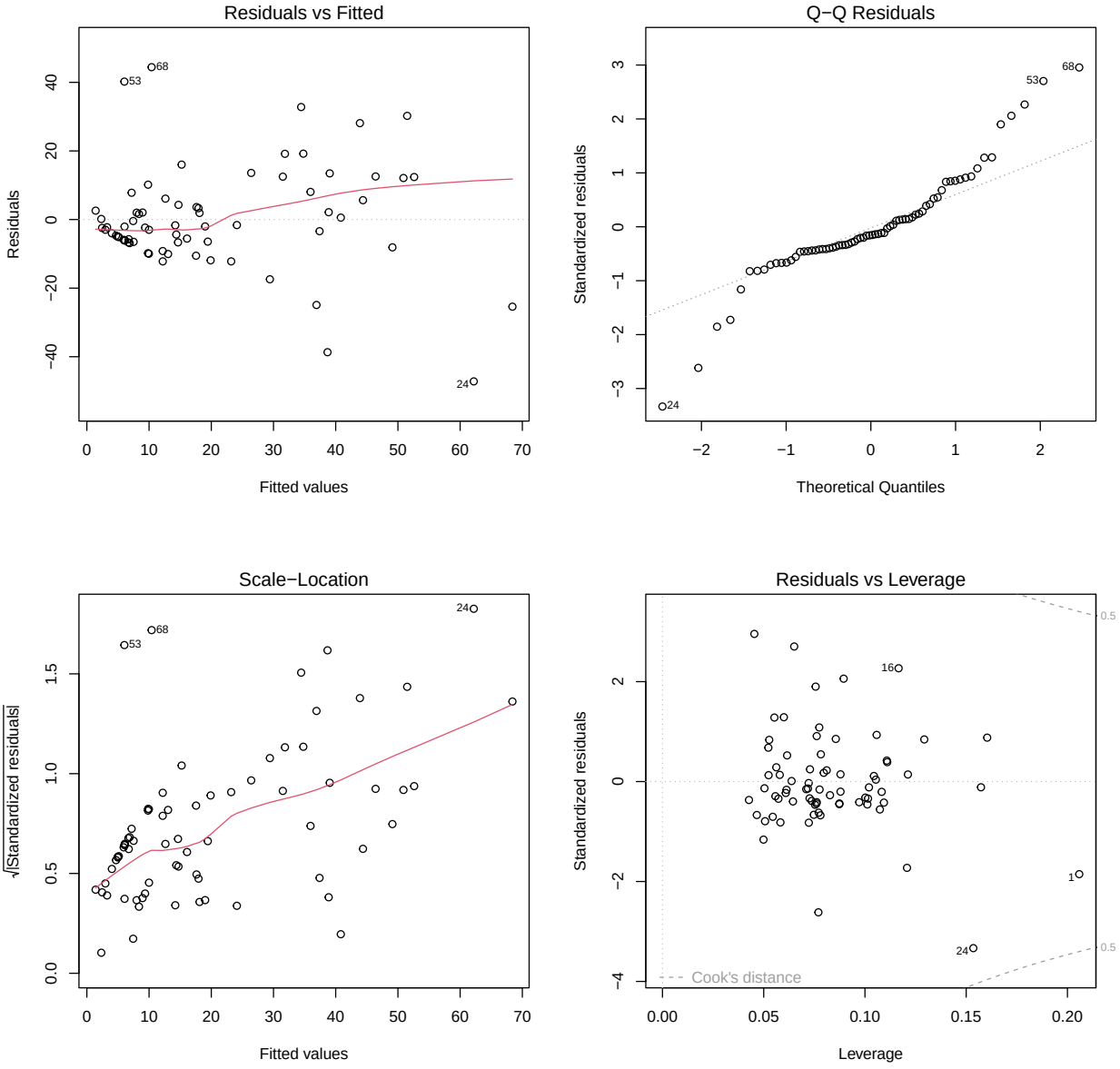
```
confint(model_c_asb_2_with_cov_1)
```

```
##                               2.5 %     97.5 %
```

```
## (Intercept)             -36.6422905 77.5923256
## ASB_sum_D1                0.5054891  0.8120293
## Training_ConditionTRAIN  -6.4635070  8.2216409
## IQ_score                 -0.2782250  0.3846155
## age_screening            -5.2212566  2.4079017
## FSM_eligibilityYes       -9.4356191  5.5781063
```

2) compare to original model - cannot do this as not every child has FSM eligiblity data

3) Get Diagnostics



4) Bootstrap:

```r
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition+ IQ_score + age_screening + FSM_eligibility,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide,
                      statistic = boot_coef_c_asb,
                      R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##        original       bias    std. error
## t1* 20.47501757 -1.760524518  24.5024958
## t2*  0.65875919  0.010343966   0.1128080
## t3*  0.87906698  0.008358534   3.7087444
## t4*  0.05319522  0.005658393   0.1617387
## t5* -1.40667741  0.079218249   1.5049478
## t6* -1.92875639  0.194518931   3.5640942
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-24.81,  70.62 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
```

```
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4145,  0.8640 )
## Calculations and Intervals on Original Scale

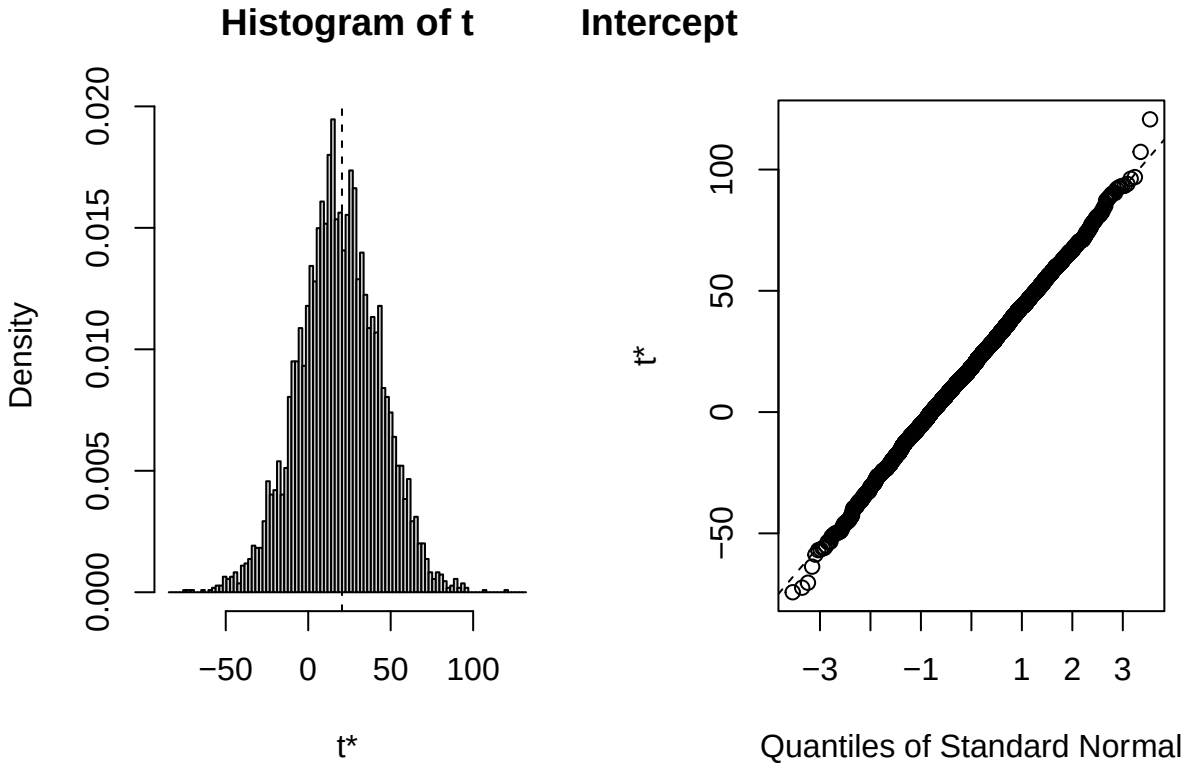
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.2196,  8.3062 )
## Calculations and Intervals on Original Scale

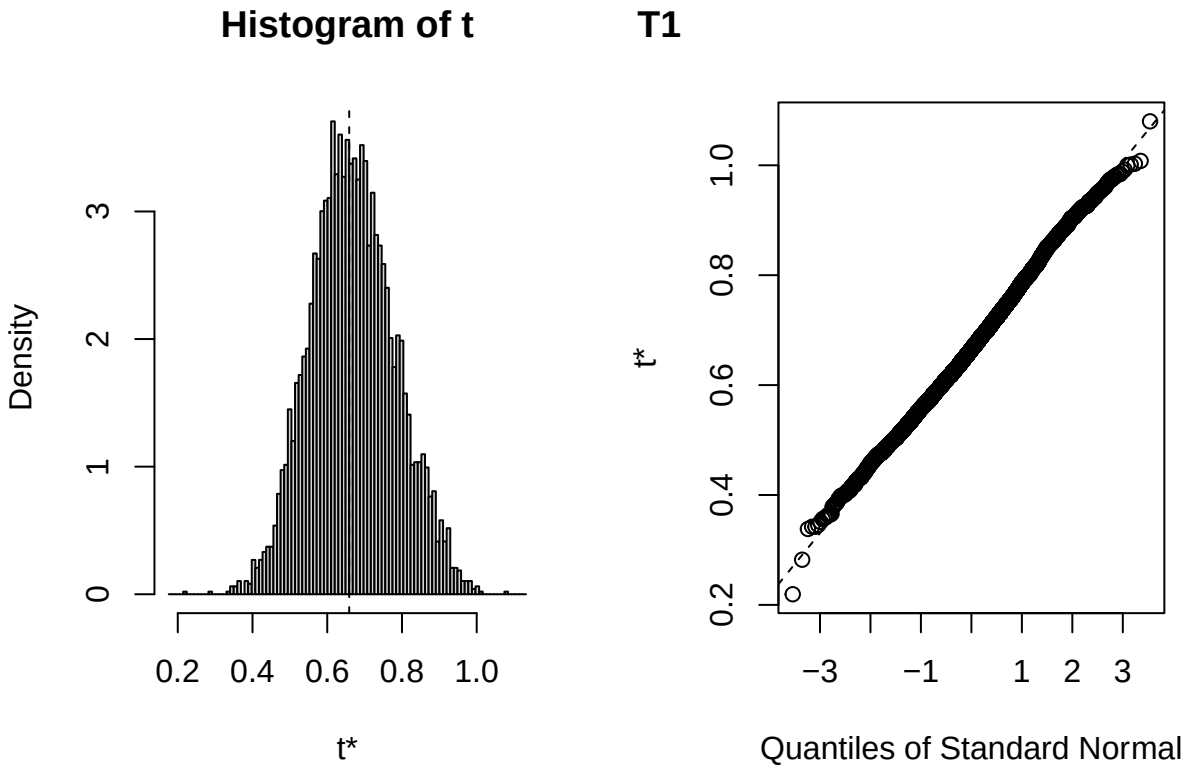
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-0.236,  0.412 )
## Calculations and Intervals on Original Scale

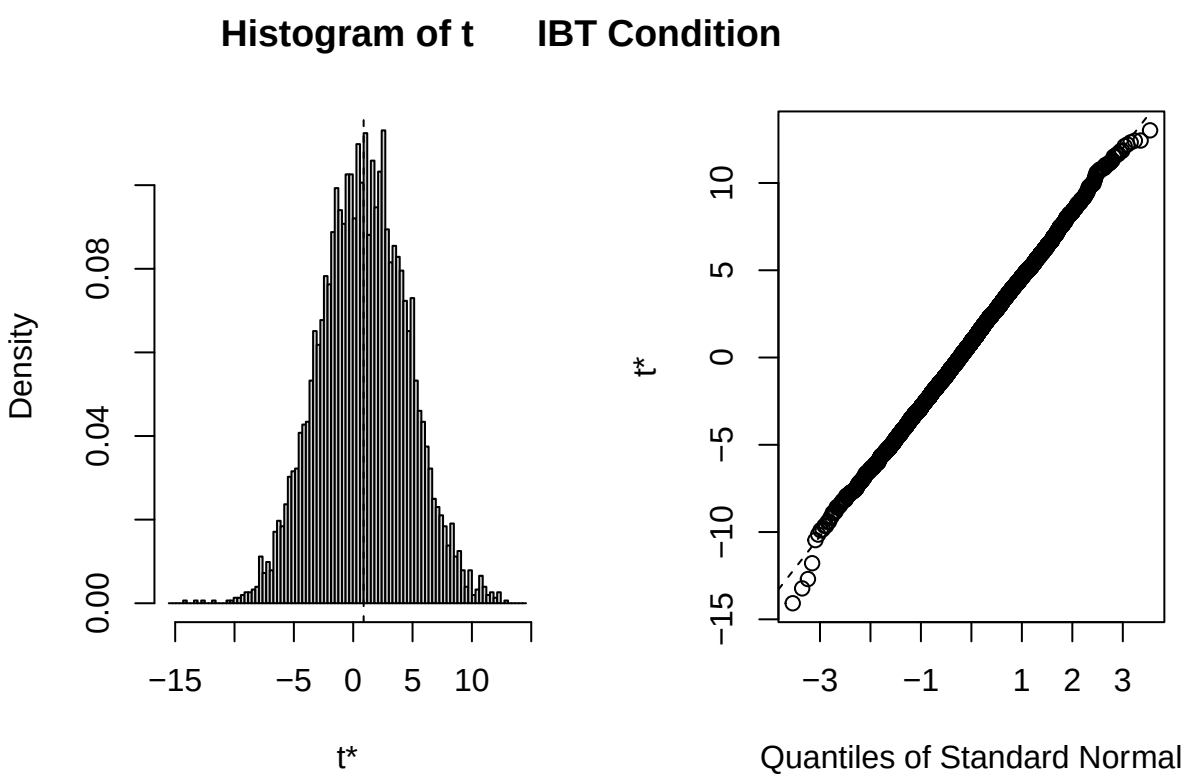
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-4.388,  1.633 )
## Calculations and Intervals on Original Scale

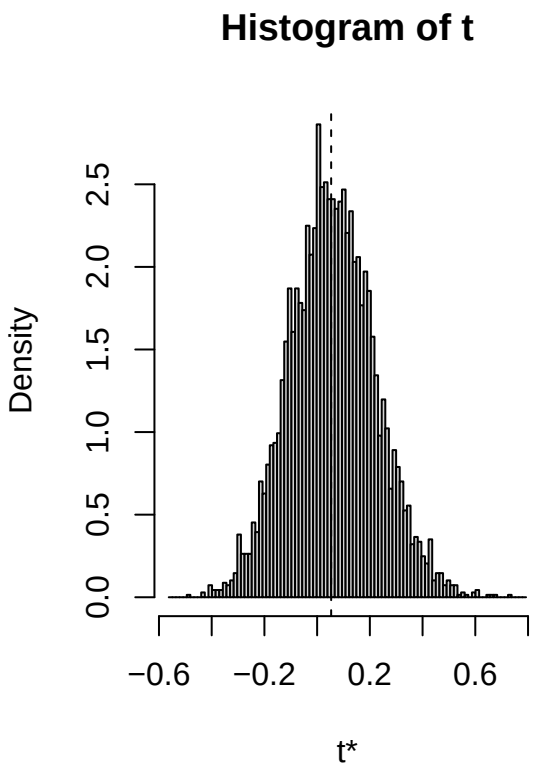
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 6)
##
## Intervals :
## Level       BCa
## 95%   (-9.547,  4.418 )
## Calculations and Intervals on Original Scale
```

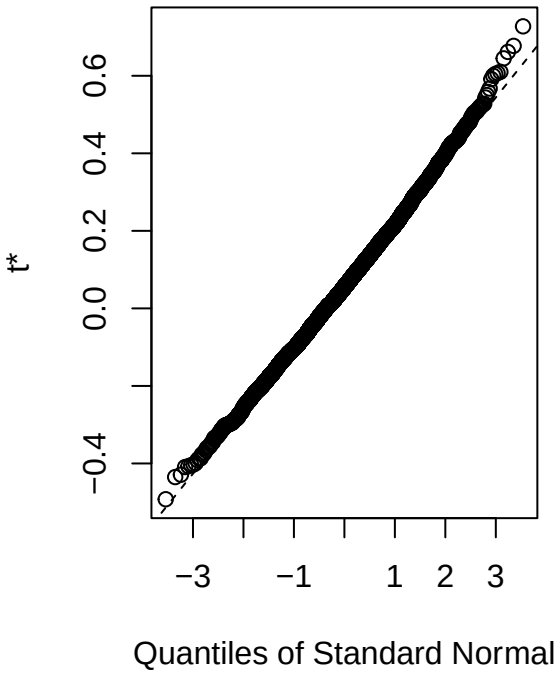
Bootstrap Diagnostics

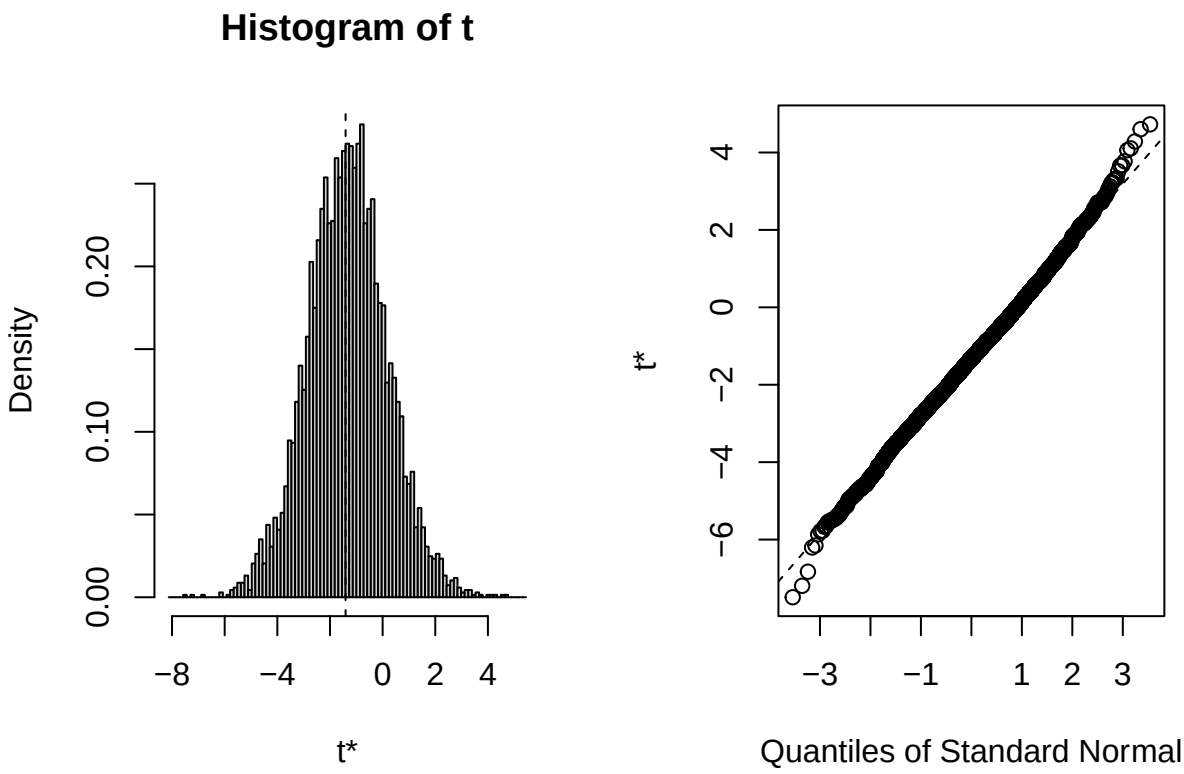

Histogram of t
Density
0.000
0.005
0.010
0.015
0.020
−50
0
50
100
t*
Intercept
t*
−50
0
50
100
−3
−1
1
2
3
Quantiles of Standard Normal

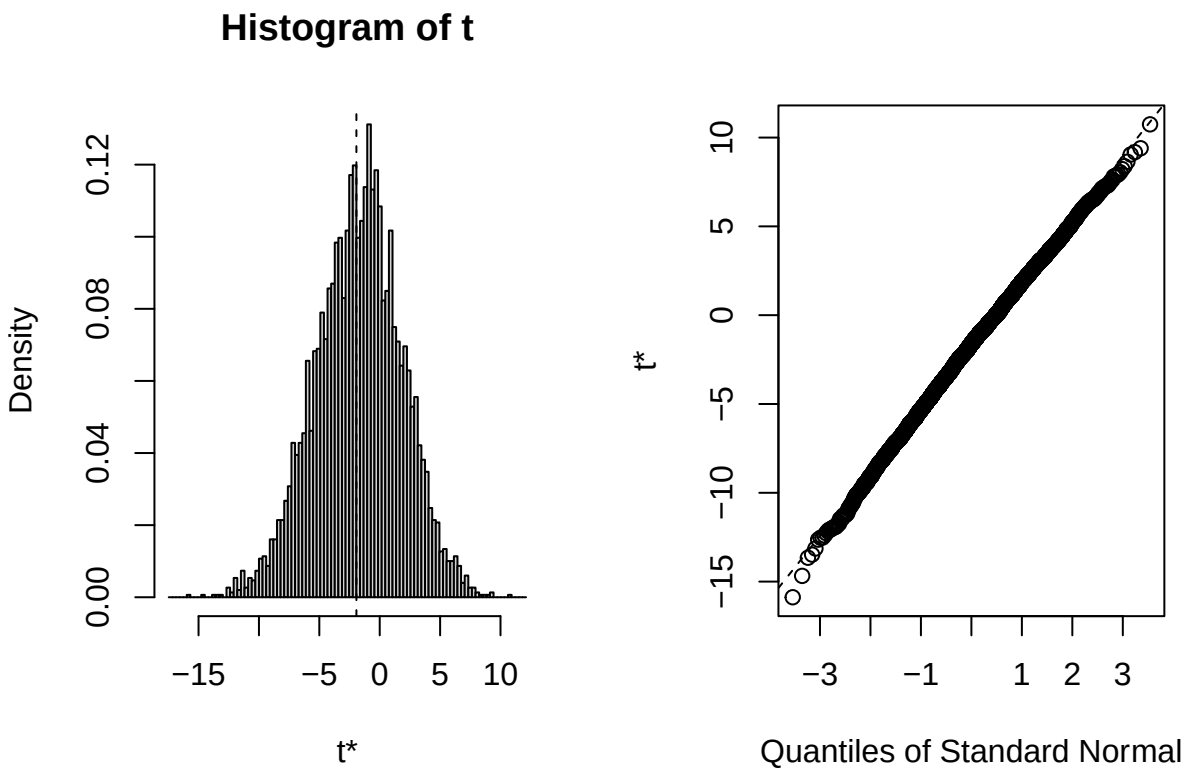

Histogram of t
Density
t*
T1
t*
Quantiles of Standard Normal


Histogram of t IBT Condition
Density
0.00
0.04
0.08
−15
−5
0
5
10
t*
t*
−15
−10
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.5
1.0
1.5
2.0
2.5
−0.6
−0.2
0.2
0.6
t*


t*
−0.4
0.0
0.2
0.4
0.6
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.10
0.20
−8
−4
0
2
4
t*
t*
−6
−4
−2
0
2
4
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.04
0.08
0.12
−15
−5
0
5
10
t*
t*
−15
−10
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal

4.1.2) Teacher

1) Define and summarise model

```
model_t_asb_3_with_cov_1 <- lm(ASB_mean_D2~ ASB_mean_D1 + Training_Condition + IQ_score + age_screening
                      data =  teach_asb_d1d2_wide)


summary(model_t_asb_3_with_cov_1)
```

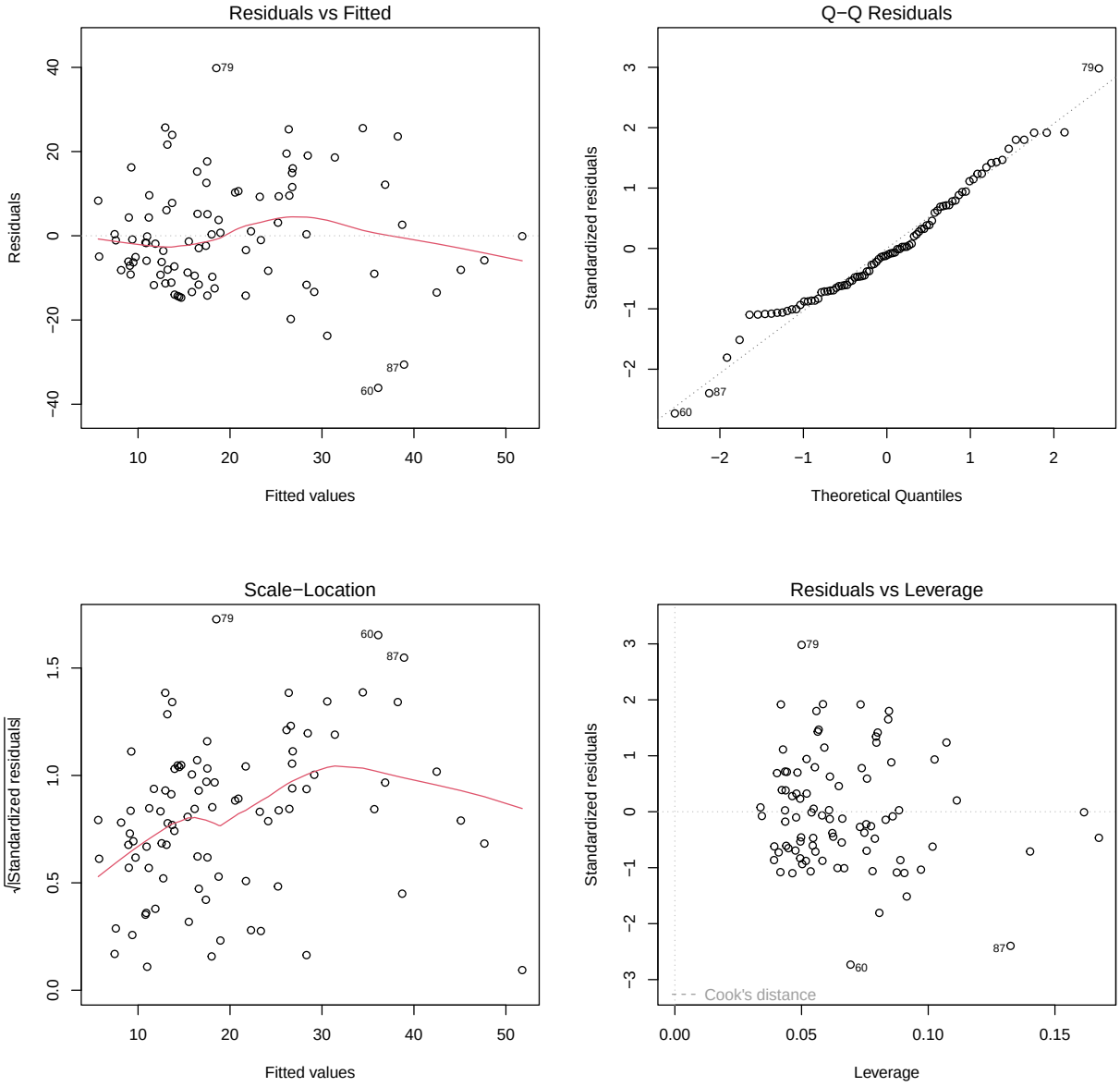
```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition +
##     IQ_score + age_screening + FSM_eligibility, data = teach_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.114  -9.152  -1.496   9.341  39.819
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)              -28.53360   22.28472  -1.280   0.2039
## ASB_mean_D1                0.50781    0.07857   6.463 6.39e-09 ***
## Training_ConditionTRAIN   -2.07231    2.97629  -0.696   0.4882
## IQ_score                   0.06482    0.13673   0.474   0.6367
## age_screening              2.65479    1.46155   1.816   0.0729 .
## FSM_eligibilityYes        -2.74263    2.97429  -0.922   0.3591
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.7 on 84 degrees of freedom
##   (12 observations deleted due to missingness)
## Multiple R-squared:  0.3672, Adjusted R-squared:  0.3296
## F-statistic:  9.75 on 5 and 84 DF,  p-value: 2.269e-07
```

```
confint(model_t_asb_3_with_cov_1)
```

```
##                                2.5 %     97.5 %
## (Intercept)              -72.8492170 15.7820104
## ASB_mean_D1                0.3515604  0.6640563
## Training_ConditionTRAIN   -7.9909864  3.8463604
## IQ_score                  -0.2070816  0.3367236
## age_screening             -0.2516680  5.5612457
## FSM_eligibilityYes        -8.6573261  3.1720641
```

2) compare to original model

3) Get Diagnostics



4) Bootstrap

```
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition+ IQ_score + age_screening + FSM_eligibility,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
```

```r
boot_results_c_asb <- boot(data = teach_asb_d1d2_wide,
                     statistic = boot_coef_c_asb,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##         original        bias    std. error
## t1* -28.53360333  0.2196763304 21.78208545
## t2*   0.50780836  0.0060320147  0.08771502
## t3*  -2.07231302 -0.0794260727  3.00950143
## t4*   0.06482098  0.0006380507  0.13459139
## t5*   2.65478886 -0.0240934352  1.53039628
## t6*  -2.74263099  0.0470282001  3.05154356
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-70.17,  15.33 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3366,  0.6816 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
```

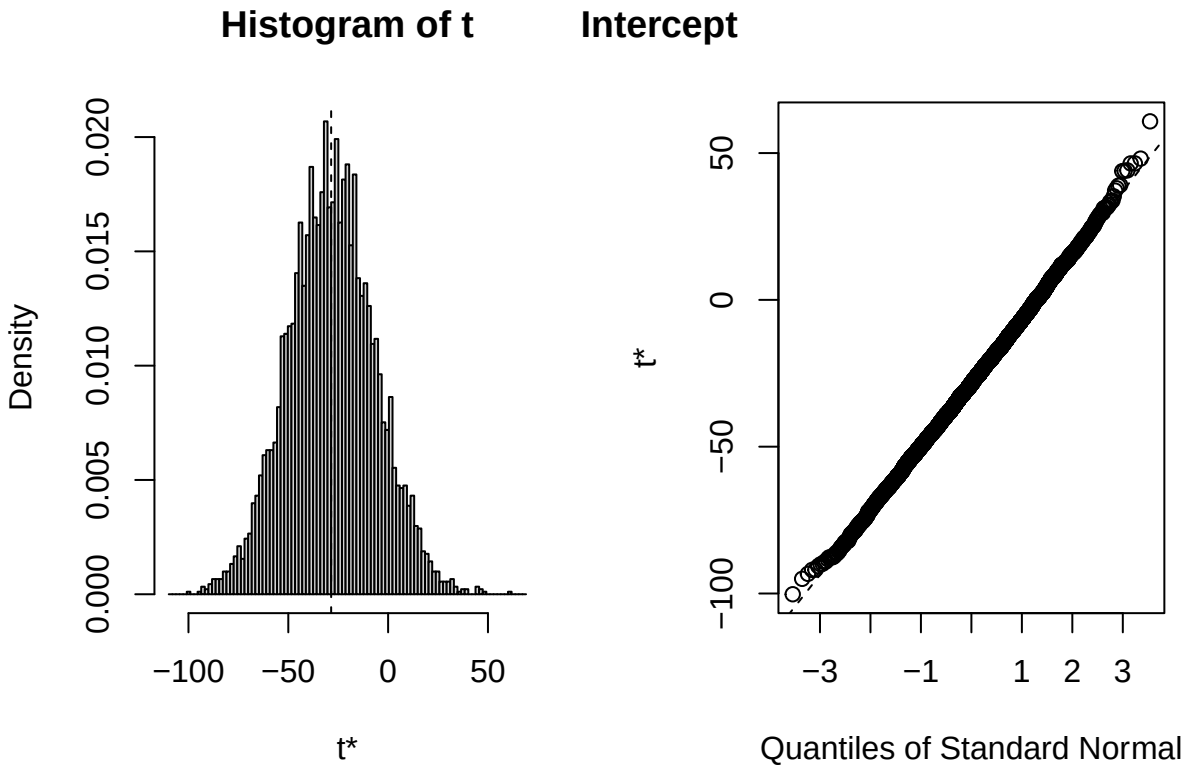
```
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-7.742,  3.991 )
## Calculations and Intervals on Original Scale

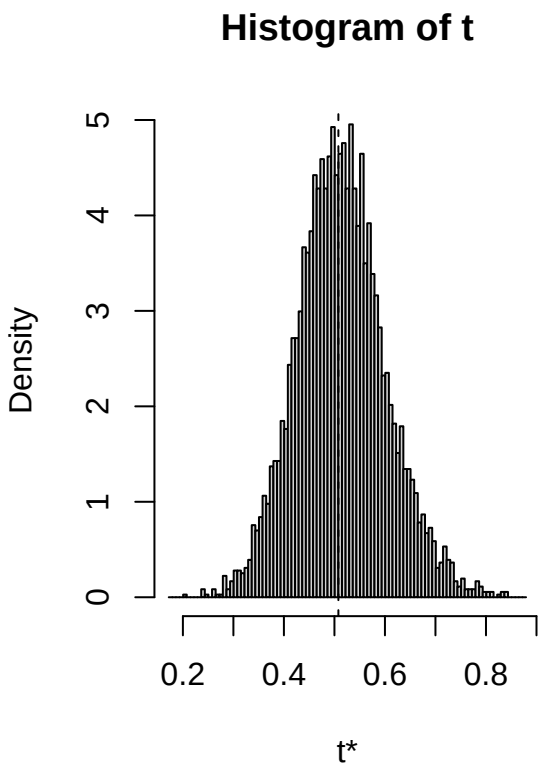
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-0.2201,  0.3165 )
## Calculations and Intervals on Original Scale

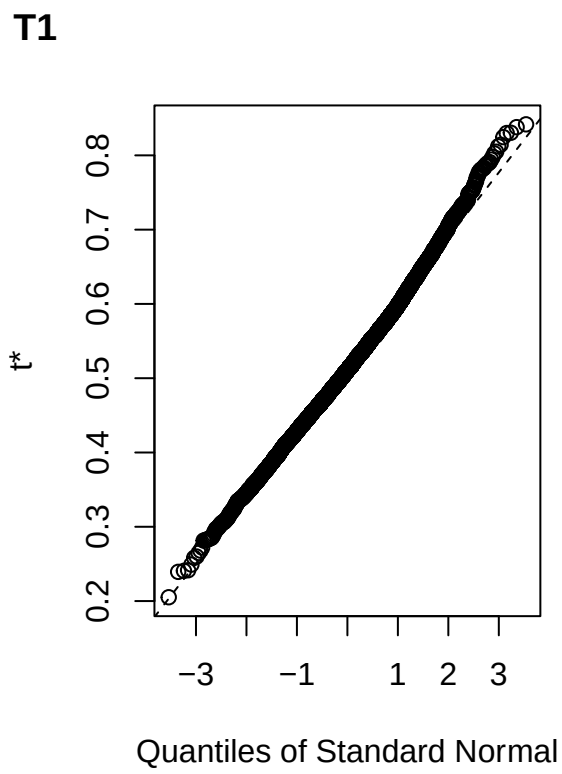
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-0.229,  5.927 )
## Calculations and Intervals on Original Scale

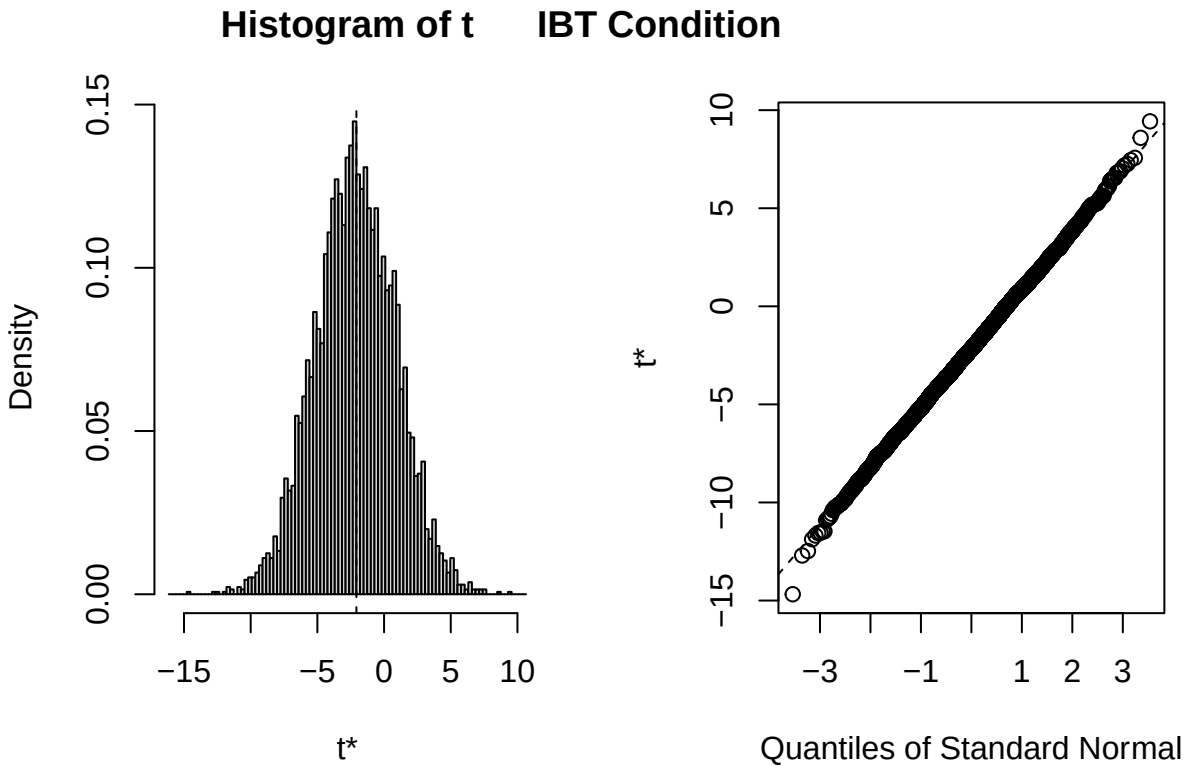
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 6)
##
## Intervals :
## Level       BCa
## 95%   (-9.189,  2.719 )
## Calculations and Intervals on Original Scale
```

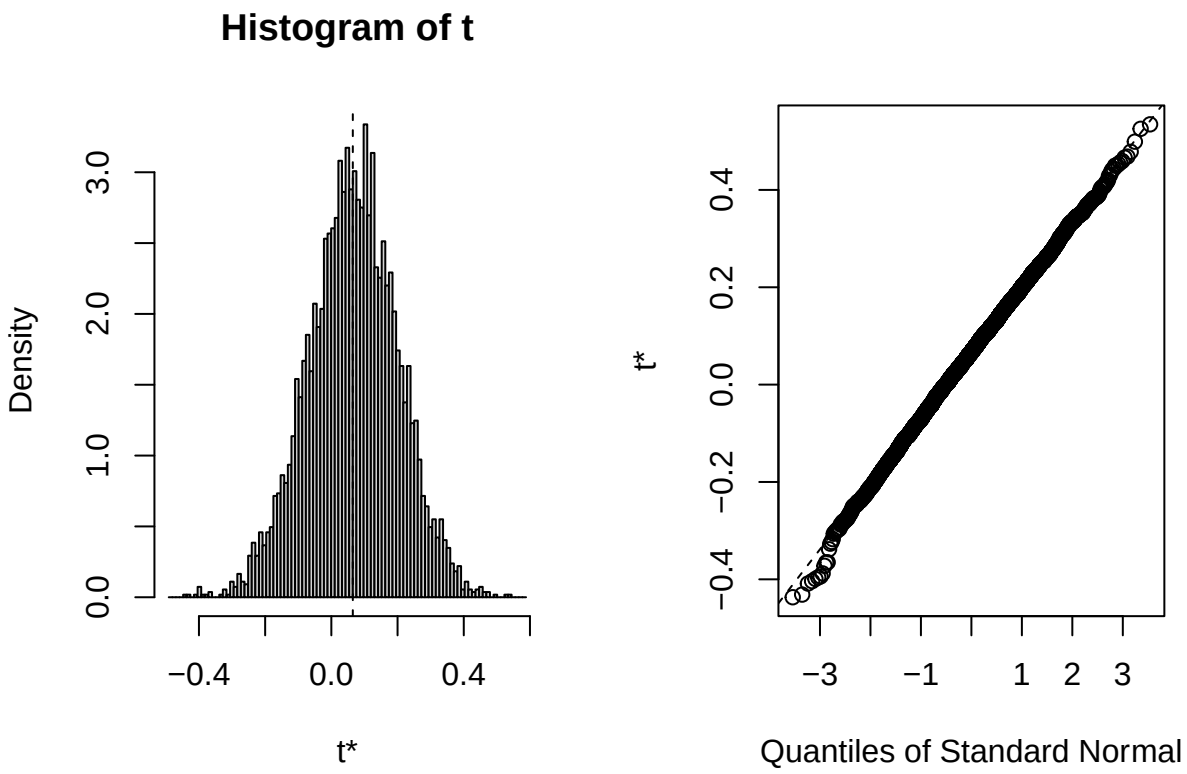
5) Bootstrap Diagnostics

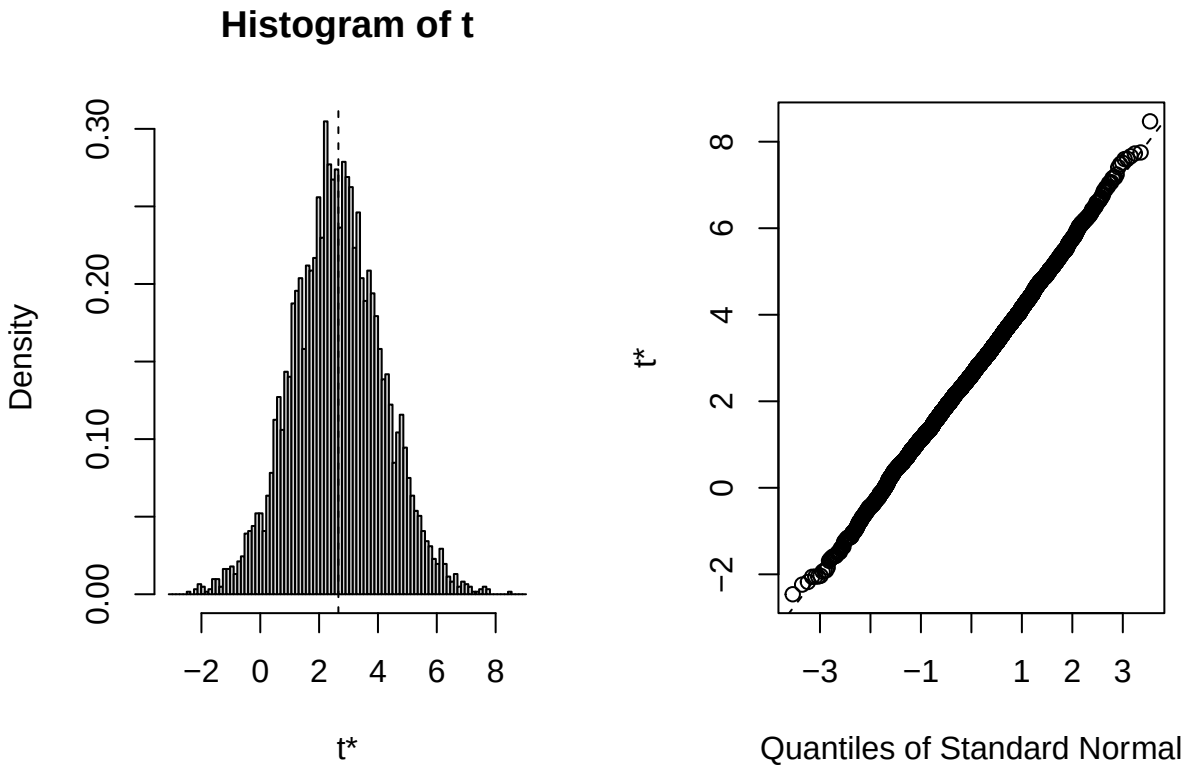

Histogram of t
Density
0.000
0.005
0.010
0.015
0.020
−100
−50
0
50
t*
Intercept
t*
50
0
−50
−100
−3
−1
1
2
3
Quantiles of Standard Normal

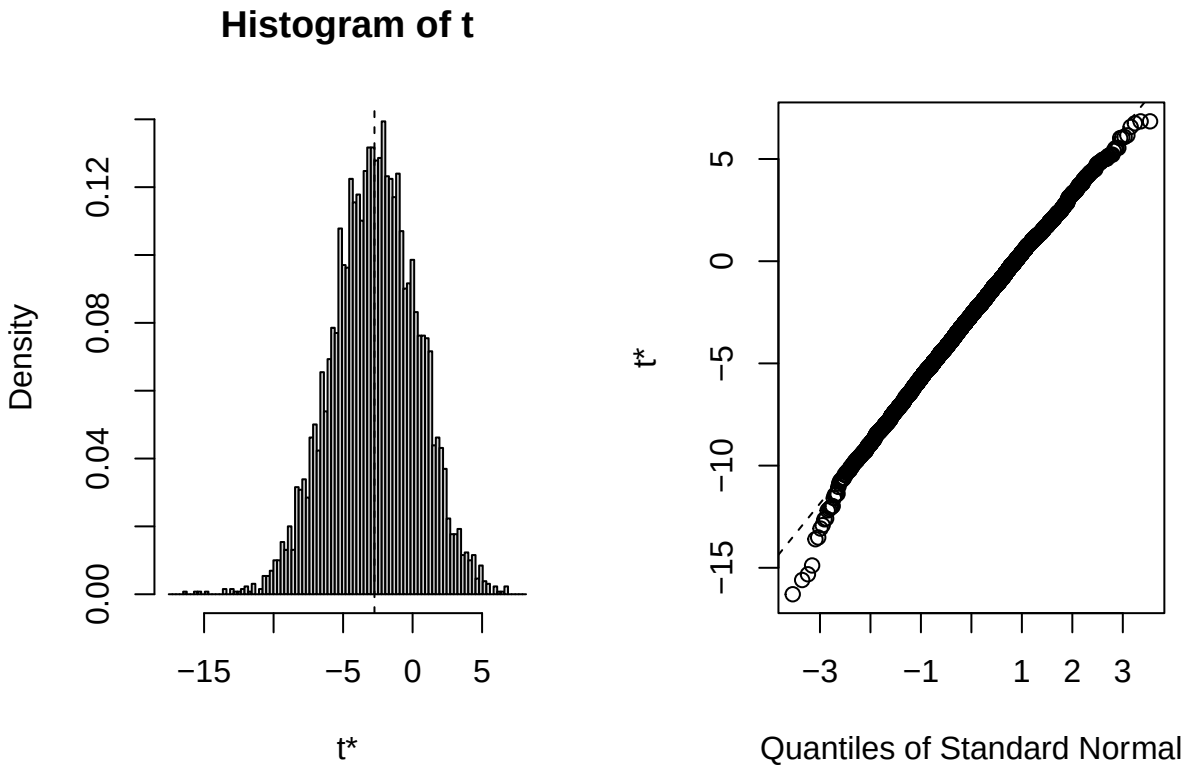

Histogram of t
Density
t*
0
1
2
3
4
5
0.2
0.4
0.6
0.8


T1
t*
Quantiles of Standard Normal
0.2
0.3
0.4
0.5
0.6
0.7
0.8
−3
−1
1
2
3


Histogram of t IBT Condition
Density
0.00
0.05
0.10
0.15
−15
−5
0
5
10
t*
t*
10
5
0
−5
−10
−15
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
1.0
2.0
3.0
−0.4
0.0
0.4
t*
t*
−0.4
−0.2
0.0
0.2
0.4
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.10
0.20
0.30
−2
0
2
4
6
8
t*
t*
−2
0
2
4
6
8
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.04
0.08
0.12
−15
−5
0
5
t*
t*
5
0
−5
−10
−15
−3
−1
1
2
3
Quantiles of Standard Normal

4.2) AUDIT, DUDIT, SDQ

4.2.1) Child

1) Define and summarise model

```
#making important variable factor:


model_c_asb_2_with_cov_2 <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + AUDIT_risk + DUDIT_risk +
                               data =  child_asb_d1d2_wide)


summary(model_c_asb_2_with_cov_2)
```

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + AUDIT_risk +
##     DUDIT_risk + SDQ_EmotionalProblems + SDQ_Hyperactivity +
##     SDQ_PeerProblems + SDQ_Prosocial, data = child_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -45.894  -6.352  -1.468   6.515  37.757
##
## Coefficients:
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       -0.76711    7.32455  -0.105   0.9169
## ASB_sum_D1                         0.64747    0.07533   8.595 1.19e-12 ***
## Training_ConditionTRAIN            1.13289    3.31666   0.342   0.7337
## AUDIT_riskRisky or hazardous level  8.87659   18.23996   0.487   0.6280
## DUDIT_riskPossible drug problems   -5.11299   10.47643  -0.488   0.6270
## SDQ_EmotionalProblems               0.74412    0.88982   0.836   0.4058
## SDQ_Hyperactivity                   1.27907    0.78758   1.624   0.1087
## SDQ_PeerProblems                   -2.06822    1.06778  -1.937   0.0567 .
## SDQ_Prosocial                      -0.13782    0.85415  -0.161   0.8723
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.53 on 72 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.5989, Adjusted R-squared:  0.5544
## F-statistic: 13.44 on 8 and 72 DF,  p-value: 1.079e-11
```

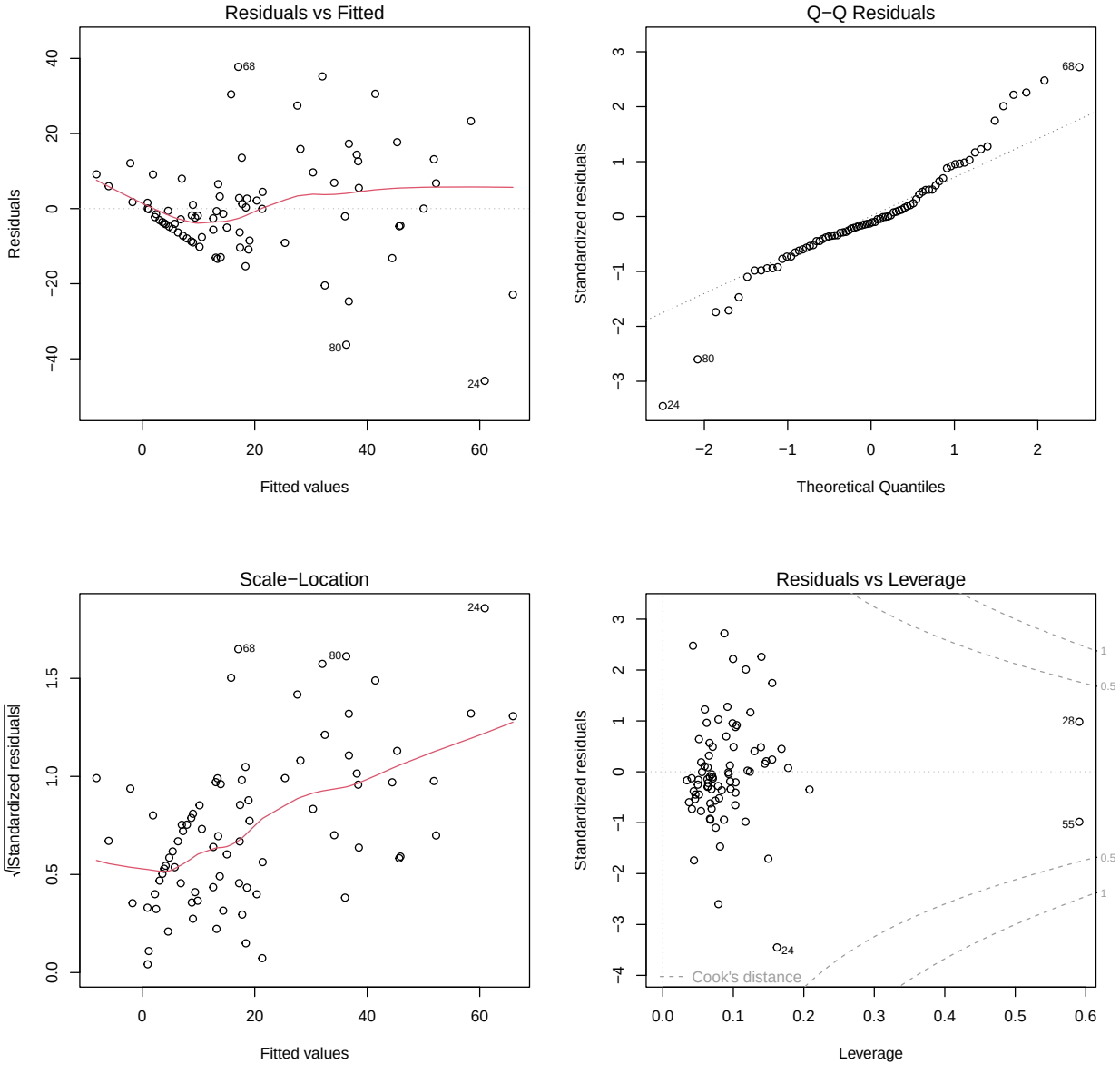
```
confint(model_c_asb_2_with_cov_2)
```

```
##                                       2.5 %      97.5 %
## (Intercept)                      -15.3683246 13.83411343
## ASB_sum_D1                         0.4973047  0.79763311
## Training_ConditionTRAIN           -5.4787593  7.74453732
## AUDIT_riskRisky or hazardous level -27.4841089 45.23728879
## DUDIT_riskPossible drug problems   -25.9973832 15.77139895
## SDQ_EmotionalProblems              -1.0297038  2.51793491
```

```
## SDQ_Hyperactivity                -0.2909514  2.84908815
## SDQ_PeerProblems                 -4.1968030  0.06036399
## SDQ_Prosocial                    -1.8405351  1.56488961
```

2) compare to original model - cannot do this as not every child has FSM eligiblity data

3) Get Diagnostics



4) Bootstrap

```
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
```

```
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + SDQ_EmotionalProblems + SDQ_Hyperactivity + S
                    data =  d)
  return(coef(fit))
}

 child_asb_d1d2_wide %>% group_by(AUDIT_risk) %>% summarise(n=n())
```

```
## # A tibble: 3 x 2
##   AUDIT_risk                   n
##   <fct>                    <int>
## 1 Low risk                    80
## 2 Risky or hazardous level     1
## 3 <NA>                         1
```

```
child_asb_d1d2_wide %>% group_by(DUDIT_risk) %>% summarise(n=n())
```

```
## # A tibble: 3 x 2
##   DUDIT_risk                 n
##   <fct>                  <int>
## 1 Low risk                  78
## 2 Possible drug problems     3
## 3 <NA>                       1
```

```
str(child_asb_d1d2_wide$sdq_)
```

```
##  NULL
```

```
# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide,
                      statistic = boot_coef_c_asb,
                      R = 5000)
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* -1.0305705 -0.207416443   7.3585098
## t2*  0.6631593  0.007157962   0.1005889
## t3*  0.8989575  0.171565190   3.4204756
## t4*  0.7451475 -0.016421883   0.8456914
## t5*  1.2991844 -0.012746544   0.6399557
## t6* -2.0538535  0.078991572   1.0987954
## t7* -0.1647084 -0.036753035   0.8839345
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-13.109,  16.342 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4569,  0.8533 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.2450,  7.2454 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-0.8220,  2.5187 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 5)
##
## Intervals :
```

```
## Level       BCa
## 95%   ( 0.037,  2.590 )
## Calculations and Intervals on Original Scale

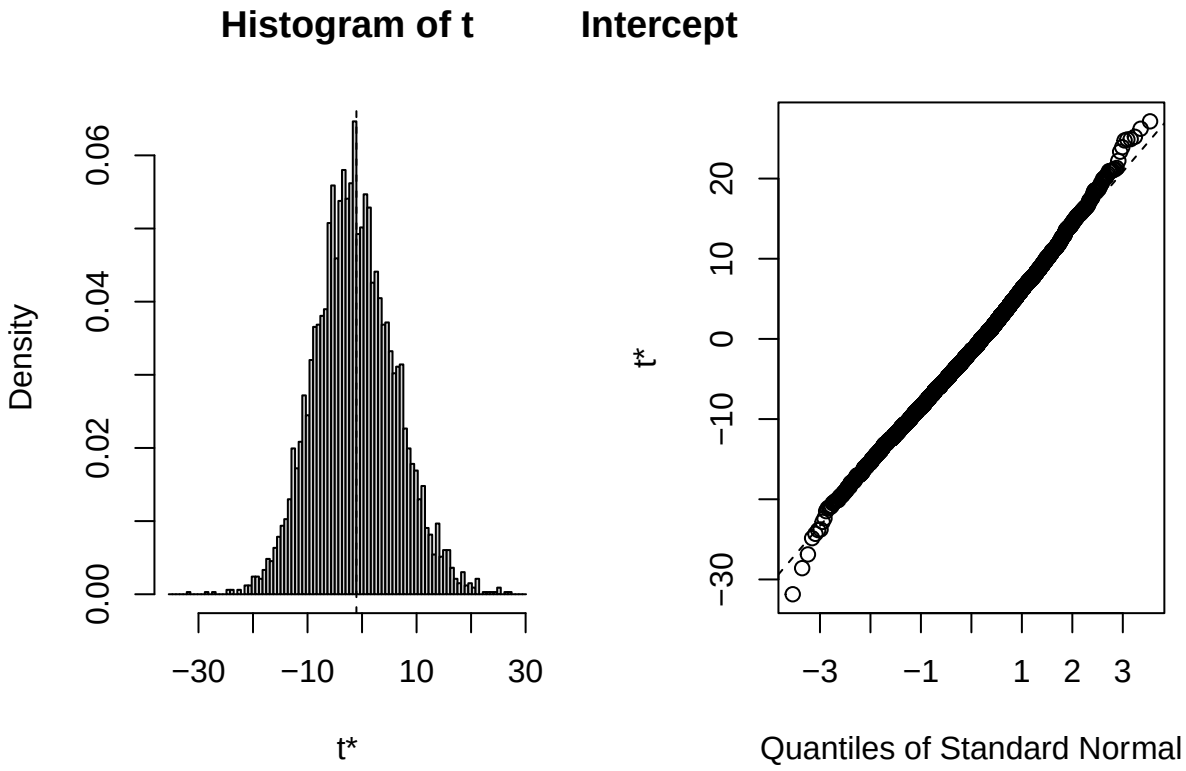
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 6)
##
## Intervals :
## Level       BCa
## 95%   (-4.632, -0.261 )
## Calculations and Intervals on Original Scale

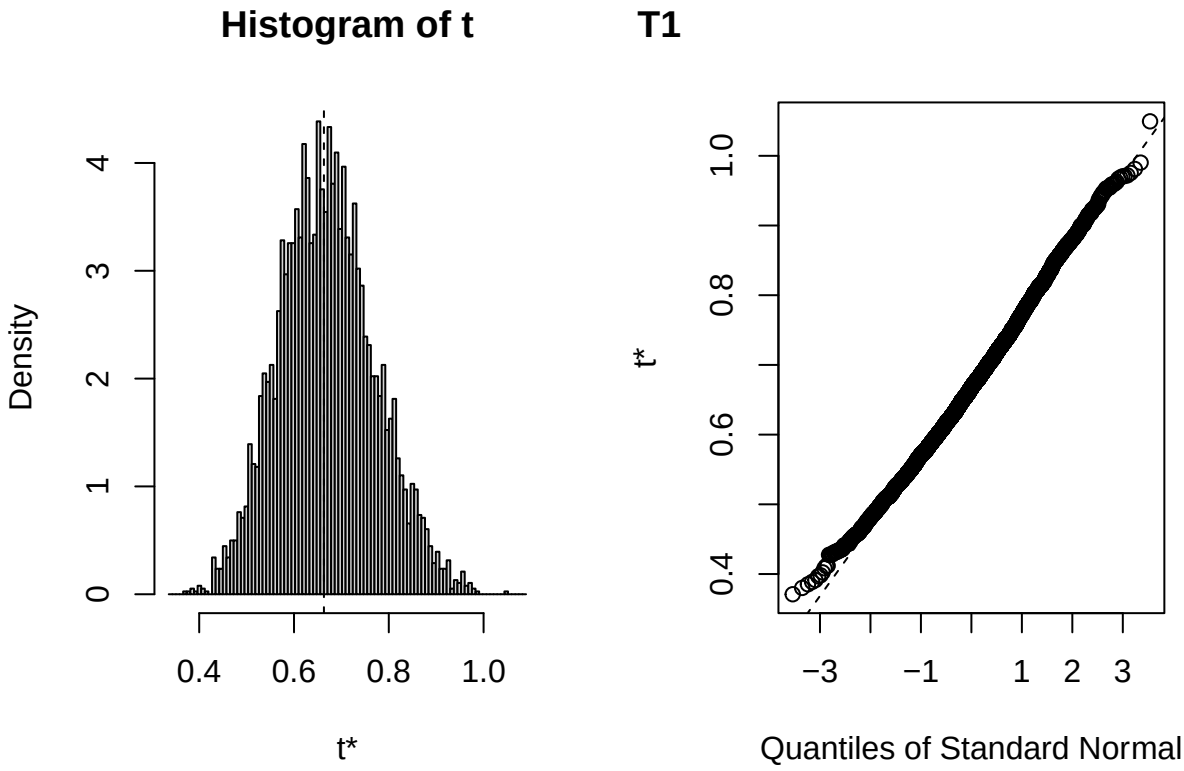
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 7)
##
## Intervals :
## Level       BCa
## 95%   (-2.0732,  1.3886 )
## Calculations and Intervals on Original Scale

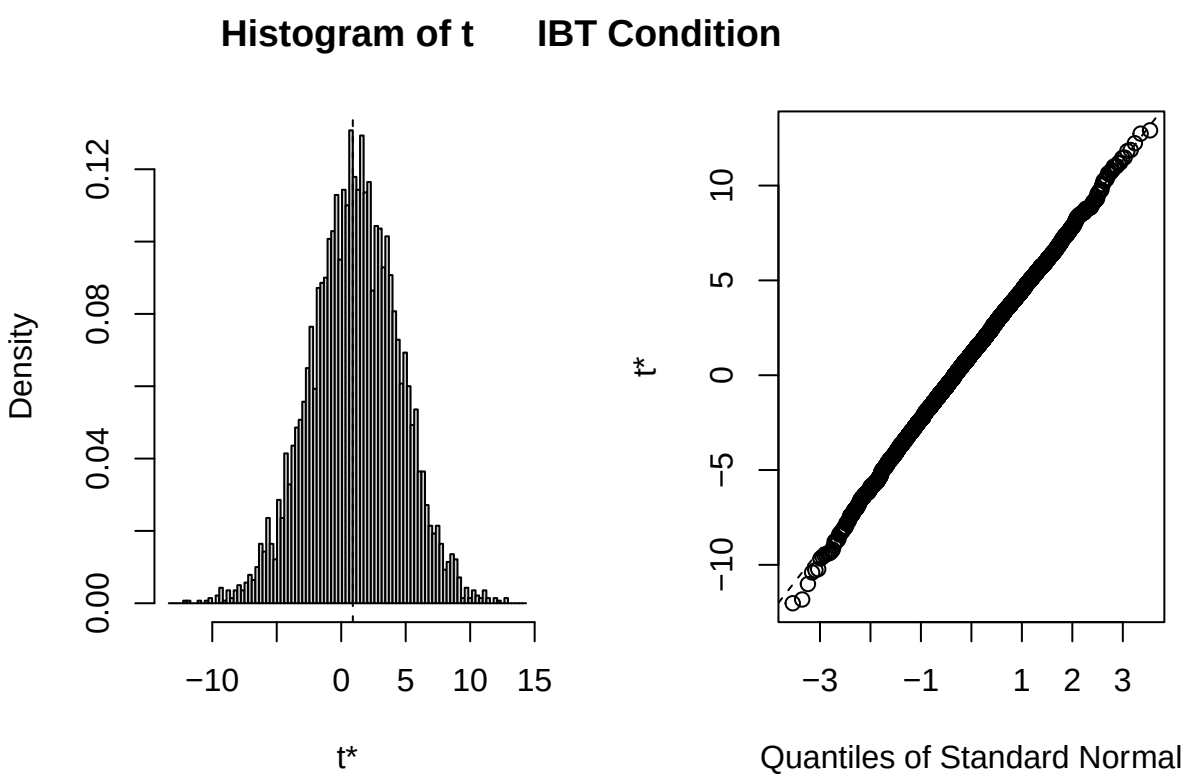
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot::boot.ci(boot.out = boot_results_c_asb, conf = 0.995, type = "bca",
##     index = 5)
##
## Intervals :
## Level       BCa
## 99.5%   (-0.565,  3.108 )
## Calculations and Intervals on Original Scale
## Some BCa intervals may be unstable

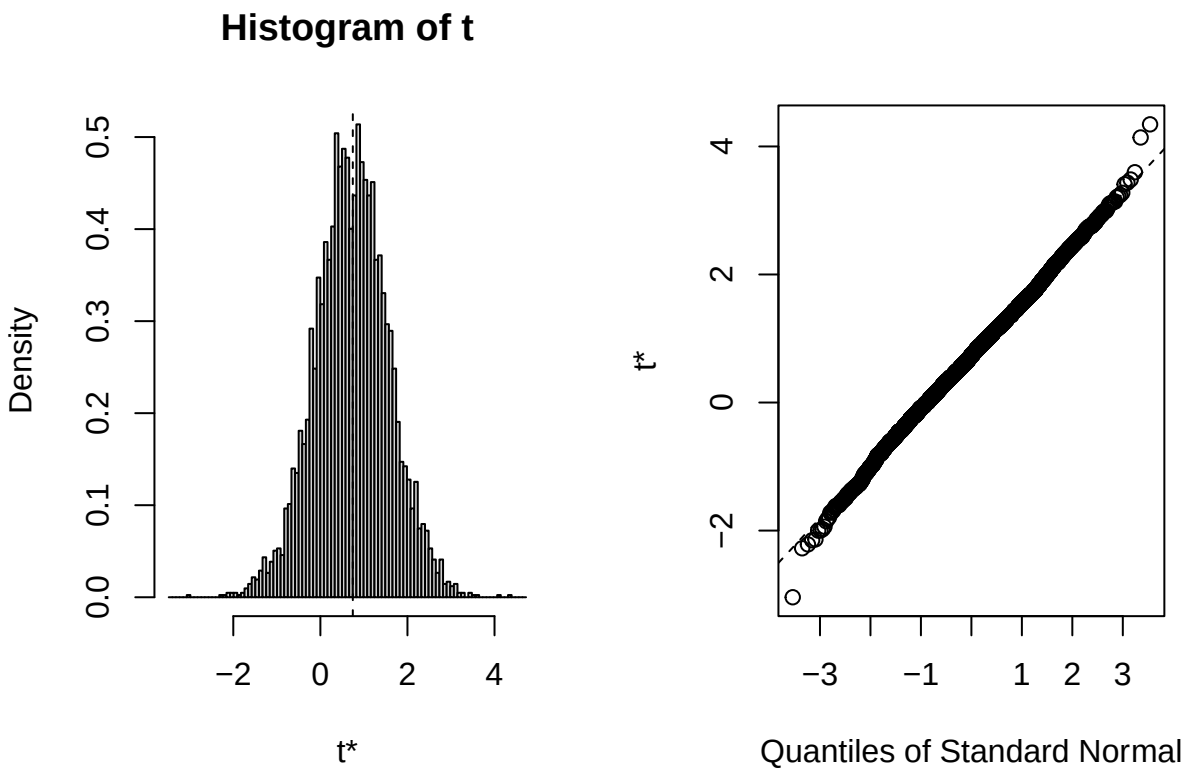
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, conf = 0.995, type = "bca",
##     index = 6)
##
## Intervals :
## Level       BCa
## 99.5%   (-6.356,  0.526 )
## Calculations and Intervals on Original Scale
## Some BCa intervals may be unstable
```

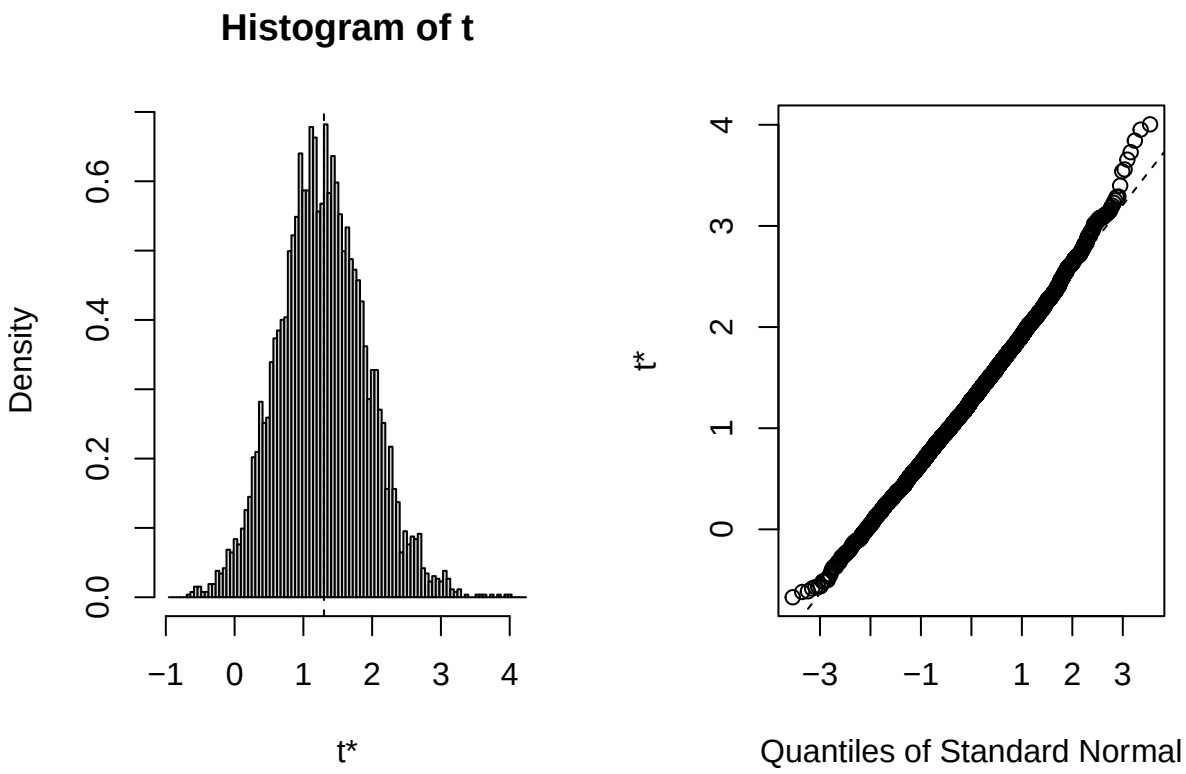
Bootstrap Diagnostics

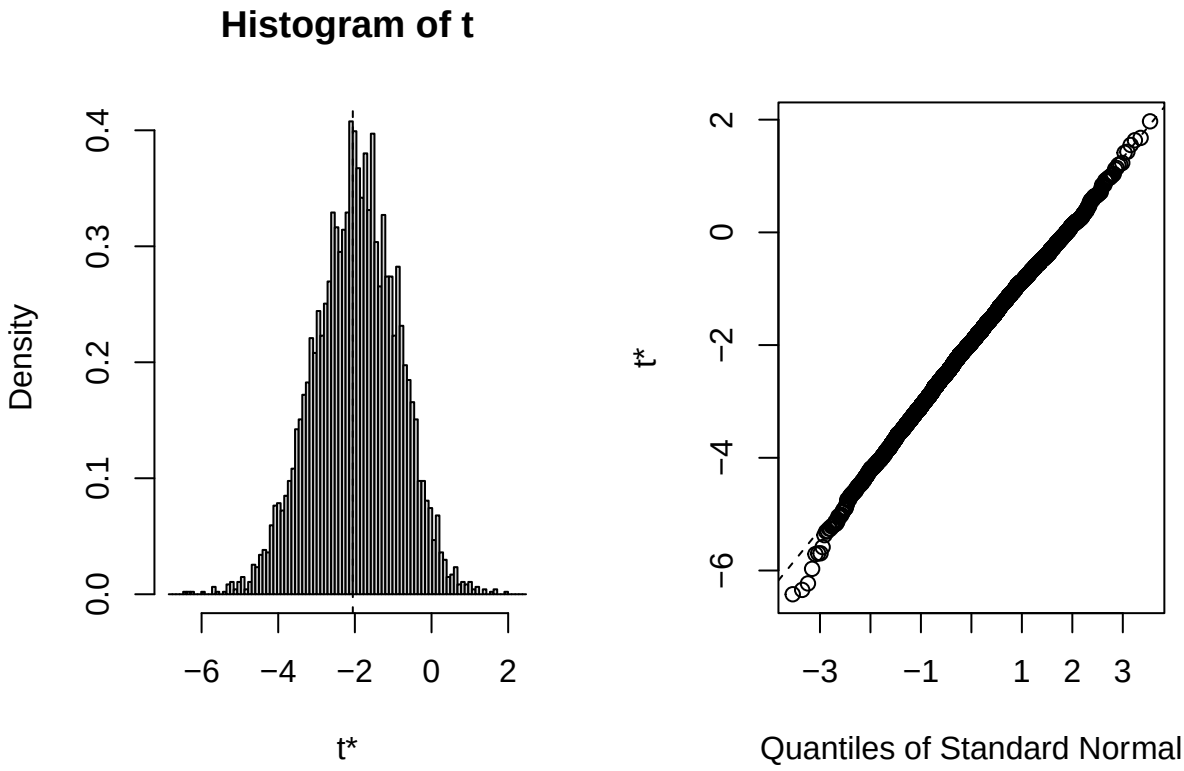

Histogram of t
Density
0.00
0.02
0.04
0.06
−30
−10
10
30
t*
Intercept
t*
−30
−10
0
10
20
−3
−1
1
2
3
Quantiles of Standard Normal

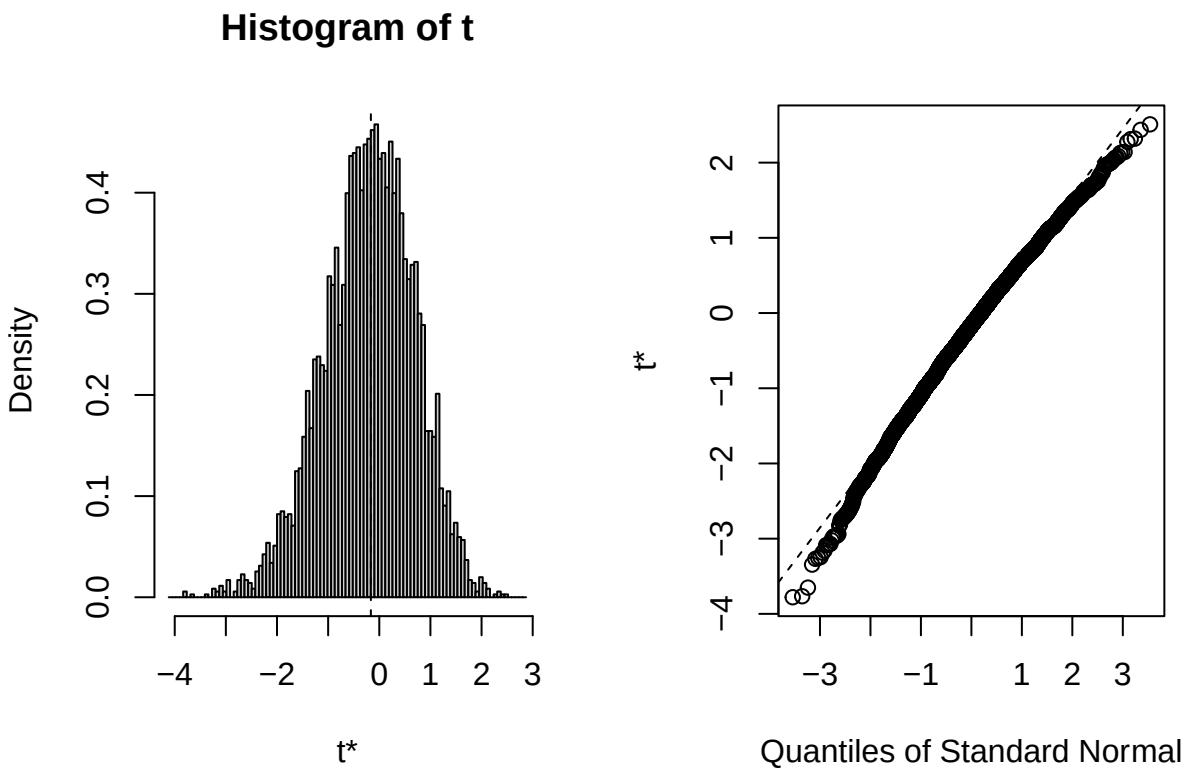

Histogram of t
Density
t*
0
1
2
3
4
0.4
0.6
0.8
1.0
T1
t*
Quantiles of Standard Normal
0.4
0.6
0.8
1.0
−3
−1
1
2
3


Histogram of t IBT Condition
Density
0.00
0.04
0.08
0.12
−10
0
5
10
15
t*
t*
10
5
0
−5
−10
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.1
0.2
0.3
0.4
0.5
−2
0
2
4
t*
t*
−2
0
2
4
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.2
0.4
0.6
−1
0
1
2
3
4
t*
t*
0
1
2
3
4
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.1
0.2
0.3
0.4
−6
−4
−2
0
2
t*
t*
2
0
−2
−4
−6
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.1
0.2
0.3
0.4
−4
−2
0
1
2
3
t*
t*
−4
−3
−2
−1
0
1
2
−3
−1
1
2
3
Quantiles of Standard Normal

4.1.2) Teacher

1) Define and summarise model

```
model_t_asb_3_with_cov_2 <- lm(ASB_mean_D2~ ASB_mean_D1 + Training_Condition + AUDIT_risk + DUDIT_risk +
                      data =  teach_asb_d1d2_wide)


summary(model_t_asb_3_with_cov_2)
```

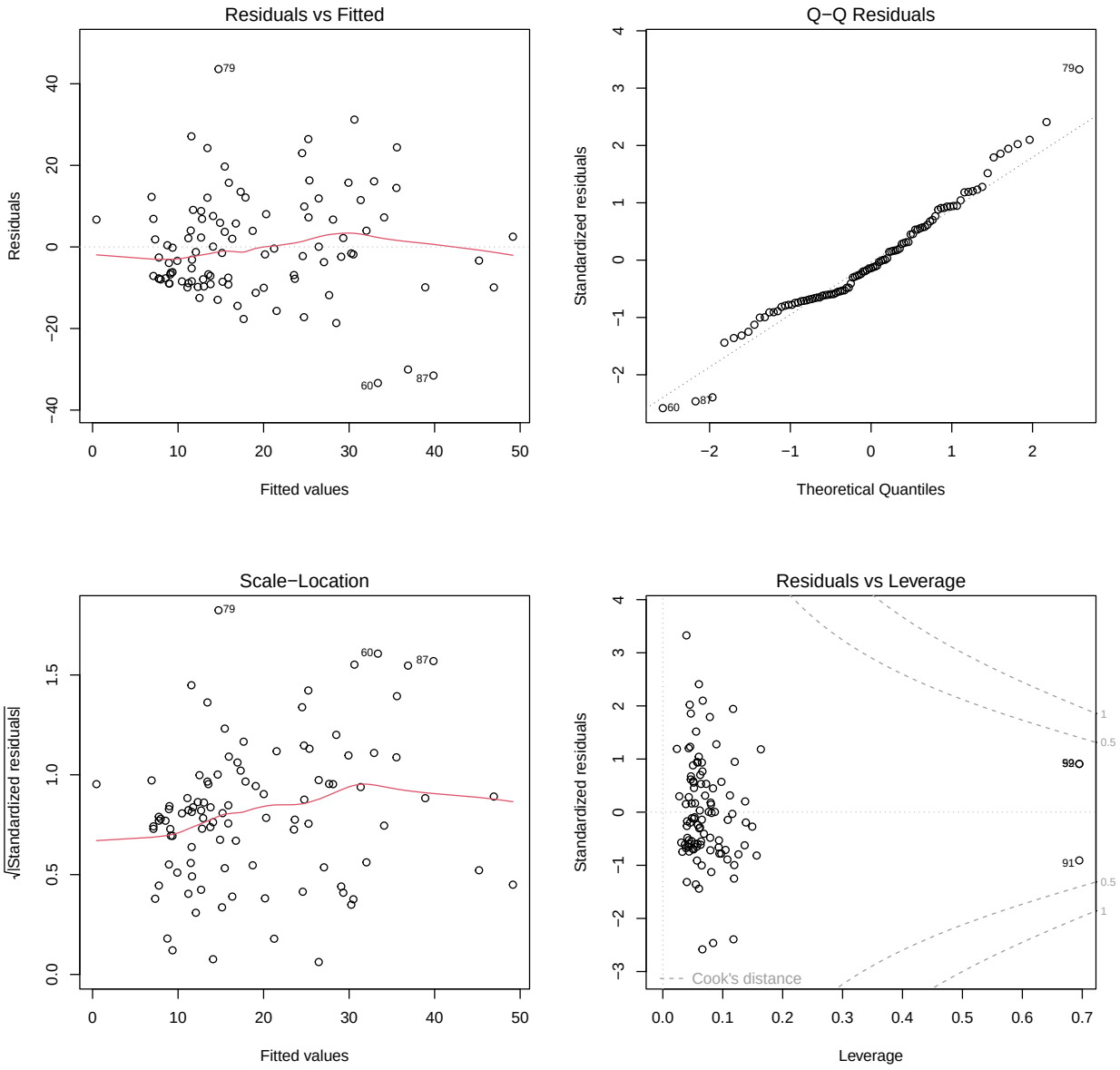
```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition +
##     AUDIT_risk + DUDIT_risk + SDQ_EmotionalProblems + SDQ_Hyperactivity +
##     SDQ_PeerProblems + SDQ_Prosocial, data = teach_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.365  -8.435  -1.833   7.237  43.608
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        9.139e+00  6.088e+00   1.501   0.1367
## ASB_mean_D1                        5.057e-01  7.502e-02   6.740 1.36e-09 ***
## Training_ConditionTRAIN           -5.356e-01  2.748e+00  -0.195   0.8459
## AUDIT_riskRisky or hazardous level  8.304e+00  1.105e+01   0.751   0.4544
## DUDIT_riskPossible drug problems   -1.966e+01  1.123e+01  -1.751   0.0833 .
## SDQ_EmotionalProblems               1.342e+00  7.311e-01   1.836   0.0696 .
## SDQ_Hyperactivity                  -1.049e-01  6.554e-01  -0.160   0.8732
## SDQ_PeerProblems                   -4.276e-01  9.629e-01  -0.444   0.6580
## SDQ_Prosocial                      -8.598e-04  6.908e-01  -0.001   0.9990
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.37 on 92 degrees of freedom
##   (1 observation deleted due to missingness)
## Multiple R-squared:  0.3812, Adjusted R-squared:  0.3273
## F-statistic: 7.083 on 8 and 92 DF,  p-value: 2.931e-07
```

```
confint(model_t_asb_3_with_cov_2)
```

```
##                                        2.5 %     97.5 %
## (Intercept)                        -2.9512968 21.2293219
## ASB_mean_D1                         0.3566784  0.6546746
## Training_ConditionTRAIN            -5.9927686  4.9215445
## AUDIT_riskRisky or hazardous level -13.6465998 30.2537282
## DUDIT_riskPossible drug problems   -41.9508696  2.6395897
## SDQ_EmotionalProblems               -0.1096949  2.7943871
## SDQ_Hyperactivity                   -1.4066475  1.1968257
## SDQ_PeerProblems                    -2.3400897  1.4848023
## SDQ_Prosocial                       -1.3727884  1.3710687
```

2) compare to original model

3) Get Diagnostics




```
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition  + SDQ_EmotionalProblems + SDQ_Hyperactivity
                    data =  d)
  return(coef(fit))
}

teach_asb_d1d2_wide %>% group_by(AUDIT_risk) %>% summarise(n=n())
```

```
## # A tibble: 3 x 2
```

```
##   AUDIT_risk                   n
##   <fct>                    <int>
## 1 Low risk                    99
## 2 Risky or hazardous level     2
## 3 <NA>                         1
```

```
teach_asb_d1d2_wide %>% group_by(DUDIT_risk) %>% summarise(n=n())
```

```
## # A tibble: 3 x 2
##   DUDIT_risk                 n
##   <fct>                  <int>
## 1 Low risk                  99
## 2 Possible drug problems     2
## 3 <NA>                       1
```

```
str(teach_asb_d1d2_wide$AUDIT_risk)
```

```
##  Factor w/ 2 levels "Low risk","Risky or hazardous level": 1 1 1 1 1 1 1 1 1 1 ...
```

```
# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = teach_asb_d1d2_wide,
                     statistic = boot_coef_c_asb,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original        bias    std. error
## t1* 10.2505432 -1.059798e-01  5.61751790
## t2*  0.5083181  1.775489e-05  0.08911836
## t3* -0.5840444 -3.814810e-02  2.77232367
## t4*  1.2455967  4.081905e-02  0.77772404
## t5* -0.2779169 -2.692661e-03  0.60308863
## t6* -0.2151659 -5.738300e-03  0.97586032
## t7* -0.1219459  3.844969e-02  0.62880279
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
```

```
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.58, 21.79 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3305,  0.6827 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-5.9759,  5.0536 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-0.364,  2.692 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-1.5563,  0.8749 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
```

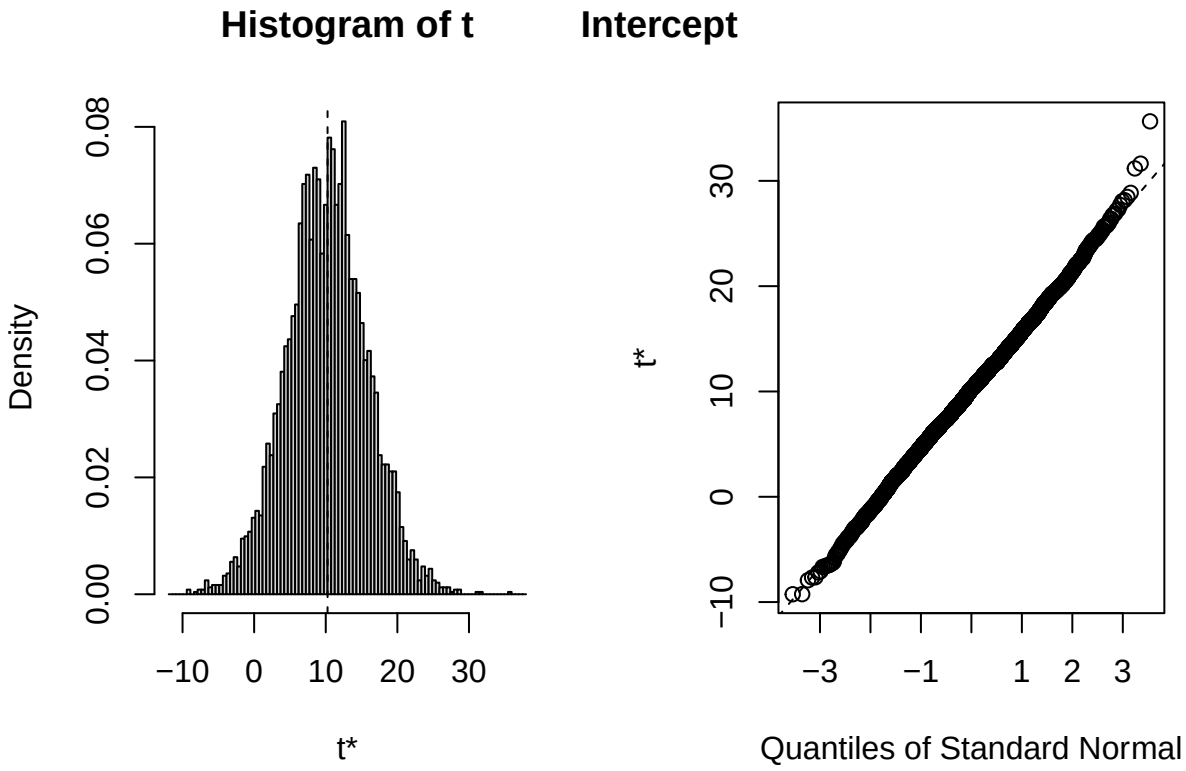
```
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 6)
##
## Intervals :
## Level       BCa
## 95%   (-2.0240,  1.7702 )
## Calculations and Intervals on Original Scale

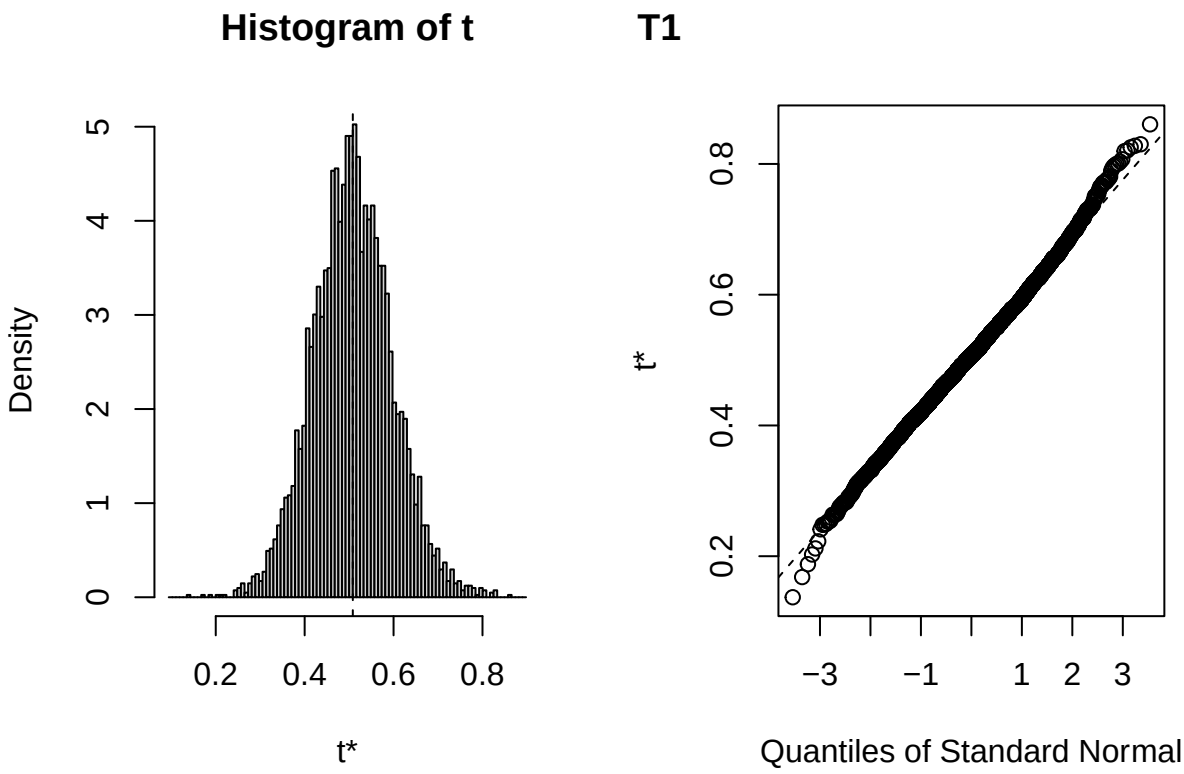
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 7)
##
## Intervals :
## Level       BCa
## 95%   (-1.3561,  1.1193 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 5,
##     level = 0.995)
##
## Intervals :
## Level       BCa
## 95%   (-1.5563,  0.8749 )
## Calculations and Intervals on Original Scale

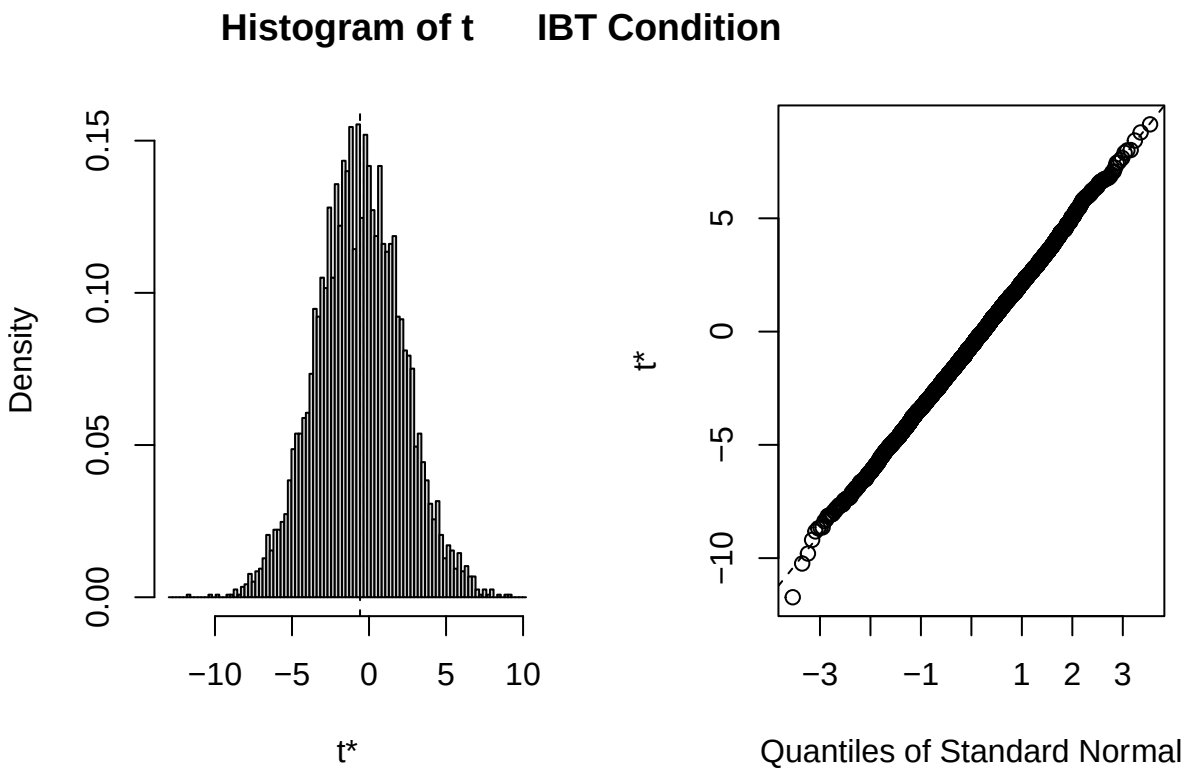
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 6,
##     level = 0.995)
##
## Intervals :
## Level       BCa
## 95%   (-2.0240,  1.7702 )
## Calculations and Intervals on Original Scale
```

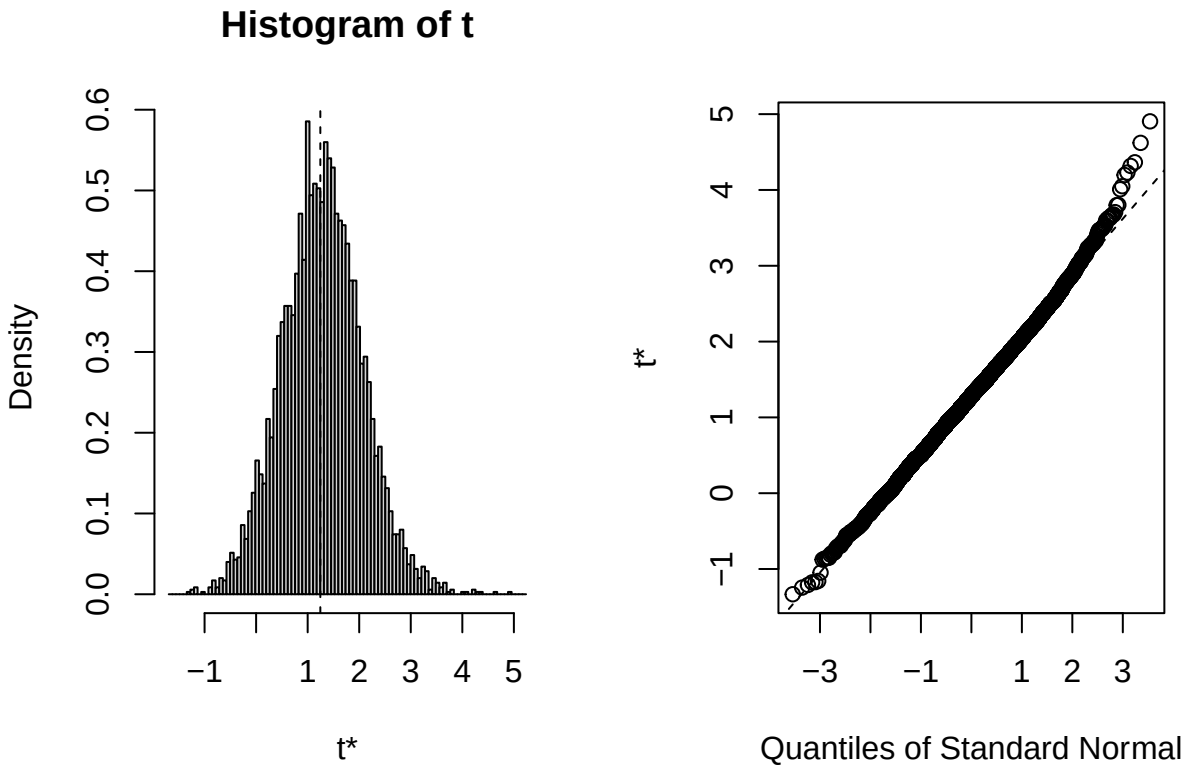
Bootstrap Diagnostics

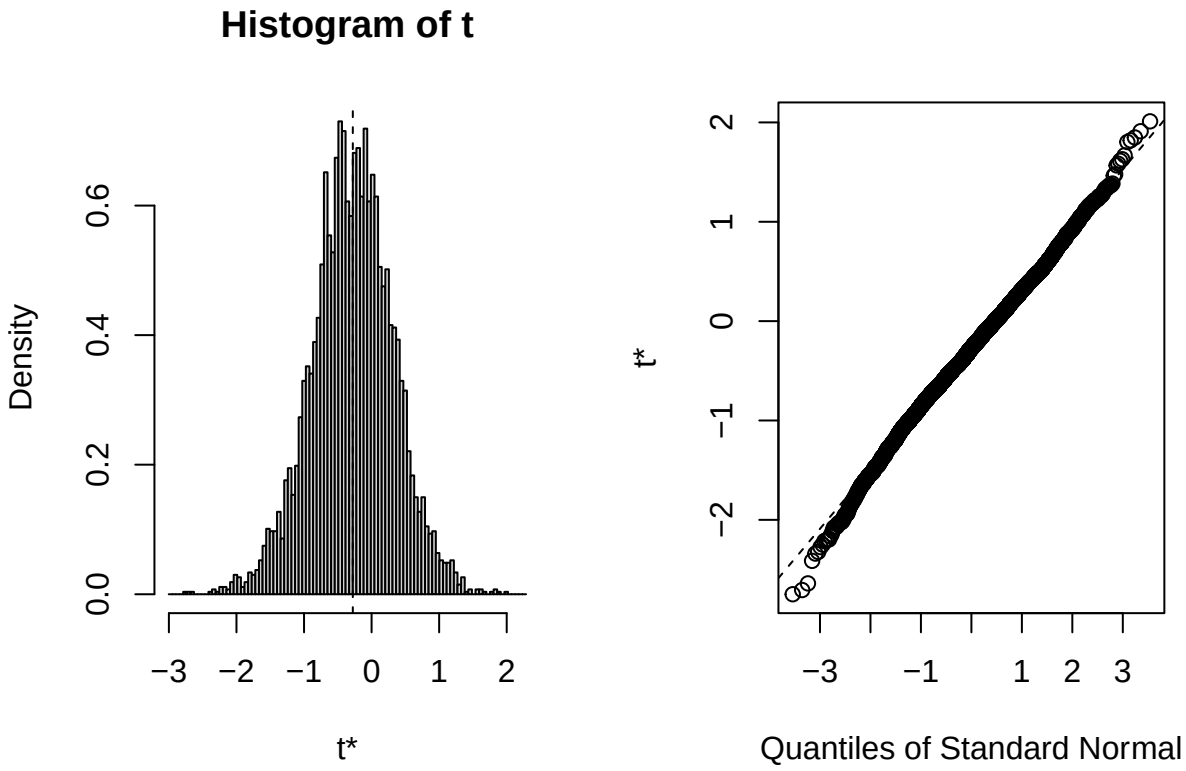

Histogram of t
Density
0.00
0.02
0.04
0.06
0.08
−10
0
10
20
30
t*
Intercept
t*
−10
0
10
20
30
−3
−1
1
2
3
Quantiles of Standard Normal

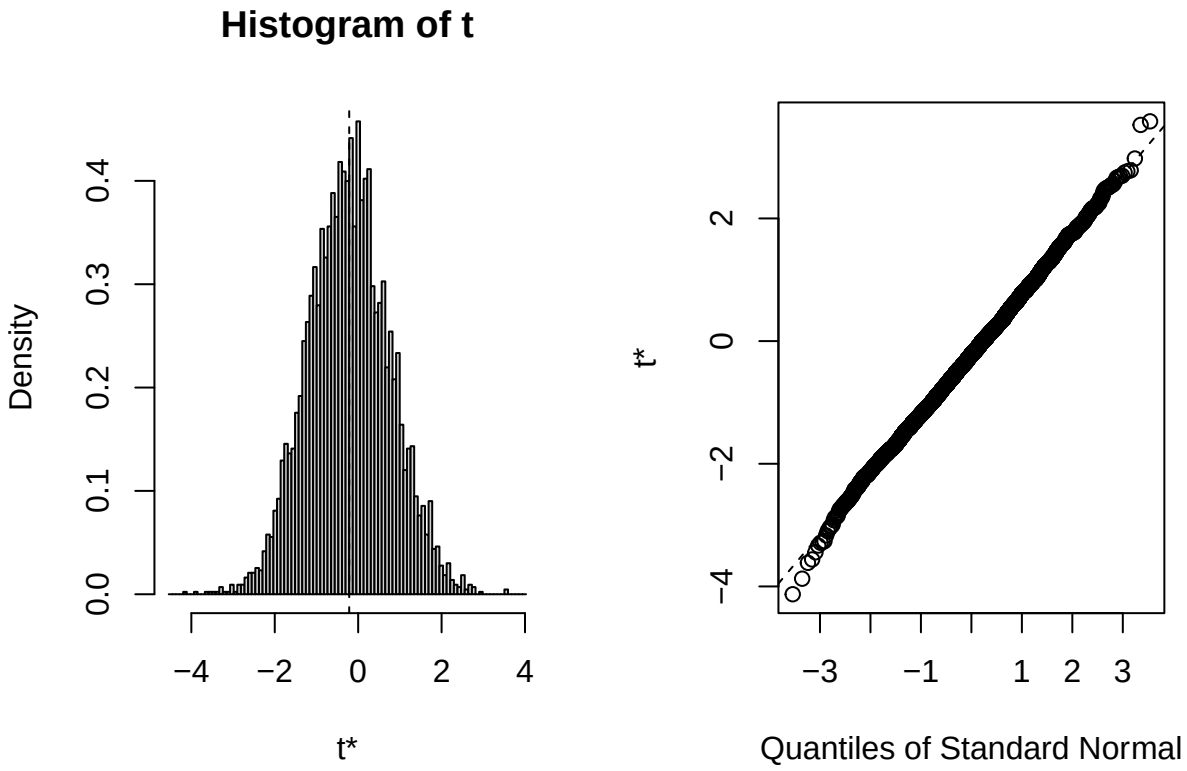

Histogram of t
Density
t*
0
1
2
3
4
5
0.2
0.4
0.6
0.8

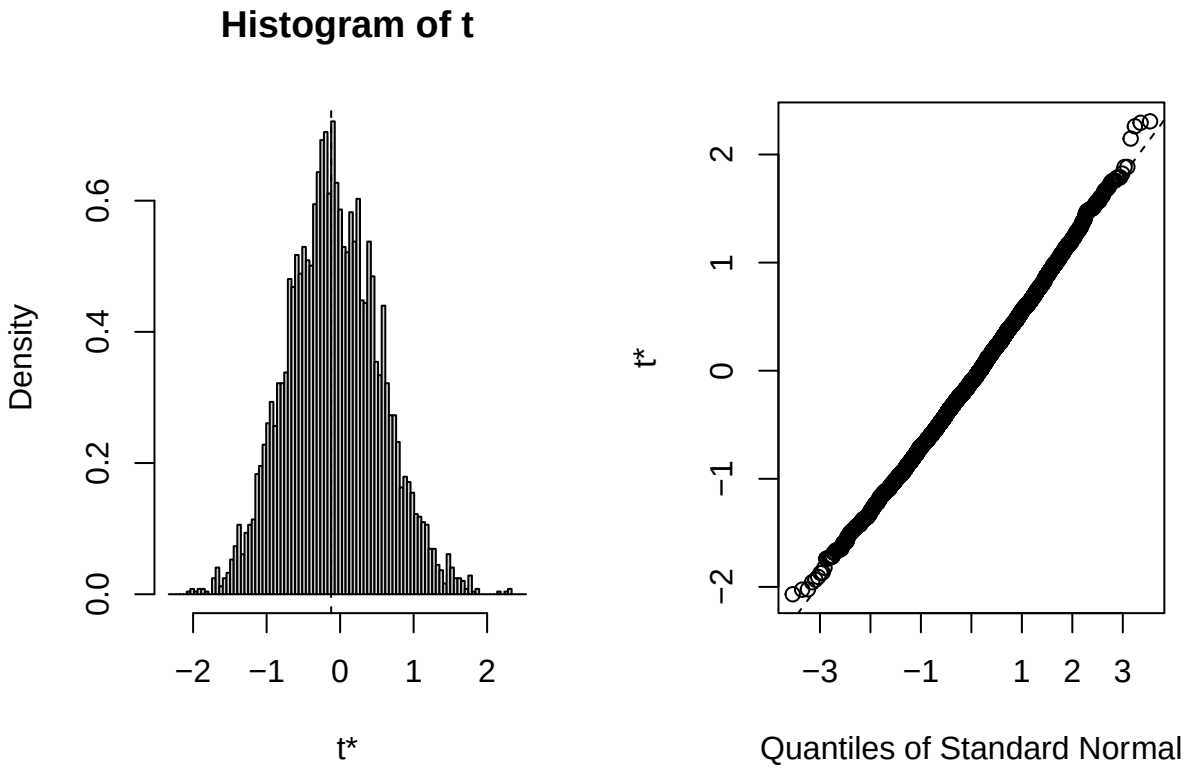
T1
t*
Quantiles of Standard Normal
0.2
0.4
0.6
0.8
−3
−1
1
2
3


Histogram of t IBT Condition
Density
0.00
0.05
0.10
0.15
−10
−5
0
5
10
t*
t*
−10
−5
0
5
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.1
0.2
0.3
0.4
0.5
0.6
−1
1
2
3
4
5
t*
t*
−1
0
1
2
3
4
5
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.2
0.4
0.6
−3
−2
−1
0
1
2
t*
t*
−2
−1
0
1
2
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.0
0.1
0.2
0.3
0.4
−4
−2
0
2
4
t*
t*
−4
−2
0
2
−3
−1
1
2
3
Quantiles of Standard Normal



5. Moderation Analyses (reviewer requested)

5.1) New model: moderation, adjusting for baseline balance point

5.1.1) Child model:

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition * bp,
##     data = child_asb_d1d2_wide_nep)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -51.106  -6.278  -2.707   5.658  43.190
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                -10.86168   15.10220  -0.719    0.474
## ASB_sum_D1                   0.68790    0.06871  10.012 1.37e-15 ***
## Training_ConditionTRAIN      9.14067   18.40737   0.497    0.621
## bp                           1.93871    1.88559   1.028    0.307
## Training_ConditionTRAIN:bp  -1.12850    2.28698  -0.493    0.623
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 14.66 on 77 degrees of freedom
## Multiple R-squared:  0.5825, Adjusted R-squared:  0.5608
## F-statistic: 26.86 on 4 and 77 DF,  p-value: 5.82e-14
```

```
##                                 2.5 %     97.5 %
## (Intercept)                -40.9340020 19.2106498
## ASB_sum_D1                   0.5510834  0.8247092
## Training_ConditionTRAIN    -27.5130884 45.7944210
## bp                          -1.8159796  5.6933928
## Training_ConditionTRAIN:bp  -5.6824595  3.4254560
```

4) Bootstrap:

```
child_asb_d1d2_wide_nep$bp_standardised <- standardise(child_asb_d1d2_wide_nep$bp)
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition*bp,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide_nep,
                     statistic = boot_coef_c_asb,
                     R = 5000)

boot_coef_c_asb_std <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2_standardised ~ ASB_sum_D1_standardised + Training_Condition_standardised*bp_stan
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb_std <- boot(data = child_asb_d1d2_wide_nep,
                     statistic = boot_coef_c_asb_std,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide_nep, statistic = boot_coef_c_asb,
##     R = 5000)
##
```

```
##
## Bootstrap Statistics :
##        original       bias    std. error
## t1* -10.8616761  0.752097703  14.9024689
## t2*   0.6878963  0.007476218   0.1041122
## t3*   9.1406663 -0.805294877  16.5701408
## t4*   1.9387066 -0.114320202   1.9058983
## t5*  -1.1285018  0.119382719   2.1403266

##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide_nep, statistic = boot_coef_c_asb_std,
##     R = 5000)
##
##
## Bootstrap Statistics :
##         original        bias    std. error
## t1* -0.002360284  0.0005751771  0.08560575
## t2*  0.749184219  0.0081423099  0.11338804
## t3*  0.008204342  0.0064421628  0.14945600
## t4*  0.135194819 -0.0079720670  0.13290695
## t5* -0.078695554  0.0083250993  0.14925469
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.1669,  0.1675 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.5007,  0.9531 )
## Calculations and Intervals on Original Scale

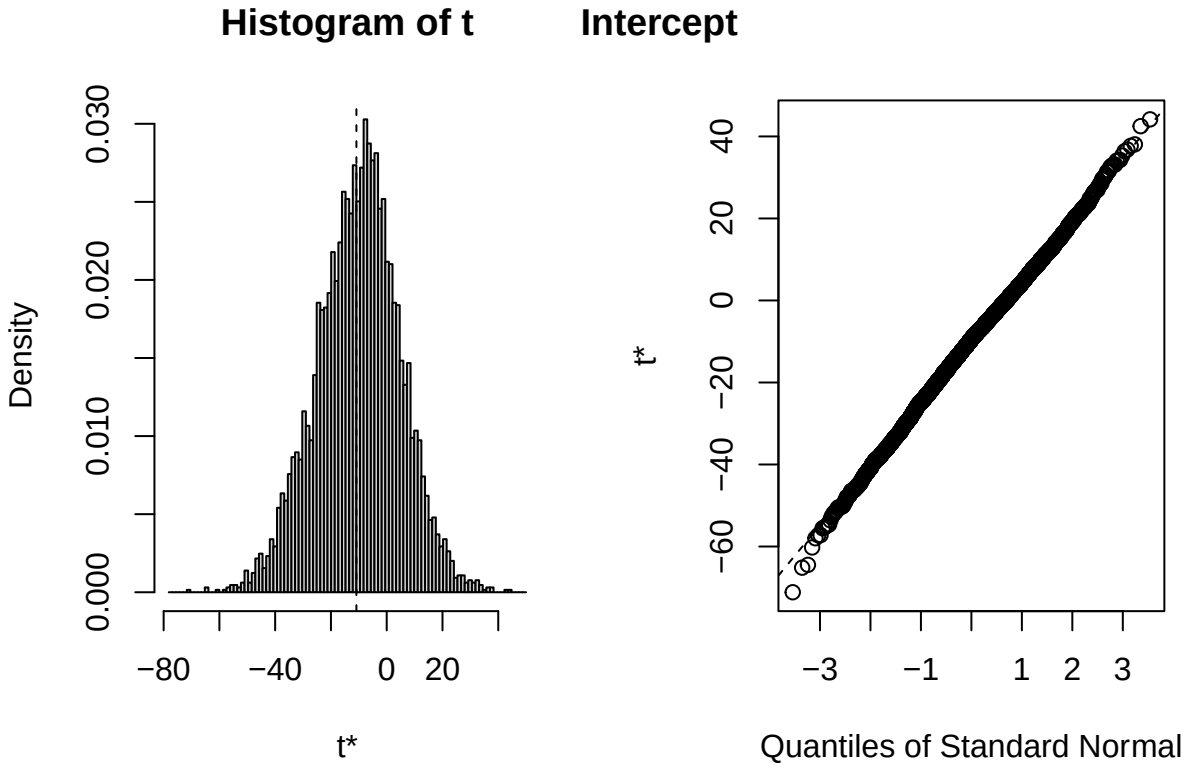
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
```

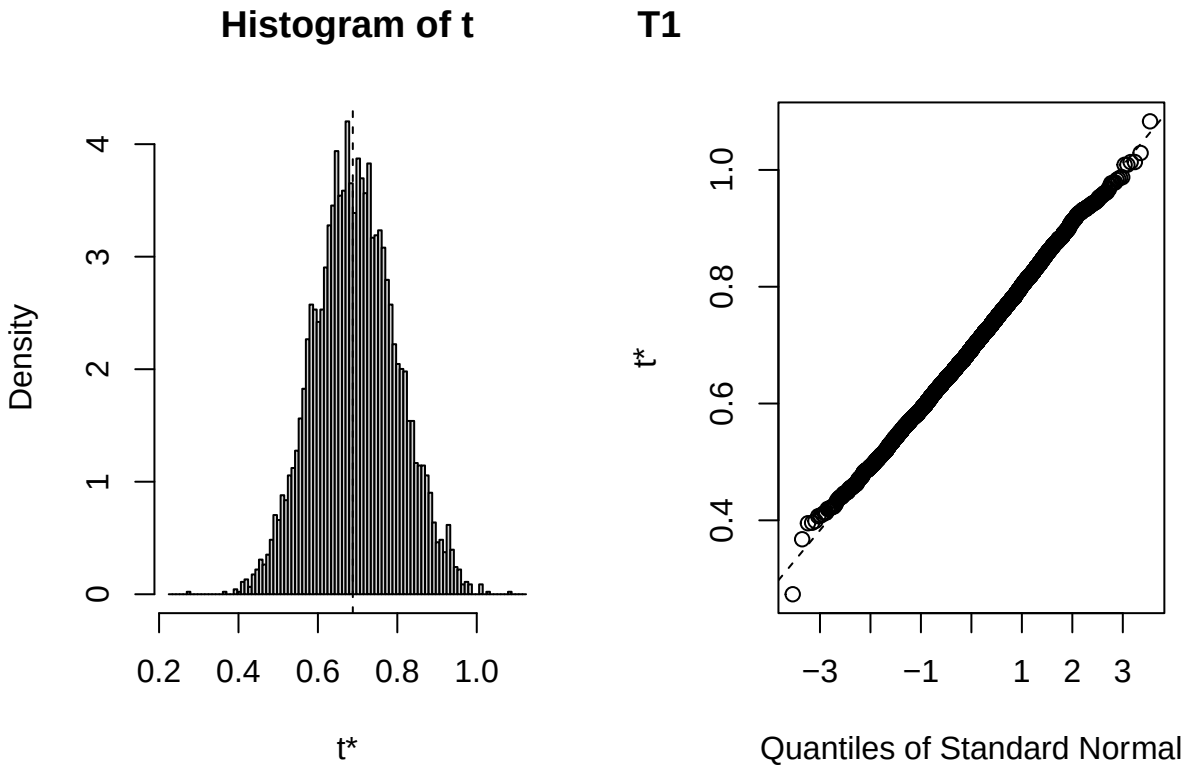
```
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-0.2992,  0.2864 )
## Calculations and Intervals on Original Scale

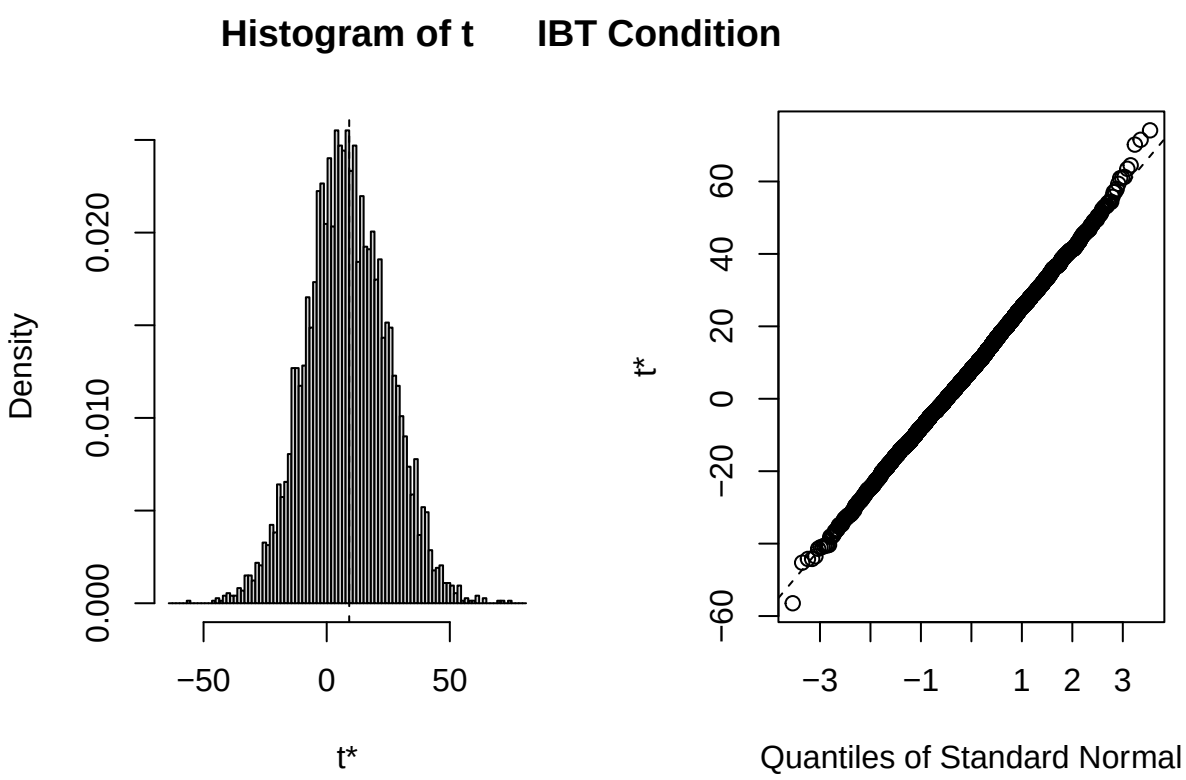
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-0.0939,  0.4309 )
## Calculations and Intervals on Original Scale

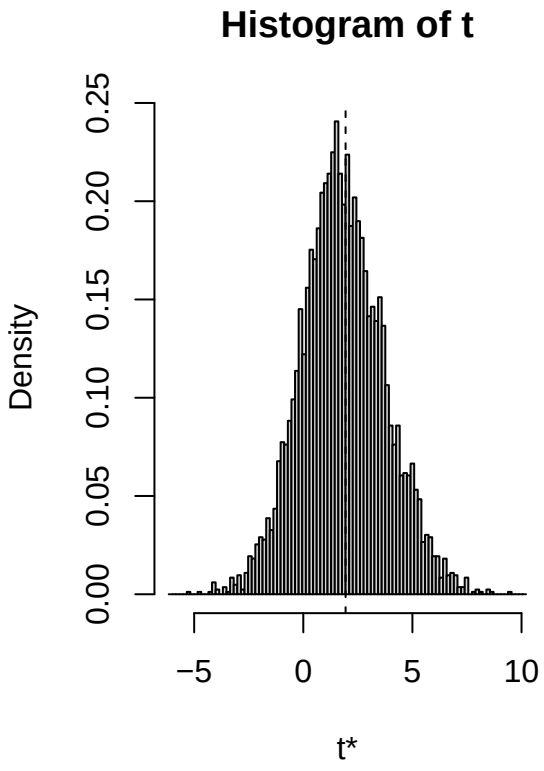
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-0.3865,  0.1939 )
## Calculations and Intervals on Original Scale
```

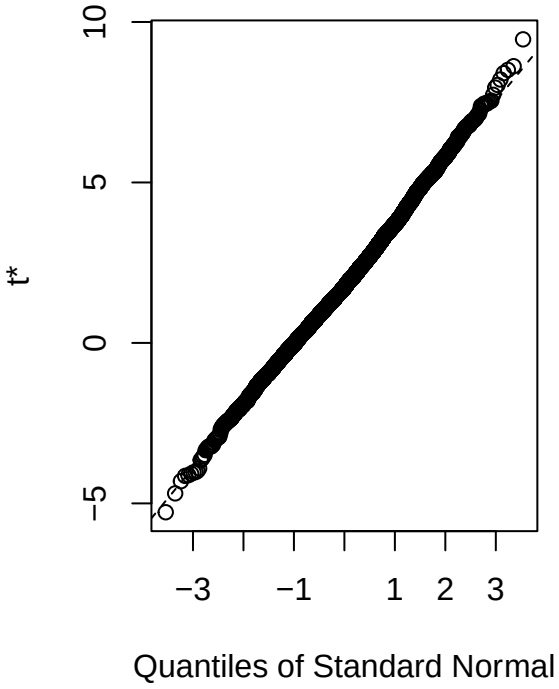
Bootstrap Diagnostics


Histogram of t
Density
0.000
0.010
0.020
0.030
−80
−40
0
20
t*
Intercept
t*
−60
−40
−20
0
20
40
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
t*
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
T1
t*
Quantiles of Standard Normal
0.4
0.6
0.8
1.0
−3
−1
1
2
3


Histogram of t IBT Condition
Density
0.000
0.010
0.020
−50
0
50
t*
t*
−60
−20
0
20
40
60
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.05
0.10
0.15
0.20
0.25
−5
0
5
10
t*


t*
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal

5.1.2) Teacher

1) Define and summarise model

```
model_t_asb_3_with_bp <- lm(ASB_mean_D2~ ASB_mean_D1 + Training_Condition*bp,
                            data =  teach_asb_d1d2_wide_nep)


summary(model_t_asb_3_with_bp)
```

```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition *
##     bp, data = teach_asb_d1d2_wide_nep)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.397  -8.708  -1.231   6.897  41.060
##
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)                26.41865   12.37397   2.135   0.0353 *
## ASB_mean_D1                 0.52992    0.07311   7.249 1.02e-10 ***
## Training_ConditionTRAIN    -9.77005   15.22801  -0.642   0.5227
## bp                         -2.11297    1.56412  -1.351   0.1799
## Training_ConditionTRAIN:bp  1.05981    1.88695   0.562   0.5756
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.34 on 97 degrees of freedom
## Multiple R-squared:  0.3535, Adjusted R-squared:  0.3269
## F-statistic: 13.26 on 4 and 97 DF,  p-value: 1.174e-08
```

```
confint(model_t_asb_3_with_bp)
```

```
##                                2.5 %     97.5 %
## (Intercept)                1.8597589 50.9775501
## ASB_mean_D1                0.3848214  0.6750111
## Training_ConditionTRAIN  -39.9934429 20.4533409
## bp                        -5.2173231  0.9913783
## Training_ConditionTRAIN:bp -2.6852571  4.8048806
```

4) Bootstrap:

```
teach_asb_d1d2_wide_nep$bp_standardised <- standardise(teach_asb_d1d2_wide_nep$bp)

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition*bp,
                    data =  d)
```

```r
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = teach_asb_d1d2_wide_nep,
                     statistic = boot_coef_c_asb,
                     R = 5000)

boot_coef_c_asb_std <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2_standardised ~ ASB_mean_D1_standardised + Training_Condition_standardised*bp_sta
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb_std <- boot(data = teach_asb_d1d2_wide_nep,
                     statistic = boot_coef_c_asb_std,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide_nep, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 26.4186545  0.445880812 16.91888042
## t2*  0.5299163  0.001729726  0.08631804
## t3* -9.7700510 -0.408623679 18.60507527
## t4* -2.1129724 -0.068779055  2.10888518
## t5*  1.0598117  0.052919847  2.32494354

##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide_nep, statistic = boot_coef_c_asb_std,
##     R = 5000)
##
##
## Bootstrap Statistics :
##        original        bias    std. error
## t1*  0.03817133 -0.0042291664   0.1261329
```

```
## t2*  0.61505964  0.0020076464   0.1001870
## t3* -0.08265483  0.0007469331   0.1682916
## t4* -0.19477772 -0.0063401809   0.1944009
## t5*  0.09769541  0.0048782497   0.2143176
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.1891,  0.3017 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4070,  0.8029 )
## Calculations and Intervals on Original Scale

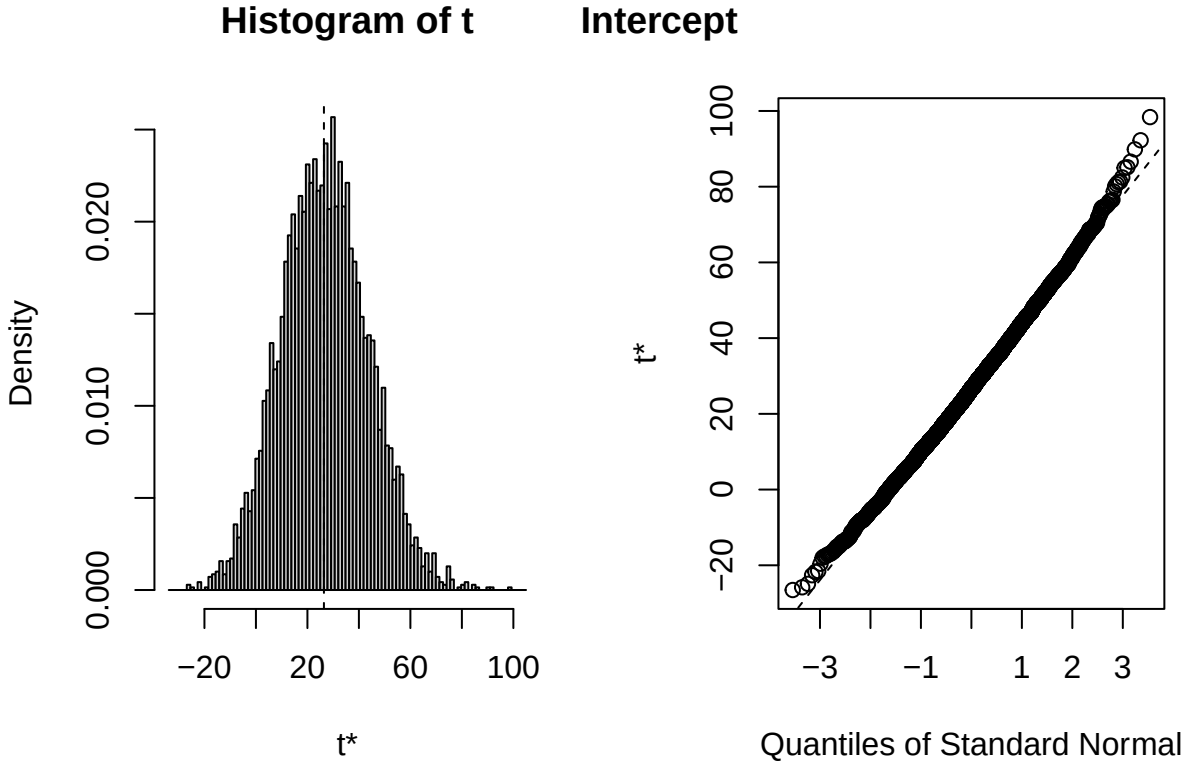
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-0.4138,  0.2474 )
## Calculations and Intervals on Original Scale

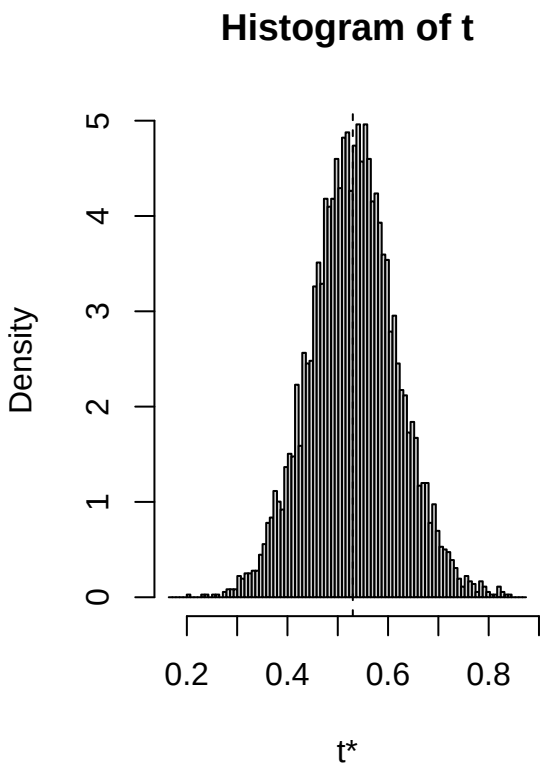
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-0.5806,  0.1699 )
## Calculations and Intervals on Original Scale

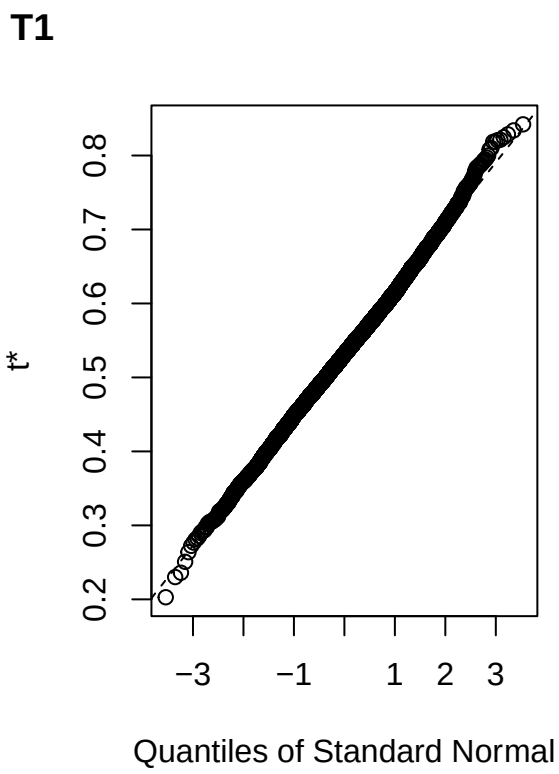
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
```

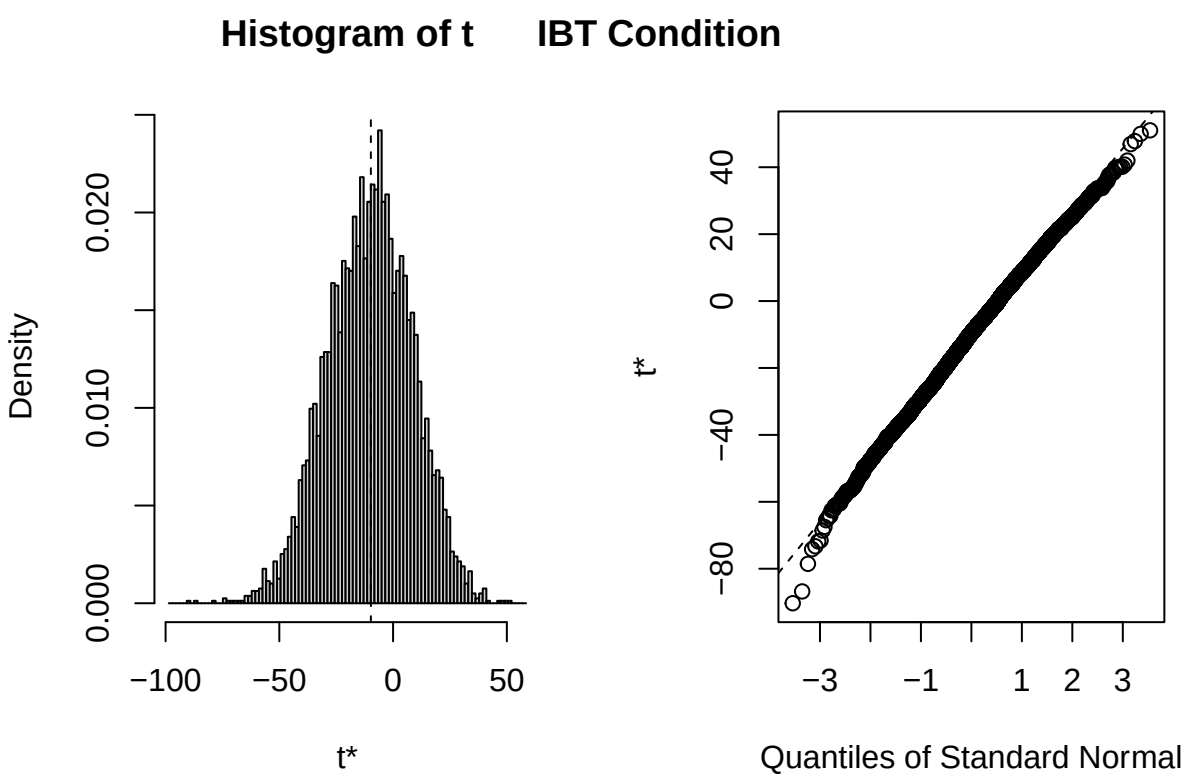
```
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-0.2917,  0.5326 )
## Calculations and Intervals on Original Scale
```

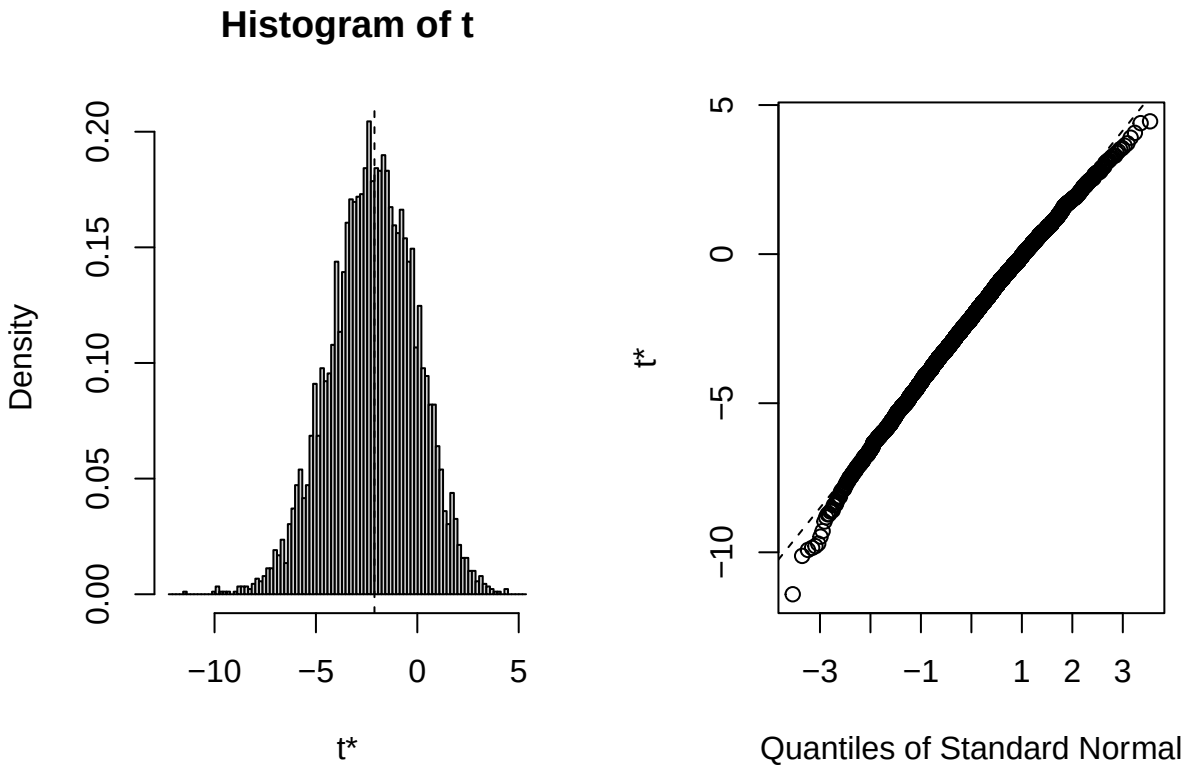
Bootstrap Diagnostics




Histogram of t
Density
t*
0
1
2
3
4
5
0.2
0.4
0.6
0.8


T1
t*
Quantiles of Standard Normal
0.2
0.3
0.4
0.5
0.6
0.7
0.8
−3
−1
1
2
3


Histogram of t IBT Condition
Density
0.000
0.010
0.020
−100
−50
0
50
t*
t*
40
20
0
−40
−80
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.05
0.10
0.15
0.20
−10
−5
0
5
t*
t*
5
0
−5
−10
−3
−1
1
2
3
Quantiles of Standard Normal

5.2) Adjusting for extent of feedback

5.2.1 Child

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition * shifted_trials,
##     data = child_asb_d1d2_wide_nep)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.236  -5.661  -3.215   6.609  42.896
##
## Coefficients: (1 not defined because of singularities)
##                                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)                             4.24354    2.68728   1.579    0.118
## ASB_sum_D1                              0.69610    0.06778  10.270 3.84e-16 ***
## Training_ConditionTRAIN                 2.12242    8.44533   0.251    0.802
## shifted_trials                         -0.04974    0.21648  -0.230    0.819
## Training_ConditionTRAIN:shifted_trials       NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.69 on 78 degrees of freedom
## Multiple R-squared:  0.5749, Adjusted R-squared:  0.5586
## F-statistic: 35.17 on 3 and 78 DF,  p-value: 1.758e-14

## Model :
## ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition * shifted_trials
##
## Complete :
##                                        (Intercept) ASB_sum_D1
## Training_ConditionTRAIN:shifted_trials 0           0
##                                        Training_ConditionTRAIN shifted_trials
## Training_ConditionTRAIN:shifted_trials 0                       1

## $CONTROL
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       0       0       0       0       0       0
##
## $TRAIN
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      16      28      40      36      48      48

##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + pct_genuine_training,
##     data = subset(child_asb_d1d2_wide_nep, Training_Condition ==
##         "TRAIN"))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -40.624  -8.240  -1.971   6.771  41.118
```

```
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)           22.8959     8.2423   2.778  0.00874 **
## ASB_sum_D1             0.6282     0.1100   5.711 1.86e-06 ***
## pct_genuine_training  -3.2679     1.4947  -2.186  0.03557 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.98 on 35 degrees of freedom
## Multiple R-squared:  0.5015, Adjusted R-squared:  0.473
## F-statistic:  17.6 on 2 and 35 DF,  p-value: 5.124e-06
```

```
##                           2.5 %     97.5 %
## (Intercept)           6.1630785 39.6286486
## ASB_sum_D1            0.4048573  0.8514467
## pct_genuine_training -6.3024492 -0.2334464
```

```
child_asb_d1d2_wide_nep$pct_genuine_training_standardised <- standardise(child_asb_d1d2_wide_nep$pct_ge

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + pct_genuine_training,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = subset(child_asb_d1d2_wide_nep, Training_Condition == "TRAIN"),
                     statistic = boot_coef_c_asb,
                     R = 5000)
```

Standardised model:

```
child_asb_d1d2_wide_nep$pct_genuine_training_standardised <- standardise(child_asb_d1d2_wide_nep$pct_ge

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb_standardised <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2_standardised ~ ASB_sum_D1_standardised + pct_genuine_training_standardised,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb_standardised <- boot(data = subset(child_asb_d1d2_wide_nep, Training_Condition == "T
```

```
                    statistic = boot_coef_c_asb_standardised,
                    R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = subset(child_asb_d1d2_wide_nep, Training_Condition ==
##     "TRAIN"), statistic = boot_coef_c_asb, R = 5000)
##
##
## Bootstrap Statistics :
##      original      bias    std. error
## t1* 22.895864 -0.74566788   9.4735050
## t2*  0.628152  0.01823699   0.1495482
## t3* -3.267948  0.10505297   1.5921708

##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = subset(child_asb_d1d2_wide_nep, Training_Condition ==
##     "TRAIN"), statistic = boot_coef_c_asb_standardised, R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1*  0.4155150 -0.003854714   0.2223041
## t2*  0.6841170  0.019861806   0.1628722
## t3* -0.4196946  0.013491697   0.2044786
```

Confidence Intervals

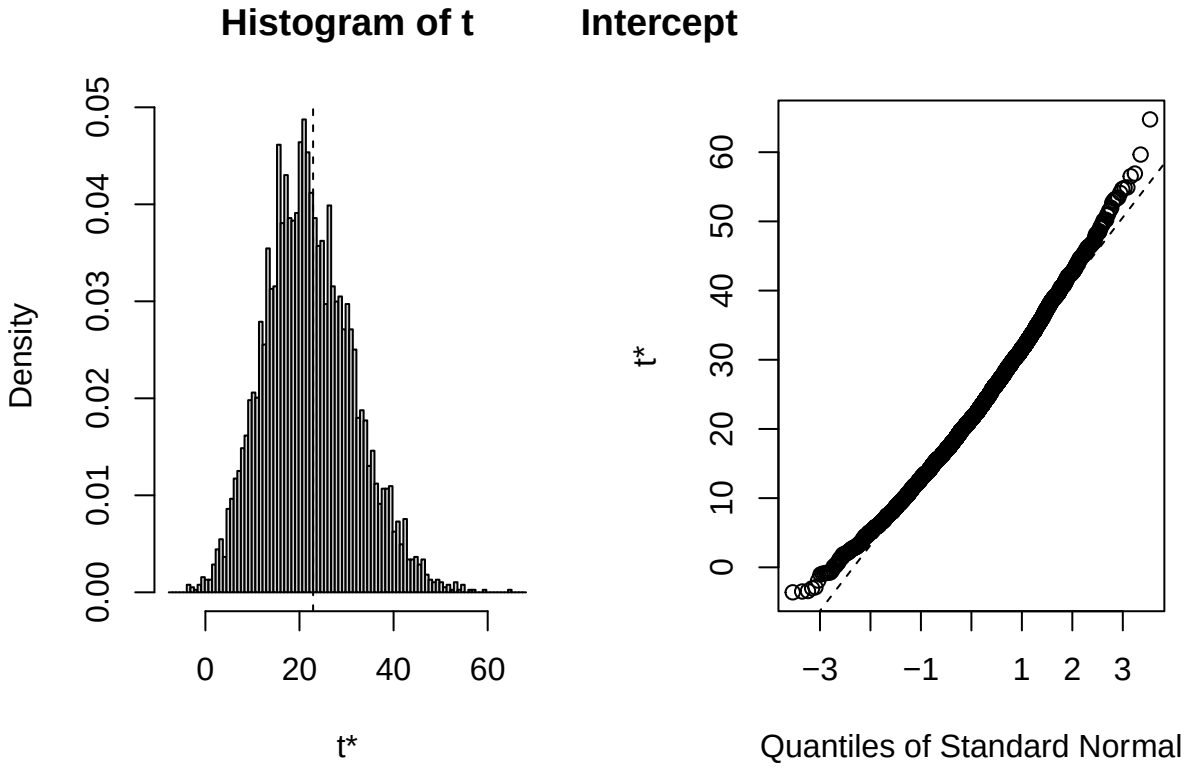
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_standardised, type = "bca",
##     index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 0.0705,  0.9793 )
## Calculations and Intervals on Original Scale

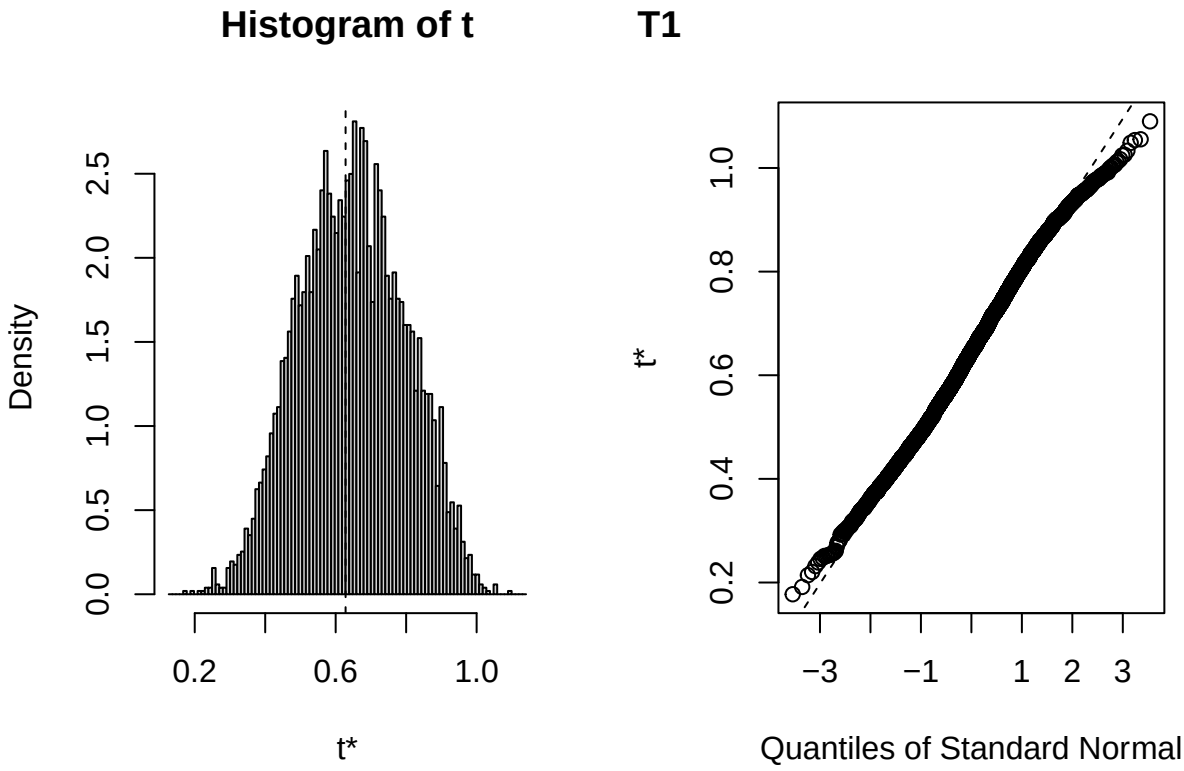
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
```

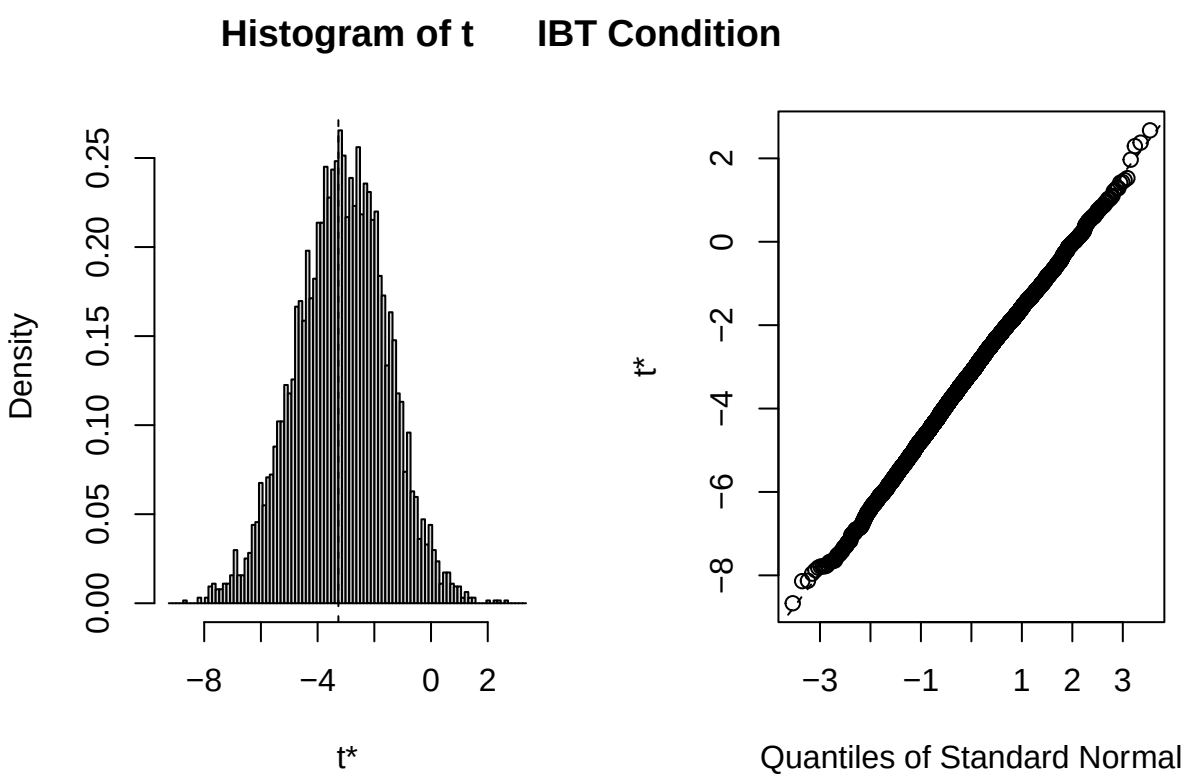
```
## CALL :
## boot.ci(boot.out = boot_results_c_asb_standardised, type = "bca",
##     index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3034,  0.9520 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_standardised, type = "bca",
##     index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-0.8838, -0.0703 )
## Calculations and Intervals on Original Scale
```

Bootstrap Diagnostics




Histogram of t
Density
0.0
0.5
1.0
1.5
2.0
2.5
0.2
0.6
1.0
t*
T1
t*
0.2
0.4
0.6
0.8
1.0
−3
−1
1
2
3
Quantiles of Standard Normal



5.2.2 Teacher

```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + pct_genuine_training,
##     data = subset(teach_asb_d1d2_wide_nep, Training_Condition ==
##         "TRAIN"))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -32.104  -8.378  -2.252   6.628  29.176
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)           16.16916    5.97405   2.707  0.00951 **
## ASB_mean_D1            0.46991    0.08943   5.255 3.72e-06 ***
## pct_genuine_training -1.25678    1.01356  -1.240  0.22128
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.34 on 46 degrees of freedom
## Multiple R-squared:  0.3885, Adjusted R-squared:  0.3619
## F-statistic: 14.61 on 2 and 46 DF,  p-value: 1.223e-05

##                            2.5 %     97.5 %
```

```
## (Intercept)           4.1440306 28.1942912
## ASB_mean_D1           0.2898994  0.6499229
## pct_genuine_training -3.2969757  0.7834116
```

```
####
teach_asb_d1d2_wide_nep$pct_genuine_training_standardised <- standardise(teach_asb_d1d2_wide_nep$pct_ge

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_t_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2 ~ ASB_mean_D1 + pct_genuine_training,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_t_asb_shift <- boot(data = teach_asb_d1d2_wide_nep,
                     statistic = boot_coef_t_asb,
                     R = 5000)


# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_t_asb_std <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2_standardised ~ ASB_mean_D1_standardised + pct_genuine_training_standardised,
                    data =  d)
  return(coef(fit))
}

str(teach_asb_d1d2_wide_nep$Training_Condition)
```

```
##  Factor w/ 2 levels "CONTROL","TRAIN": 1 1 1 1 2 1 2 1 1 2 ...
```

```
# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_t_asb_shift_std <- boot(data = subset(teach_asb_d1d2_wide_nep, Training_Condition_standard
                     statistic = boot_coef_t_asb_std,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide_nep, statistic = boot_coef_t_asb,
##     R = 5000)
##
##
```

```
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 10.5049313 -0.036801873  1.95823207
## t2*  0.5074506  0.003793286  0.08391956
## t3* -0.4287369  0.003151260  0.46246245

##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = subset(teach_asb_d1d2_wide_nep, Training_Condition_standardised ==
##     "TRAIN"), statistic = boot_coef_t_asb_std, R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1*  0.1761256 -0.007924275   0.1888071
## t2*  0.5454133  0.003322767   0.1116889
## t3* -0.2254803  0.010098234   0.1660960
```

Confidence Intervals

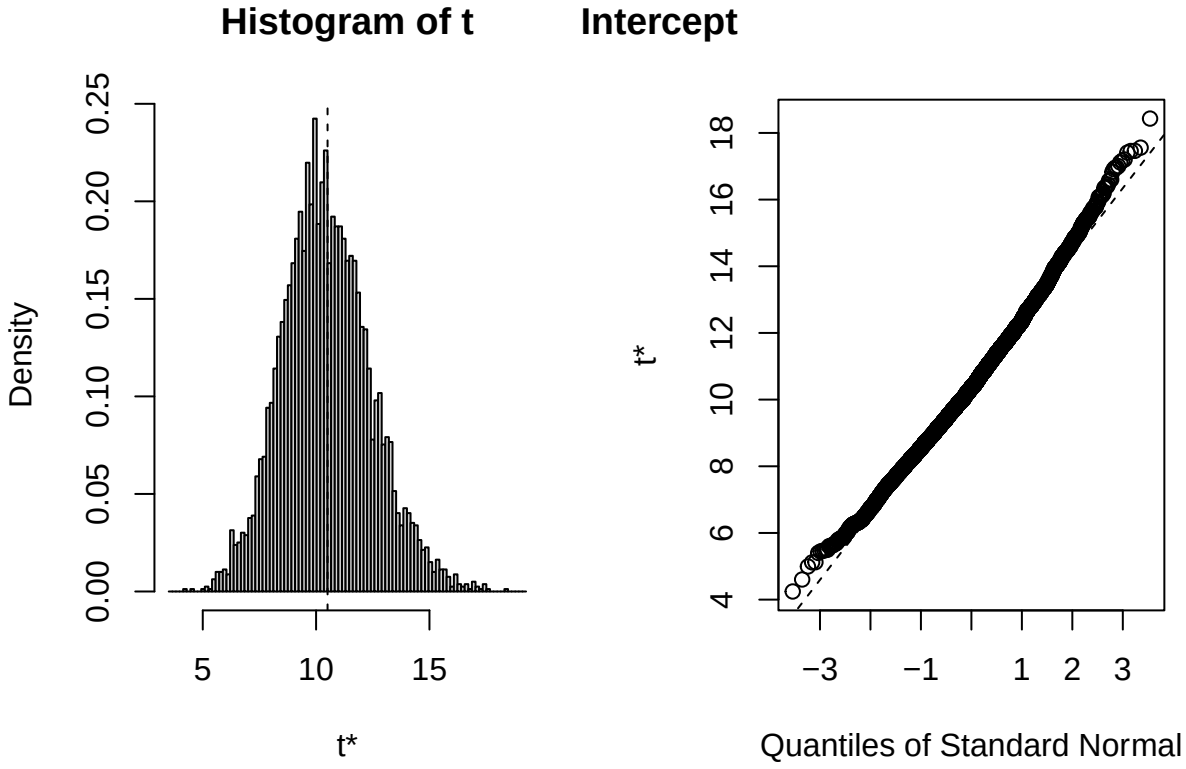
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_shift_std, type = "bca",
##     index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.1807,  0.5663 )
## Calculations and Intervals on Original Scale

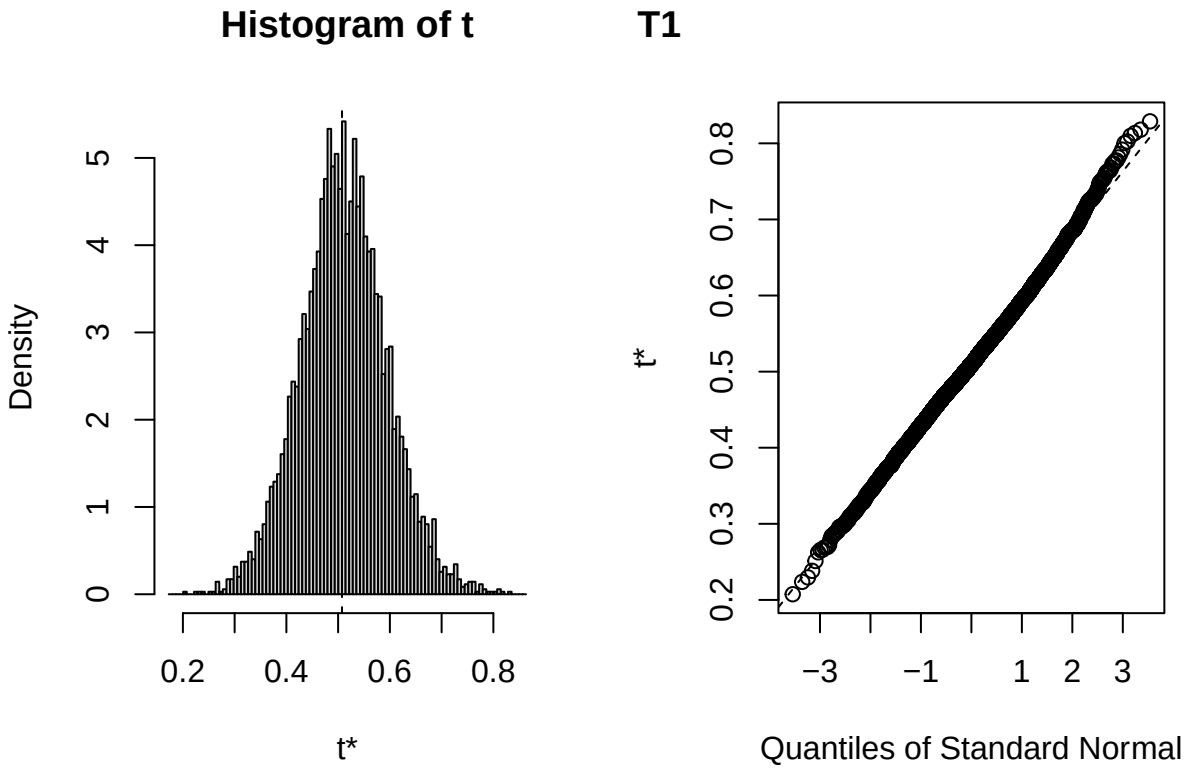
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_shift_std, type = "bca",
##     index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.2884,  0.7422 )
## Calculations and Intervals on Original Scale

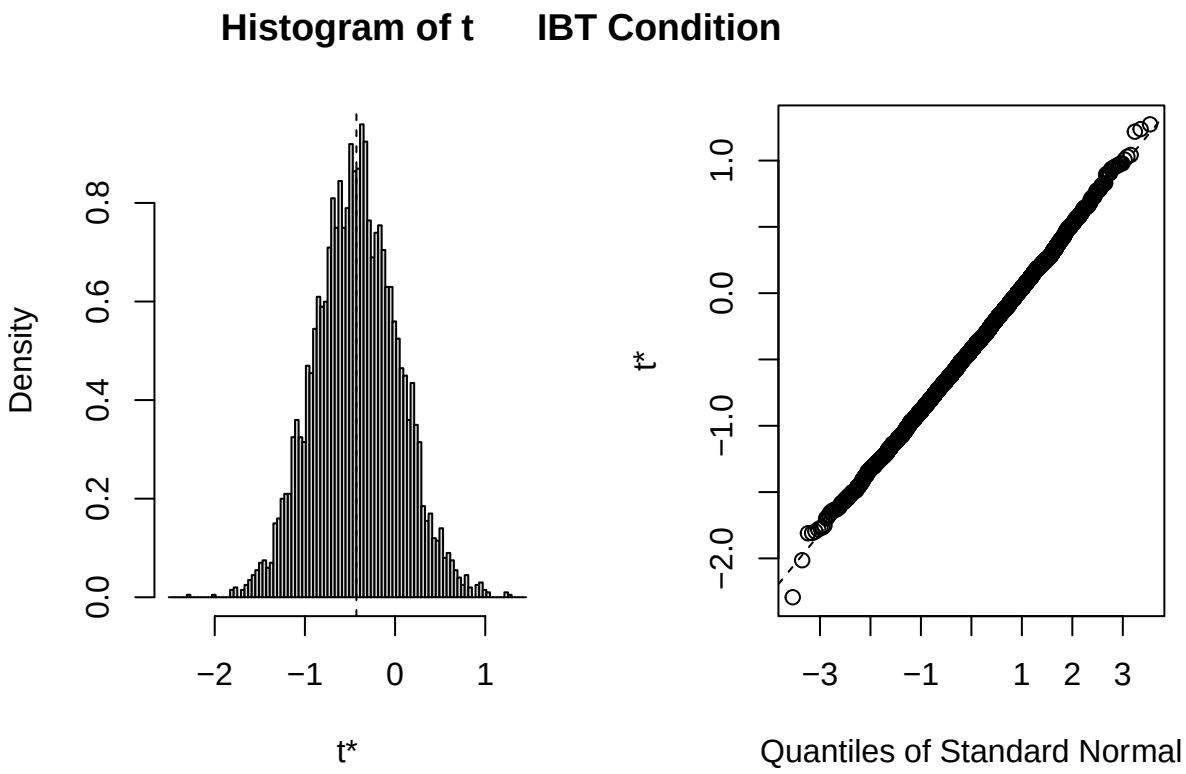
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb_shift_std, type = "bca",
##     index = 3)
```

```
##
## Intervals :
## Level       BCa
## 95%   (-0.5490,  0.1122 )
## Calculations and Intervals on Original Scale
```

Bootstrap Diagnostics




Histogram of t
Density
t*
T1
t*
Quantiles of Standard Normal



5.3) Models with Number of Training Sessions Completed

5.3.1) Child

1) Define and summarise model

```
model_c_asb_2_with_Ntrain <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition*trainSessions,
                                data =  child_asb_d1d2_wide)


summary(model_c_asb_2_with_Ntrain)
```

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition * trainSessions,
##     data = child_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -54.977  -6.723  -1.334   6.065  40.195
##
## Coefficients:
##                                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         -10.28534   12.32363  -0.835    0.407
```

```
## ASB_sum_D1                                0.70951    0.06652  10.665   <2e-16 ***
## Training_ConditionTRAIN                -10.82504   17.13826  -0.632    0.529
## trainSessions                            2.60847    2.23287   1.168    0.246
## Training_ConditionTRAIN:trainSessions    2.16660    3.12647   0.693    0.490
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.22 on 77 degrees of freedom
## Multiple R-squared:  0.6069, Adjusted R-squared:  0.5865
## F-statistic: 29.72 on 4 and 77 DF,  p-value: 5.954e-15
```

```
confint(model_c_asb_2_with_Ntrain)
```

```
##                                             2.5 %     97.5 %
## (Intercept)                            -34.8248280 14.2541405
## ASB_sum_D1                               0.5770426  0.8419756
## Training_ConditionTRAIN                -44.9516800 23.3016023
## trainSessions                           -1.8377482  7.0546785
## Training_ConditionTRAIN:trainSessions   -4.0589883  8.3921937
```

4) Bootstrap:

```
child_asb_d1d2_wide$trainSessions_std <- standardise(child_asb_d1d2_wide$trainSessions)

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition*trainSessions,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide,
                      statistic = boot_coef_c_asb,
                      R = 5000)

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb_std <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2_standardised ~ ASB_sum_D1_standardised + Training_Condition_standardised*trainSes
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb_std <- boot(data = child_asb_d1d2_wide,
                      statistic = boot_coef_c_asb_std,
                      R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##        original      bias    std. error
## t1* -10.2853437 -0.36228970    7.150411
## t2*   0.7095091  0.01266701    0.109673
## t3* -10.8250388  1.19289709   16.619449
## t4*   2.6084652  0.01231979    1.396190
## t5*   2.1666027 -0.17450575    3.178344
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb_std,
##     R = 5000)
##
##
## Bootstrap Statistics :
##        original        bias    std. error
## t1* -0.01450222 -0.0004916986  0.08202038
## t2*  0.77272263  0.0137955700  0.11944432
## t3*  0.03858759  0.0114056020  0.14209412
## t4*  0.12117462  0.0005723083  0.06485914
## t5*  0.10064818 -0.0081065554  0.14764800
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.1709,  0.1472 )
## Calculations and Intervals on Original Scale
```

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
```

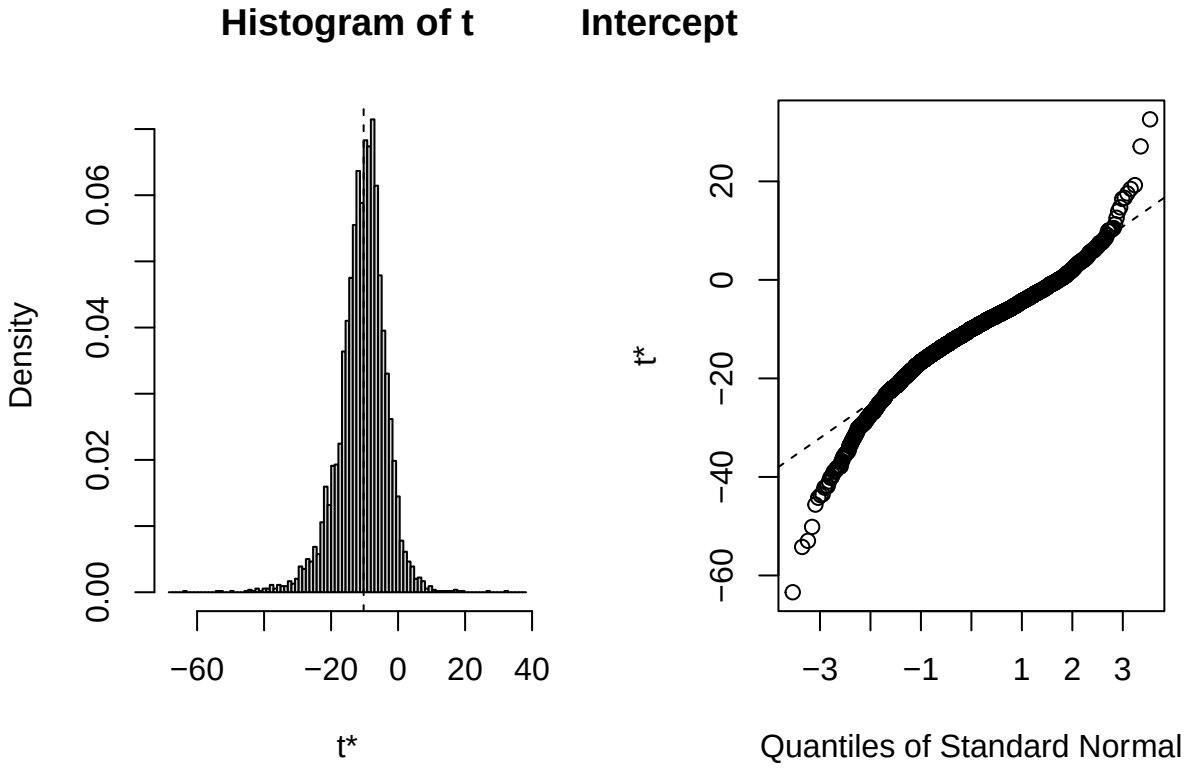
```
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4970,  0.9737 )
## Calculations and Intervals on Original Scale

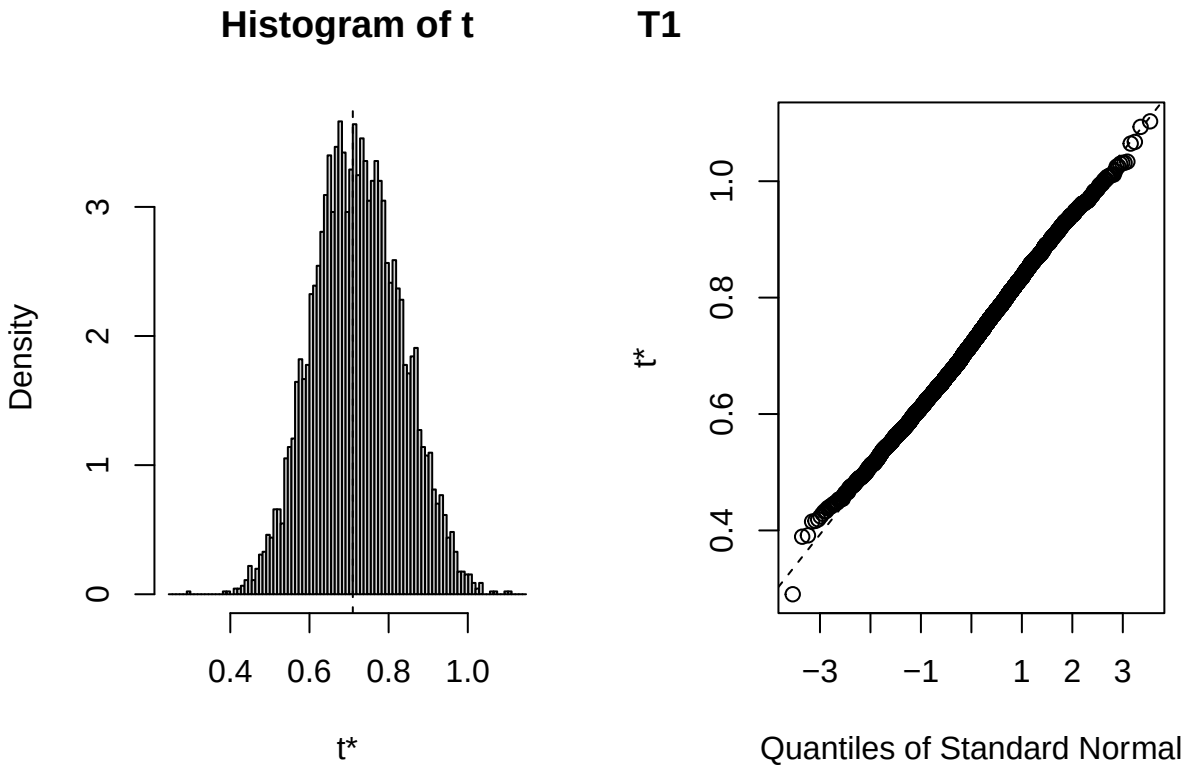
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-0.2569,  0.3007 )
## Calculations and Intervals on Original Scale

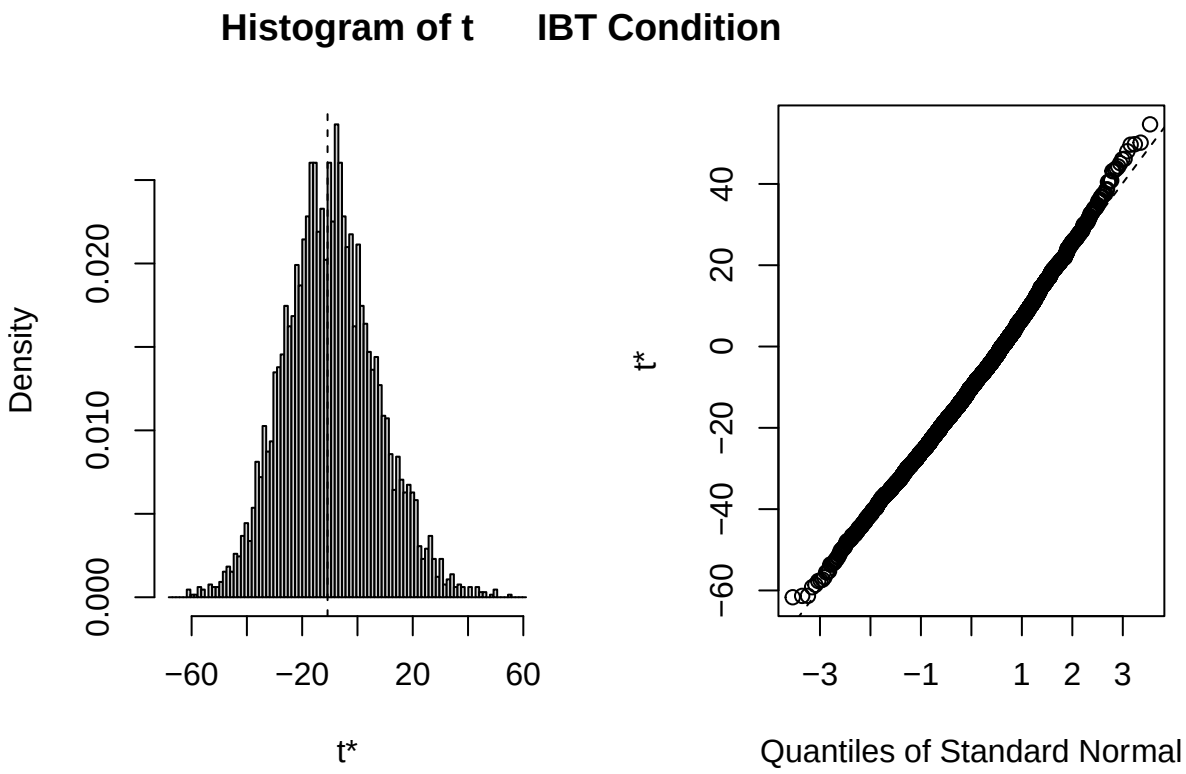
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   ( 0.0132,  0.2773 )
## Calculations and Intervals on Original Scale

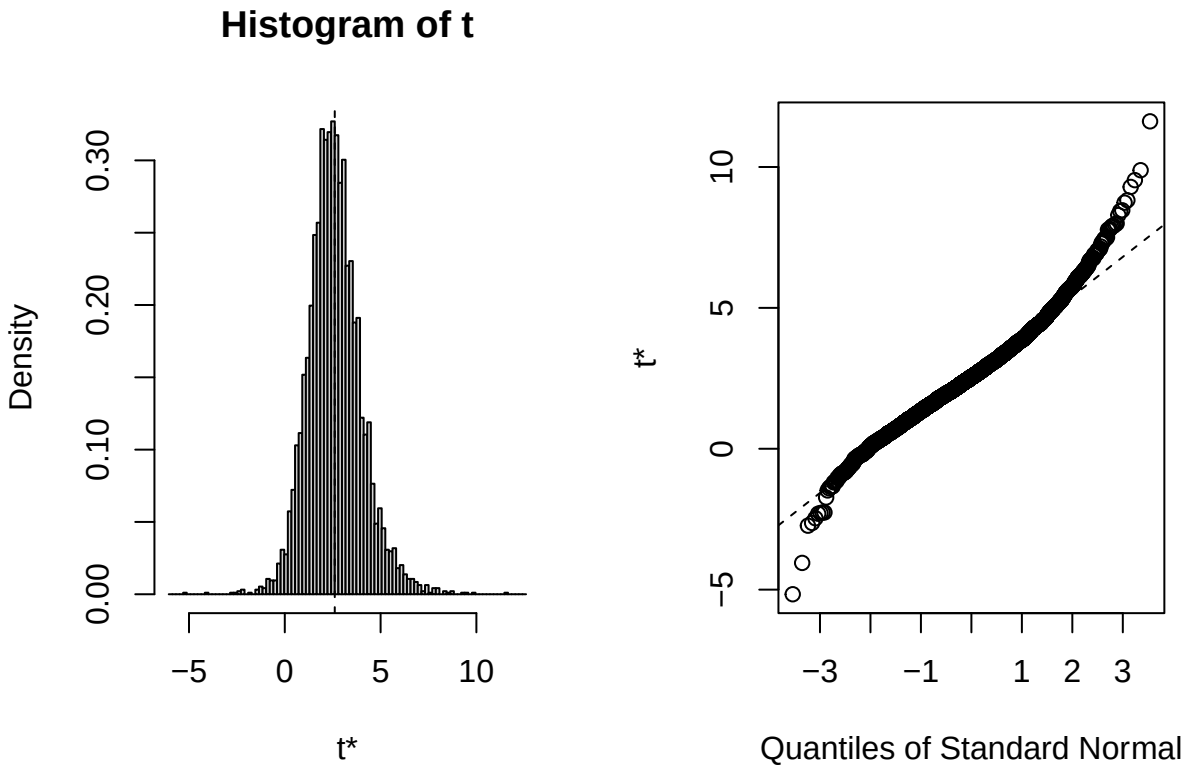
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-0.2025,  0.3775 )
## Calculations and Intervals on Original Scale
```

Bootstrap Diagnostics


Histogram of t
Density
0.00
0.02
0.04
0.06
−60
−20
0
20
40
t*
Intercept
t*
20
0
−20
−40
−60
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0
1
2
3
0.4
0.6
0.8
1.0
t*
T1
t*
0.4
0.6
0.8
1.0
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t IBT Condition
Density
0.000
0.010
0.020
−60
−20
20
60
t*
t*
40
20
0
−20
−40
−60
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.10
0.20
0.30
−5
0
5
10
t*
t*
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal

5.2) Teacher

1) Define and summarise model

```
model_t_asb_3_with_Ntrain <- lm(ASB_mean_D2~ ASB_mean_D1 + Training_Condition*trainSessions,
                                data =  teach_asb_d1d2_wide)


summary(model_t_asb_3_with_Ntrain)
```

```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition *
##     trainSessions, data = teach_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.257  -8.753  -1.811   7.906  45.841
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         17.32611   10.75020   1.612   0.1103
## ASB_mean_D1                          0.52905    0.07243   7.304 7.84e-11 ***
## Training_ConditionTRAIN            -25.47083   14.60730  -1.744   0.0844 .
## trainSessions                       -1.39344    1.93571  -0.720   0.4733
## Training_ConditionTRAIN:trainSessions  4.54231    2.67953   1.695   0.0932 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.29 on 97 degrees of freedom
## Multiple R-squared:  0.3583, Adjusted R-squared:  0.3318
## F-statistic: 13.54 on 4 and 97 DF,  p-value: 8.316e-09
```

```
confint(model_t_asb_3_with_Ntrain)
```

```
##                                         2.5 %     97.5 %
## (Intercept)                         -4.0100647 38.6622928
## ASB_mean_D1                          0.3852861  0.6728055
## Training_ConditionTRAIN            -54.4622847  3.5206155
## trainSessions                       -5.2352827  2.4484041
## Training_ConditionTRAIN:trainSessions  -0.7758120  9.8604303
```

4) Bootstrap:

```
teach_asb_d1d2_wide_nep$trainSessions_standardised <- standardise(teach_asb_d1d2_wide_nep$trainSessions)

# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition*trainSessions,
                    data =  d)
```

```
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = teach_asb_d1d2_wide,
                     statistic = boot_coef_c_asb,
                     R = 5000)



# Define a function that returns coefficients:
# taken from(https://www.datacamp.com/doc/r/bootstrapping)
boot_coef_c_asb_std <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_mean_D2_standardised ~ ASB_mean_D1_standardised + Training_Condition_standardised*train
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
set.seed(123)  # For reproducibility
boot_results_c_asb_std <- boot(data = teach_asb_d1d2_wide_nep,
                     statistic = boot_coef_c_asb_std,
                     R = 5000)
```

Results

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##        original       bias    std. error
## t1*  17.3261141 -0.350717420 12.22754237
## t2*   0.5290458  0.006798376  0.08526978
## t3* -25.4708346  0.647708323 16.10144283
## t4*  -1.3934393  0.051035553  2.20335800
## t5*   4.5423092 -0.127789324  2.92346314

##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide_nep, statistic = boot_coef_c_asb_std,
##     R = 5000)
##
```

```
##
## Bootstrap Statistics :
##        original       bias    std. error
## t1*  0.05415234  0.002963820  0.13022105
## t2*  0.61404932  0.007890693  0.09897035
## t3* -0.07660585 -0.002081568  0.17124614
## t4* -0.08643107  0.003165590  0.13666802
## t5*  0.28174650 -0.007926408  0.18133409
```

Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-0.1974,  0.3112 )
## Calculations and Intervals on Original Scale

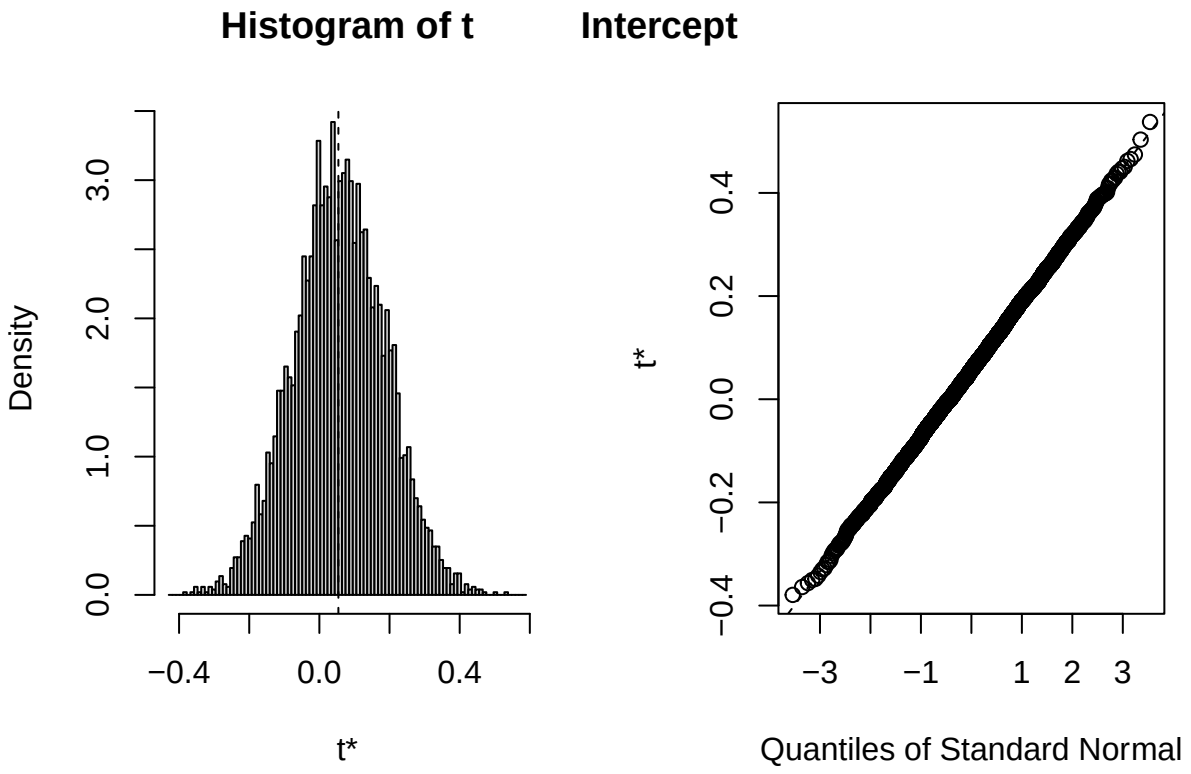
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4077,  0.8015 )
## Calculations and Intervals on Original Scale

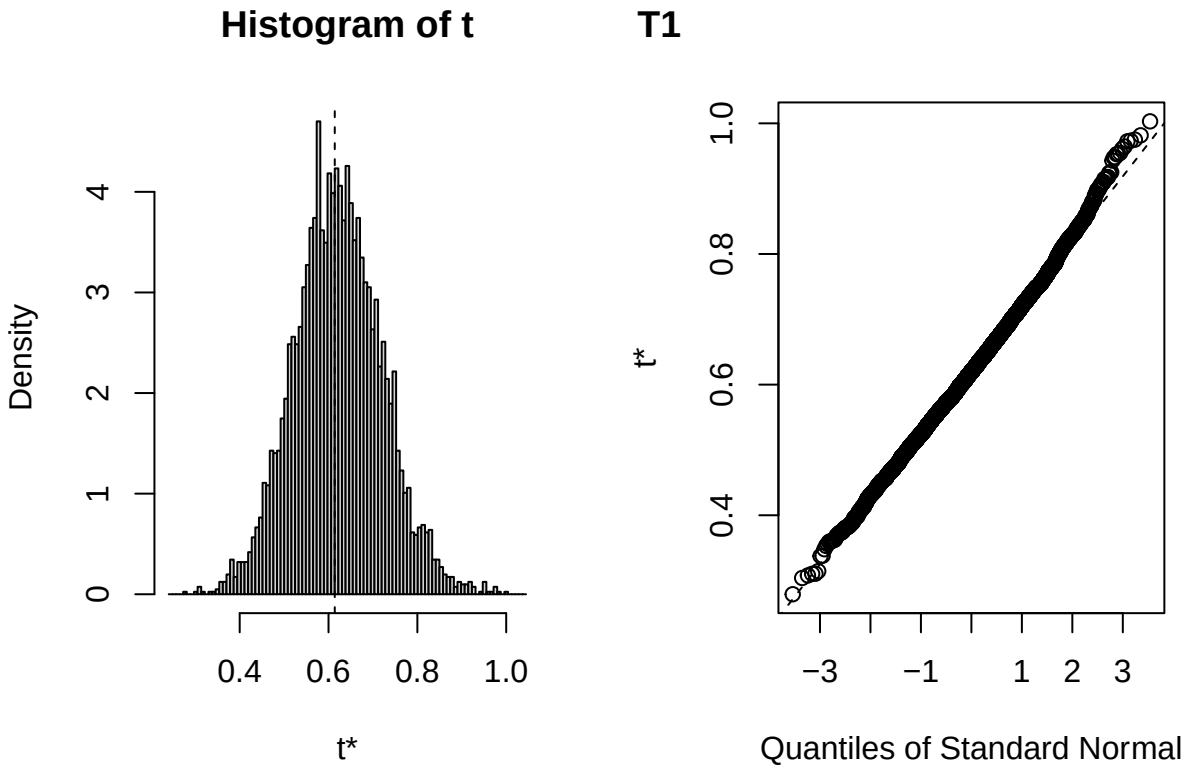
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-0.4132,  0.2703 )
## Calculations and Intervals on Original Scale

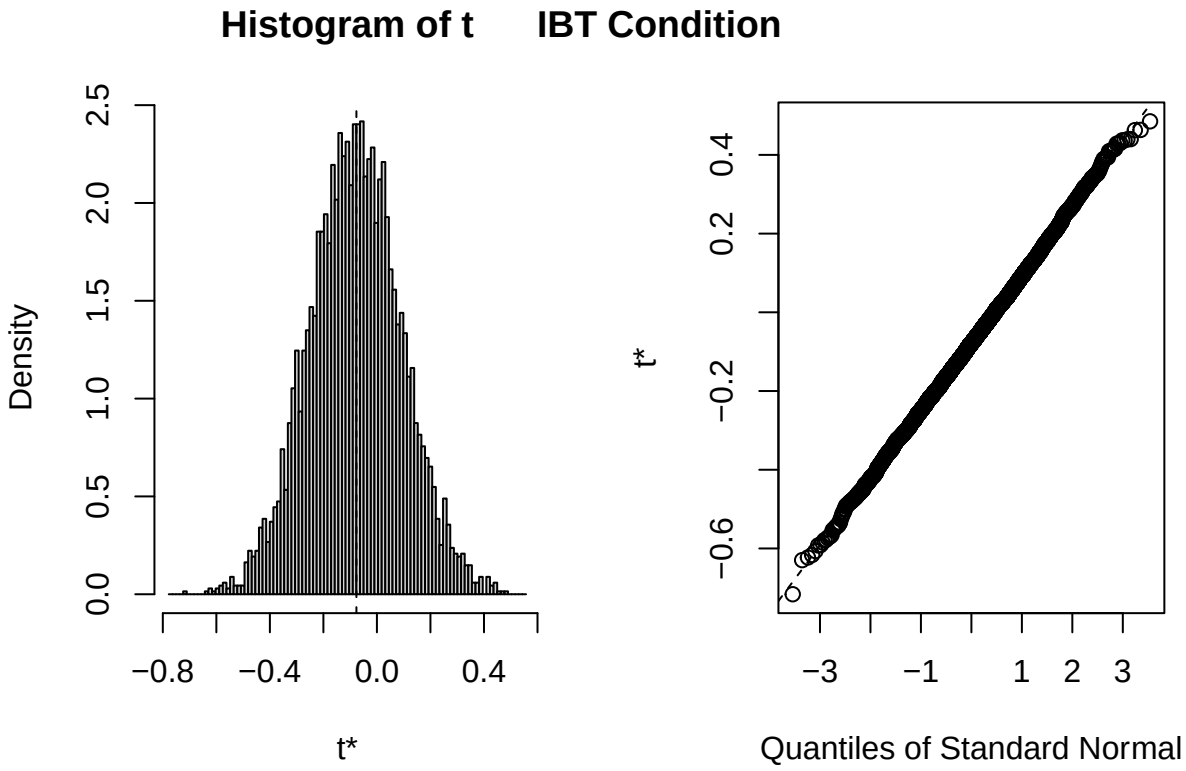
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-0.3665,  0.1843 )
## Calculations and Intervals on Original Scale
```

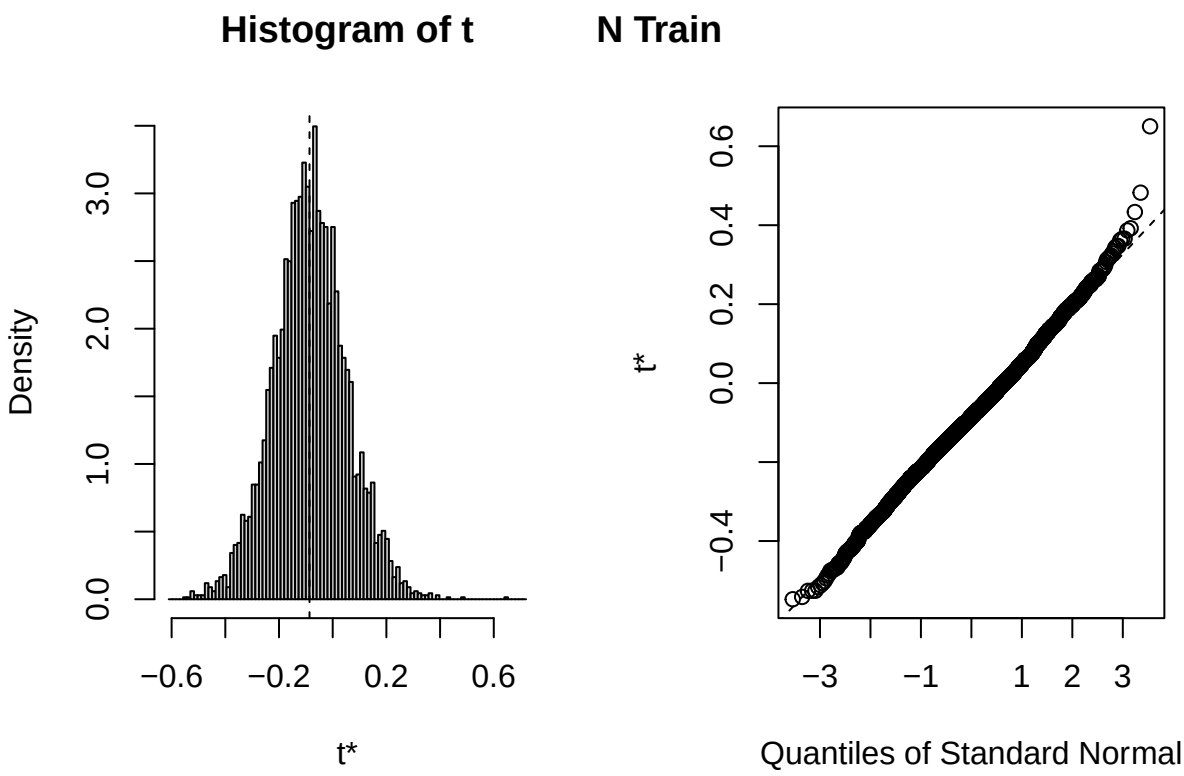
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb_std, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-0.0661,  0.6656 )
## Calculations and Intervals on Original Scale
```

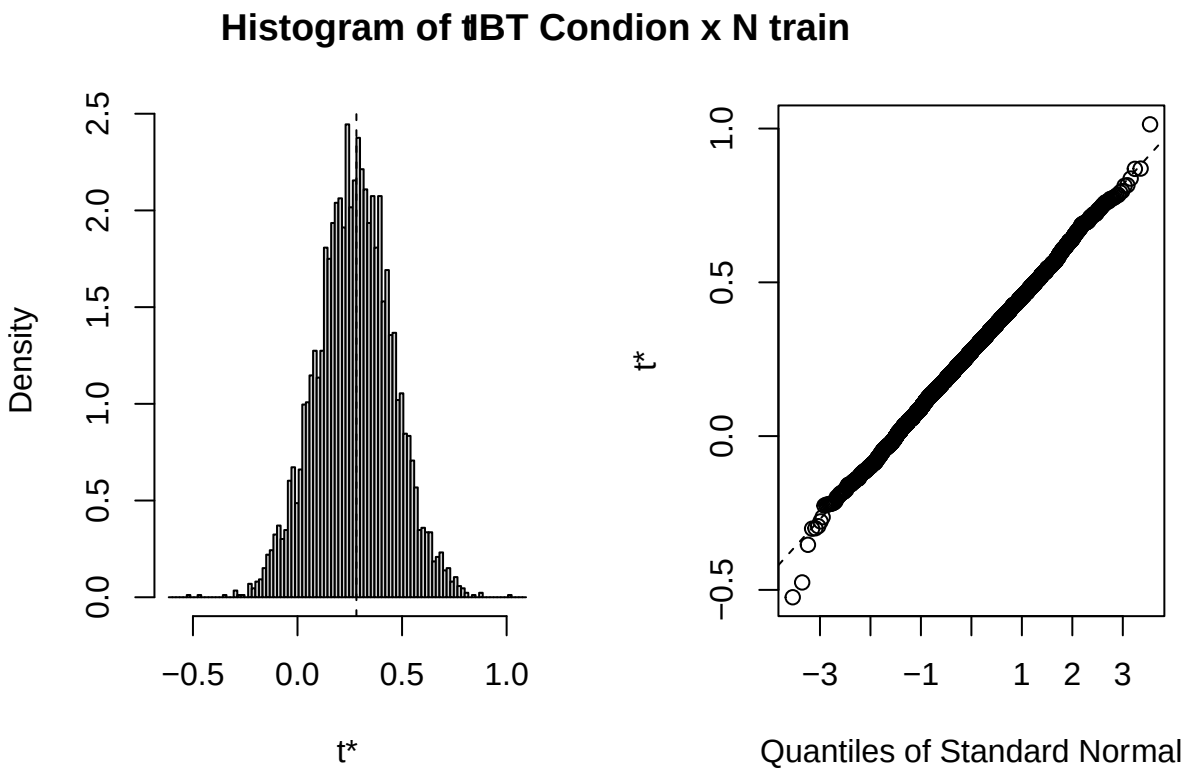
Bootstrap Diagnostics




Histogram of t
Density
0
1
2
3
4
0.4
0.6
0.8
1.0
t*
T1
t*
0.4
0.6
0.8
1.0
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t IBT Condition
Density
0.0
0.5
1.0
1.5
2.0
2.5
−0.8
−0.4
0.0
0.4
t*
t*
0.4
0.2
−0.2
−0.6
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
N Train
Density
0.0
1.0
2.0
3.0
−0.6
−0.2
0.2
0.6
t*
t*
0.0
0.2
0.4
0.6
−0.4
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of tBT Condion x N train
Density
0.0
0.5
1.0
1.5
2.0
2.5
−0.5
0.0
0.5
1.0
t*
t*
−0.5
0.0
0.5
1.0
−3
−1
1
2
3
Quantiles of Standard Normal